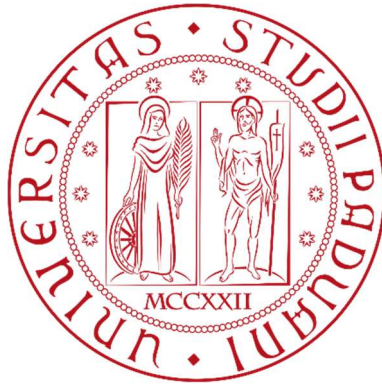




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Generative AI models: evaluations, comparison and critical aspects.

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ABSTRACT

This thesis explores the transformative potential of Generative Artificial Intelligence (AI), a cutting-edge field within AI that enables the creation of original content such as text, images, audio, and video. Generative AI has revolutionized industries by automating creative processes, offering personalized user experiences, and enhancing efficiency in tasks ranging from content creation to incident management. The aim of this study is to provide a comprehensive understanding of Generative AI, its underlying technologies, applications, and the ethical considerations surrounding its development and deployment.

The thesis is structured as follows:

- Chapter 1: introduces the foundational concepts of Generative AI, comparing it to traditional AI, and explains the key technologies that make it possible, including neural networks, machine learning, and diffusion models.
- Chapter 2: provides a historical perspective on the evolution of Generative AI, tracing its development from the early days of artificial intelligence to the advent of advanced technologies like transformers and GANs.
- Chapter 3: focuses on Large Language Models (LLMs), delving into their inner workings, evaluation benchmarks, and the debate between open-source and closed-source development models.
- Chapter 4: examines closed-source AI models, analyzing the strengths, features, and applications of leading platforms such as Claude, GPT-4 Turbo, and Gemini Ultra. A comparative analysis of their costs and functionalities is also provided.
- Chapter 5: shifts attention to open-source AI, exploring innovations like Llama 3.1, Google's Gemma, and Mistral, highlighting their role in democratizing AI access and fostering community-driven development.
- Chapter 6: expands the discussion to Generative AI applications beyond text, including image generation with tools like DALL-E and AI-powered assistance through platforms like Sora.
- Chapter 7: investigates platforms that facilitate access to Generative AI, such as Amazon Bedrock and Hugging Face, showcasing their distinct approaches to enabling developers and businesses.
- Chapter 8: concludes with case studies that demonstrate real-world applications of Generative AI. These examples illustrate how AI accelerates incident response at Google and drives rapid development across industries.

By analyzing the state of Generative AI, this thesis aims to provide insights into its capabilities, limitations, and potential, offering recommendations for leveraging this transformative technology responsibly and effectively.

1 Understanding Generative AI: Foundations and Core Technologies

This chapter provides a foundational overview of generative AI, exploring its core concepts and the key technologies that have driven its evolution. By comparing traditional AI with generative AI, it highlights the strengths and capabilities of generative models. Additionally, the chapter delves into the interconnections between artificial intelligence, machine learning, deep learning, and neural networks, offering insights into the technological frameworks that enable the creation of original content. This section serves as a starting point for understanding the theoretical and practical underpinnings that are expanded upon in subsequent chapters.

1.1 Introduction to generative AI

Generative AI, sometimes called *gen AI*, is artificial intelligence (AI) that can create original content—such as text, images, video, audio or software code—in response to a user’s prompt or request.

Generative AI relies on sophisticated machine learning models called *deep learning models*—algorithms that simulate the learning and decision-making processes of the human brain. These models work by identifying and encoding the patterns and relationships in huge amounts of data, and then using that information to understand users' natural language requests or questions and respond with relevant new content.



Figure 1 - AI representation (created by Dall-E)

1.2 Comparison between Generative AI and Traditional AI

Characteristics of traditional AI

Traditional AI, focuses on performing a specific task intelligently. It refers to systems designed to respond to a particular set of inputs. These systems have the capability to learn from data and make decisions or predictions based on that data (Marr, 2023).

The main characteristics of Traditional AI include:

- Programmed intelligence – Traditional AI works based on preprogrammed algorithms and rules. The system provides solutions and performs tasks within the limitations of its algorithm developed by programmers.
- Restricted applications – These AI models are designed with a specific set of tasks in mind, limiting their scope of potential applications.
- Data analysis – Traditional AI focuses on analyzing sets of data and making predictions based on this analysis. It can be successfully used for creating forecasts and other data analysis.
- Limited learning capabilities – The learning capabilities of Narrow AI are limited and dependent on data sets inputted by the human creator.

Imagine you're playing computer chess. The computer knows all the rules; it can predict your moves and make its own based on a pre-defined strategy. It's not inventing new ways to play chess but selecting from strategies it was programmed with. That's traditional AI - it's like a master strategist who can make smart decisions within a specific set of rules.

1.3 Generative AI: features and capabilities

Generative AI, on the other hand, can be thought of as the next generation of artificial intelligence. It's a form of AI that can create something new.

Some of the most important characteristics of generative AI are:

- Neural network generators – Generative AI utilizes neural networks like GANs and VAEs to generate relevant and original output.
- Varied applications – These AI models are adaptable and versatile, producing different types of content. They have a wide range of applications across a large number of industries.
- Creating new content using prompts – Generative AI uses data and prompts to create new content instead of simply analyzing existing data. The output is unique and relevant to the prompt used to generate it.

Suppose you have a friend who loves telling stories. But instead of a human friend, you have an AI. You give this AI a starting line, say, 'Once upon a time, in a galaxy far away...'. The AI takes that line and generates a whole space adventure story, complete

with characters, plot twists, and a thrilling conclusion. The AI creates something new from the piece of information you gave it.

Generative AI models are trained on a set of data and learn the underlying patterns to generate new data that mirrors the training set.

Key Difference: Traditional AI systems are primarily used to analyze data and make predictions, while generative AI goes a step further by creating new data similar to its training data (Wohlfarth, 2024).

1.4 Relationship among AI, machine learning, deep learning and neural networks

The easiest way to think about AI, machine learning, deep learning and neural networks is to think of them as a series of AI systems from largest to smallest, each encompassing the next (IBM, 2023).

It's the number of node layers, or depth, of neural networks that distinguishes a single neural network from a deep learning algorithm, which must have more than three.

Machine learning, deep learning, and neural networks are all sub-fields of artificial intelligence. However, neural networks is actually a sub-field of machine learning, and deep learning is a sub-field of neural networks, as we can see in the Figure 2.

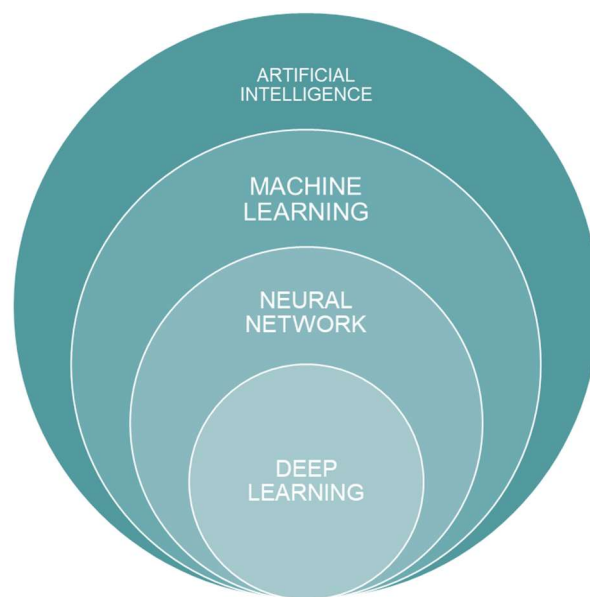


Figure 2 - Relationship between the technologies (Source: IBM)

1.4.1 Machine Learning

Machine learning is a subset of AI that allows for optimization. When set up correctly, it helps you make predictions that minimize the errors that arise from merely guessing.

For example, companies like Amazon use machine learning to recommend products to a specific customer based on what they've looked at and bought before.

Classic machine learning depends on human intervention to allow a computer system to identify patterns, learn, perform specific tasks and provide accurate results.

Human experts determine the hierarchy of features to understand the differences between data inputs, usually requiring more structured data to learn.

1.4.2 Understanding Neural Networks

A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, by using processes that mimic the way biological neurons work together to identify phenomena, weigh options and arrive at conclusions.

Every neural network consists of layers of nodes, or artificial neurons—an input layer, one or more hidden layers, and an output layer (Figure 3). Each node connects to others, and has its own associated weight and threshold. If the output of any individual node is above the specified threshold value, that node is activated, sending data to the next layer of the network. Otherwise, no data is passed along to the next layer of the network.

One of the best-known examples of a neural network is Google's search algorithm.

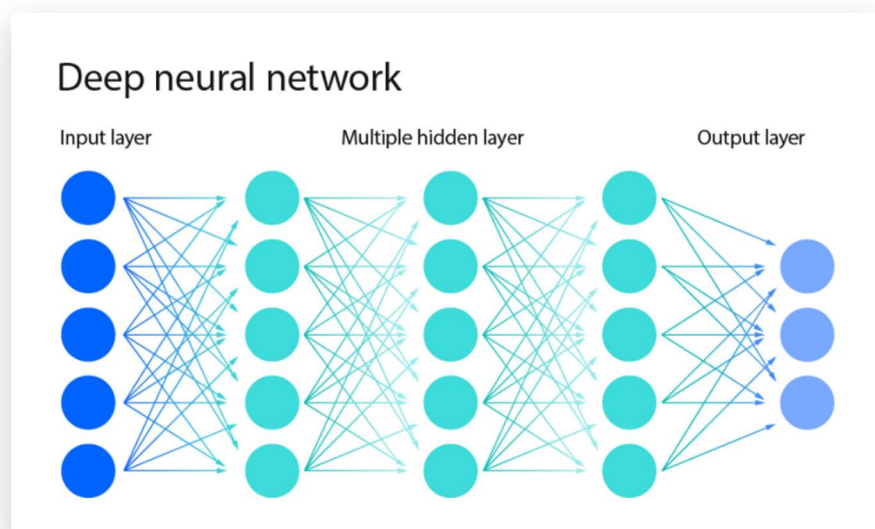


Figure 3 - Deep Neural Network representation (Source: IBM)

1.4.3 Deep Learning: Advancing Neural Network Capabilities

Deep learning is a subset of machine learning that uses multilayered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.

Deep learning models use three or more layers—but typically hundreds or thousands of layers—to train the models.

The deep learning process can ingest unstructured data in its raw form (e.g., text or images), and it can automatically determine the set of features which distinguish different categories of data from one another. This eliminates some of the human intervention required and enables the use of large amounts of data.

Deep neural networks consist of multiple layers of interconnected nodes, each building on the previous layer to refine and optimize the prediction or categorization. This progression of computations through the network is called forward propagation. The input and output layers of a deep neural network are called *visible* layers. The input layer is where the deep learning model ingests the data for processing, and the output layer is where the final prediction or classification is made.

Another process called backpropagation uses algorithms, such as gradient descent, to calculate errors in predictions, and then adjusts the weights and biases of the function by moving backwards through the layers to train the model. Together, forward propagation and backpropagation enable a neural network to make predictions and correct for any errors. Over time, the algorithm becomes gradually more accurate.

Mechanism of Forward Propagation

Forward propagation is the process in a neural network where the input data is passed through the network's layers to generate an output.

In order to generate some output, the input data should be fed in the forward direction only. The data should not flow in reverse direction during output generation otherwise it would form a cycle and the output could never be generated (anshumanm2fja, 2024).

The processing happens in two steps:

1. *Preactivation*: it is a *weighted sum of inputs* i.e. the *linear transformation of weights* w.r.t to inputs available. Based on this *aggregated sum* and *activation function* the neuron makes a decision whether to pass this information further or not.
2. *Activation*: the calculated weighted sum of inputs is passed to the activation function.

Backward propagation

In machine learning, backpropagation is an effective algorithm used to train artificial neural networks, especially in feed-forward neural networks.

Backpropagation is an iterative algorithm, that helps to minimize the cost function by determining which weights and biases should be adjusted. During every epoch, the model learns by adapting the weights and biases to minimize the loss by moving down toward the gradient of the error (Kalirane, 2023).

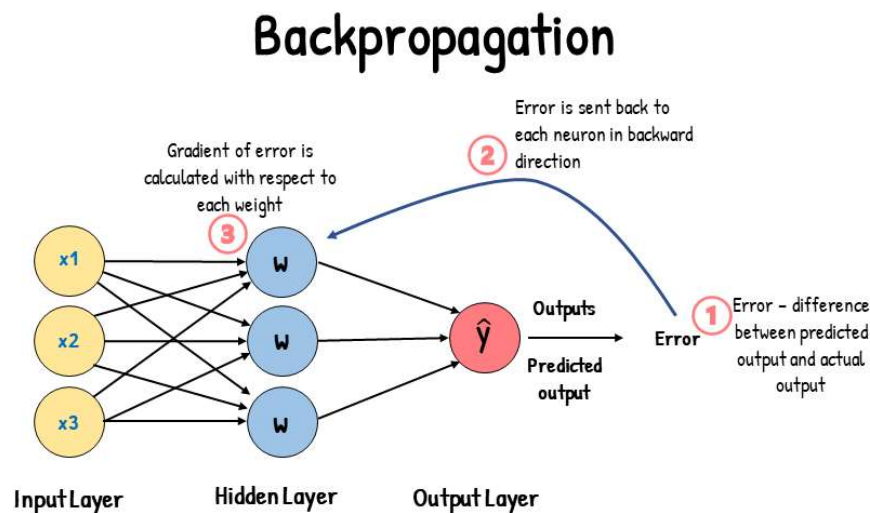


Figure 4 - Backward propagation representation (Source: Analytics Vidhya)

1.5 Introduction to Machine Learning

Machine learning (ML) is a branch of and computer science that focuses on the using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.

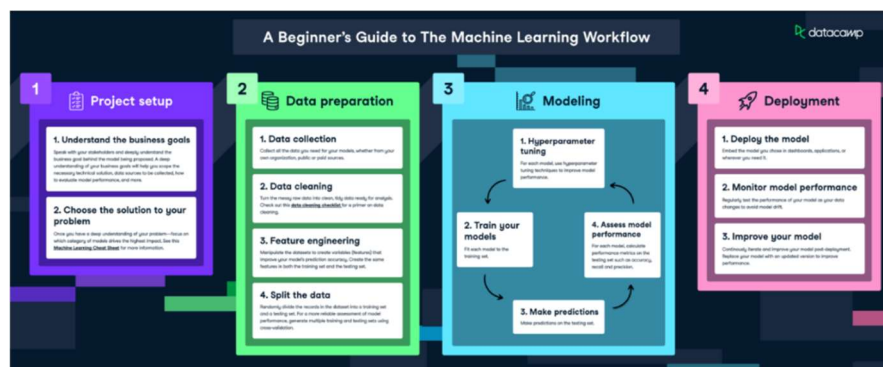


Figure 5 - Machine Learning workflow (Source: DataCamp)

How does machine learning work?

1. A Decision Process: In general, machine learning algorithms are used to make a prediction or classification. Based on some input data, which can be labeled or unlabeled, your algorithm will produce an estimate about a pattern in the data.
2. An Error Function: An error function evaluates the prediction of the model. If there are known examples, an error function can make a comparison to assess the accuracy of the model.
3. A Model Optimization Process: If the model can fit better to the data points in the training set, then weights are adjusted to reduce the discrepancy between the known example and the model estimate. The algorithm will repeat this iterative “evaluate and optimize” process, updating weights autonomously until a threshold of accuracy has been met.

Methods of Machine Learning

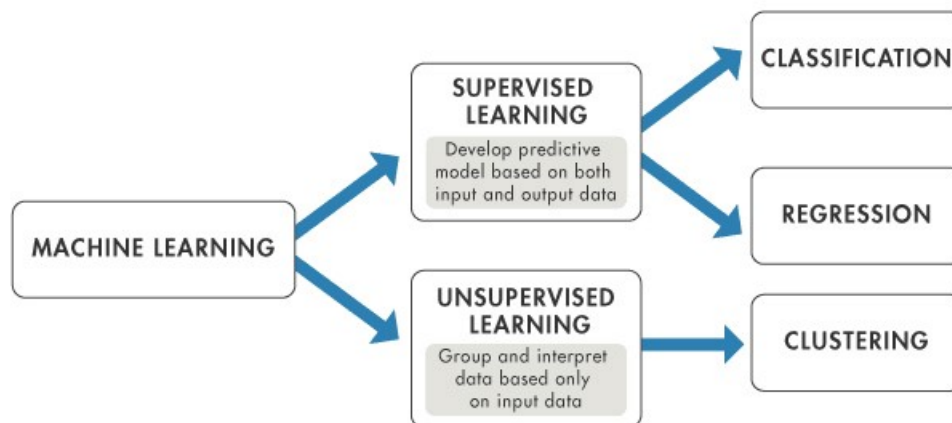


Figure 6 - Machine Learning methods

- *Supervised Learning*, also known as supervised machine learning, is defined by its use of labeled datasets to train algorithms to classify data or predict outcomes accurately. As input data is fed into the model, the model adjusts its weights until it has been fitted appropriately. This occurs as part of the cross validation process to ensure that the model avoids overfitting or underfitting. Supervised learning helps organizations solve a variety of real-world problems at scale, such as classifying spam in a separate folder from your inbox. Supervised learning can be separated into two types of problems when data mining: classification and regression:
Classification problems use an algorithm to accurately assign test data into specific categories, such as separating apples from oranges.

Regression is another type of supervised learning method that uses an algorithm to understand the relationship between dependent and independent variables.

- *Unsupervised Learning*, also known as unsupervised machine learning, uses machine learning algorithms to analyze and cluster unlabeled datasets (subsets called clusters). These algorithms discover hidden patterns or data groupings without the need for human intervention. This method's ability to discover similarities and differences in information make it ideal for exploratory data analysis, cross-selling strategies, customer segmentation, and image and pattern recognition.
- *Semi-supervised learning* offers a happy medium between supervised and unsupervised learning. During training, it uses a smaller labeled data set to guide classification and feature extraction from a larger, unlabeled data set. Semi-supervised learning can solve the problem of not having enough labeled data for a supervised learning algorithm. It also helps if it's too costly to label enough data.
- *Reinforcement learning* is an interesting learning model, with the ability not just to learn how to map an input to an output but to map a series of inputs to outputs with dependencies.
This model learns as it goes by using trial and error. A sequence of successful outcomes will be reinforced to develop the best recommendation or policy for a given problem.

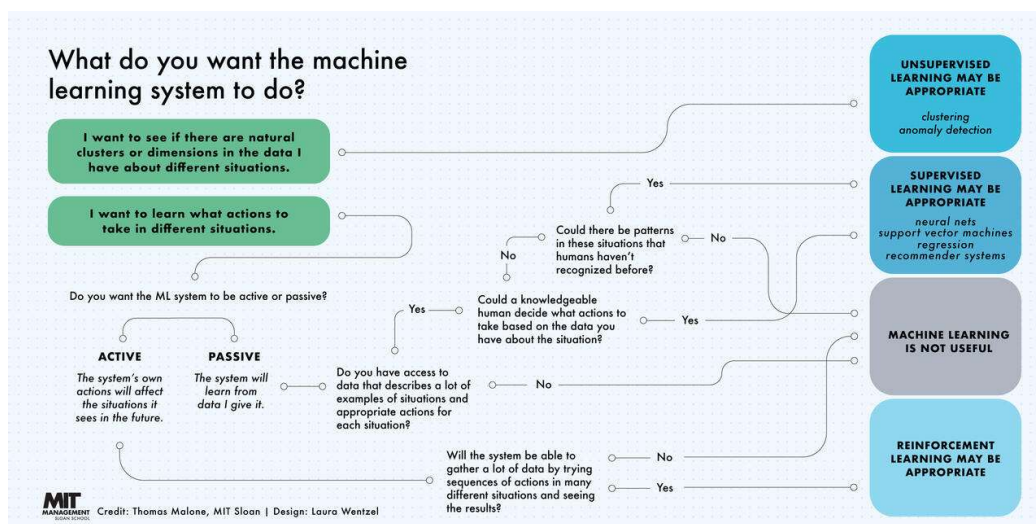


Figure 7 - Understanding the best Machine Learning models for you (Source: Thomas Malone, MIT Sloan)

1.6 How generative AI works

For the most part, generative AI operates in three phases:

- Training: to create a foundation model that can serve as the basis of multiple gen AI applications.
- Tuning: to tailor the foundation model to a specific gen AI application.
- Generation: evaluation and retuning, to assess the gen AI application's output and continually improve its quality and accuracy.

Training

Generative AI begins with a foundation model—a deep learning model that serves as the basis for multiple different types of generative AI applications. The most common foundation models today are large language models (LLMs), created for text generation applications.

To create a foundation model, practitioners train a deep learning algorithm on huge volumes of raw, unstructured, unlabeled data. Training, the algorithm performs and evaluates millions of ‘fill in the blank’ exercises, trying to predict the next element in a sequence—e.g., the next word in a sentence, the next element in an image, the next command in a line of code—and continually adjusting itself to minimize the difference between its predictions and the actual data (or ‘correct’ result).

The result of this training is a neural network of *parameters* that can generate content autonomously in response to inputs, or prompts.

Tuning

Metaphorically speaking, a foundation model is a generalist: It knows a lot about a lot of types of content, but often can’t generate specific types of output with desired accuracy or fidelity. For that, the model must be tuned to a specific content generation task. This can be done in a variety of ways.

Fine tuning

Fine tuning involves feeding the model labeled data specific to the content generation application—questions or prompts the application is likely to receive, and corresponding correct answers in the desired format. For example, if a development team is trying to create a customer service chatbot, it would create hundreds or thousands of documents containing labeled customers service questions and correct answers, and then feed those documents to the model.

Fine-tuning is labor-intensive. Developers often outsource the task to companies with large data-labeling workforces.

Reinforcement learning with human feedback (RLHF)

In RLHF, human users respond to generated content with evaluations the model can use to update the model for greater accuracy or relevance. Often, RLHF involves people ‘scoring’ different outputs in response to the same prompt. But it can be as simple as having people type or talk back to a chatbot or virtual assistant, correcting its output.

1.7 Diffusion Models in Machine Learning

They are named for their similarity to the natural diffusion process in physics, which describes how molecules move from high-concentration to low-concentration areas. In the context of machine learning, diffusion models generate new data by reversing a diffusion process, i.e., information loss due to noise intervention. The main idea here is to add random noise to data and then undo the process to get the original data distribution from the noisy data.

Diffusion models are advanced machine learning algorithms that uniquely generate high-quality data by progressively adding noise to a dataset and then learning to reverse this process. This innovative approach enables them to create remarkably accurate and detailed outputs, from lifelike images to coherent text sequences. Central to their function is the concept of gradually degrading data quality, only to reconstruct it to its original form or transform it into something new (Weng, 2021).

How diffusion models work:

The process behind diffusion models in machine learning is intricate, involving multiple steps that work together to enable the model to generate new, high-quality data. To fully grasp how diffusion models in machine learning function, it's essential to explain the process of diffusion and explore each component of the model. Understanding the process of diffusion involves examining how these components interact to achieve coherent and realistic outputs, highlighting the systematic addition and removal of noise that characterizes this approach.

Data Preprocessing

Before a diffusion model can begin generating data, it must first process the input data through a series of preprocessing steps. These steps are critical as they ensure the data is formatted correctly and can be transformed during the diffusion process. Data preprocessing typically involves:

- **Normalization:** Scaling data to a standard range to enhance model performance.
- **Noise Scheduling:** Planning the addition of noise at incremental stages, which helps manage the corruption process effectively.
- **Augmentation:** Applying transformations like rotations or cropping to diversify the data and improve model generalization.

Forward Diffusion Process

In the forward diffusion process, the model gradually transforms data by adding Gaussian noise at each step, pushing it closer to a pure noise distribution. This process is mathematically represented as a Markov chain, where each step's output depends on the previous step's noisy state.

Reverse Diffusion Process

The reverse diffusion process is where the model truly shines, as it reconstructs the data by removing noise step-by-step. This phase involves learning the noise distribution and reversing the Markov process, allowing the model to generate new data from random noise.

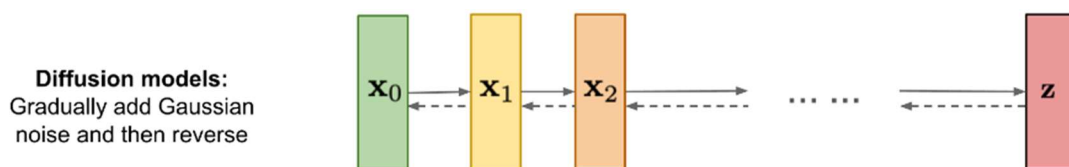


Figure 8 - Diffusion Models representation (Source: Github)

1.8 Generative Adversarial Network (GANs)

A generative adversarial network (GAN) is a deep learning architecture. It trains two neural networks to compete against each other to generate more authentic new data from a given training dataset. For instance, you can generate new images from an existing image database or original music from a database of songs. A GAN is called *adversarial* because it trains two different networks and pits them against each other (Amazon, n.d.). One network generates new data by taking an input data sample and modifying it as much as possible. The other network tries to predict whether the generated data output belongs in the original dataset. In other words, the predicting network determines whether the generated data is fake or real. The system generates newer, improved versions of fake data values until the predicting network can no longer distinguish fake from original.

How does a generative adversarial network work?

A generative adversarial network system comprises two deep neural networks—the *generator network* and the *discriminator network*. Both networks train in an adversarial game, where one tries to generate new data and the other attempts to predict if the output is fake or real data.

Technically, the GAN works as follows. A complex mathematical equation forms the basis of the entire computing process, but this is a simplistic overview:

1. The generator neural network analyzes the training set and identifies data attributes
2. The discriminator neural network also analyzes the initial training data and distinguishes between the attributes independently
3. The generator modifies some data attributes by adding noise (or random changes) to certain attributes
4. The generator passes the modified data to the discriminator
5. The discriminator calculates the probability that the generated output belongs to the original dataset
6. The discriminator gives some guidance to the generator to reduce the noise vector randomization in the next cycle

The generator attempts to maximize the probability of mistake by the discriminator, but the discriminator attempts to minimize the probability of error. In training iterations, both the generator and discriminator evolve and confront each other continuously until they reach an equilibrium state. In the equilibrium state, the discriminator can no longer recognize synthesized data. At this point, the training process is over.

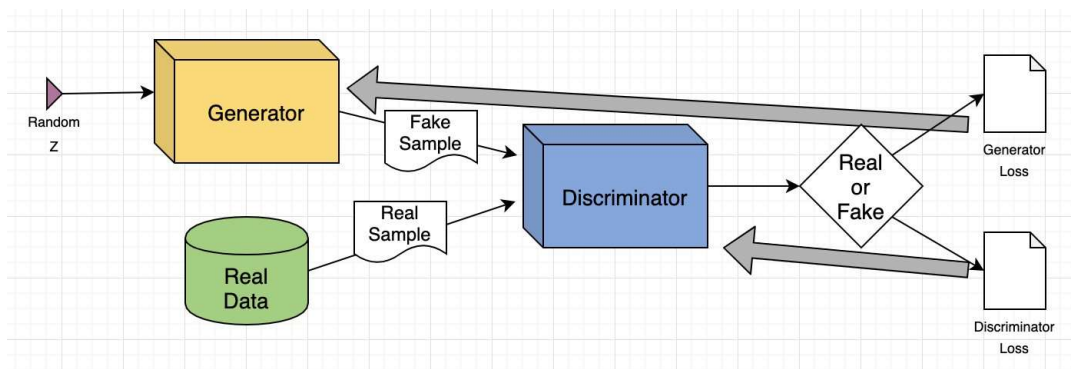


Figure 9 - GANs mechanism (Source: Amazon Web Service)

EXAMPLE: Consider that the input image is a human face that the GAN attempts to modify. For example, the attributes can be the shapes of eyes or ears. Let's say the generator changes the real images by adding sunglasses to them. The discriminator receives a set of images, some of real people with sunglasses and some generated images that were modified to include sunglasses.

If the discriminator can differentiate between fake and real, the generator updates its parameters to generate even better fake images. If the generator produces images that fool the discriminator, the discriminator updates its parameters. Competition improves both networks until equilibrium is reached.

1.9 Autoencoder and Latent Space

VAEs are a subset of the larger category of autoencoders, a neural network architecture typically used in deep learning for tasks such as data compression, image denoising, anomaly detection and facial recognition.

Autoencoders are self-supervised systems whose training goal is to compress (or *encode*) input data through dimensionality reduction and then accurately reconstruct (or *decode*) their original input by using that compressed representation (Bergmann & Stryker, 2023).

On a fundamental level, the function of an autoencoder is to effectively extract the data's most salient information—its latent variables—and discard irrelevant noise.

In training, the encoder network passes input data from the training data set through a "bottleneck" before it reaches the decoder. The decoder network, in turn, is then responsible for reconstructing the original input by using only the vector of latent variables.

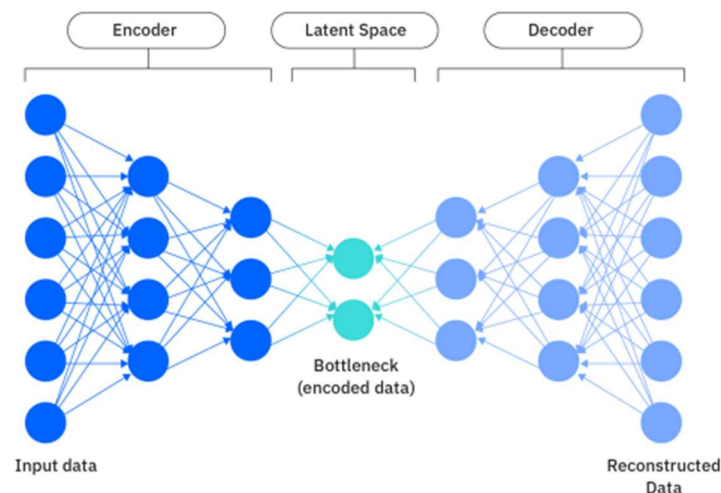


Figure 10 - Autoencoder and Latent Space

Latent Space

Latent space is a lower-dimensional space that captures the essential features of the input data. In simpler terms, it is a compressed representation of the original data where each dimension corresponds to a specific feature or characteristic. This dimensionality reduction is achieved through various techniques, such as autoencoders and variational autoencoders (VAEs), which learn to encode the most important information in the data.

Essential to understanding VAEs or any other type of autoencoders is the notion of *latent space*, the name given to the collective *latent variables* of a specific set of

input data. In short, latent variables are underlying variables of data that inform the way the data is distributed but are often not directly observable.

For a useful visualization of the concept of latent variables, imagine a bridge with a sensor that measures the weight of each passing vehicle. Naturally, there are different kinds of vehicles that use the bridge, from small, lightweight convertibles to huge, heavy trucks. Because there is no camera, we have no way to detect if a specific vehicle is a convertible, sedan, van or truck. However, we do know that the type of vehicle significantly influences that vehicle's weight.

Autoencoders model latent space through dimensionality reduction: the compression of data into a lower-dimensional space that captures the meaningful information contained in the original input.

Autoencoder structure

The encoder extracts latent variables of input data x and outputs them in the form of a vector representing latent space z . In a typical "vanilla" autoencoder, each subsequent layer of the encoder contains progressively fewer nodes than the previous layer; as data traverses each encoder layer, it's compressed through the process of "squeezing" itself into fewer dimensions (Ippolito, 2024).

The bottleneck, or "code," is both the output layer of the encoder network and the input layer of the decoder network. It contains the latent space: the fully compressed, lower-dimensional embedding of the input data. A sufficient bottleneck is necessary to help ensure that the decoder cannot simply copy or memorize the input data, which would nominally satisfy its training task but prevent the autoencoder from learning.

The decoder uses that latent representation to reconstruct the original input by essentially reversing the encoder: in a typical decoder architecture, each subsequent layer contains a progressively *larger* number of active nodes.

1.10 Transformers: Foundation of Modern AI

Transformers were first developed to solve the problem of sequence transduction, or neural machine translation, which means they are meant to solve any task that transforms an input sequence to an output sequence. This is why they are called "Transformers".

A transformer model is a neural network that learns the context of sequential data and generates new data out of it.

To put it simply: *A transformer is a type of artificial intelligence model that learns to understand and generate human-like text by analyzing patterns in large amounts of text data.* (Merritt, 2022)

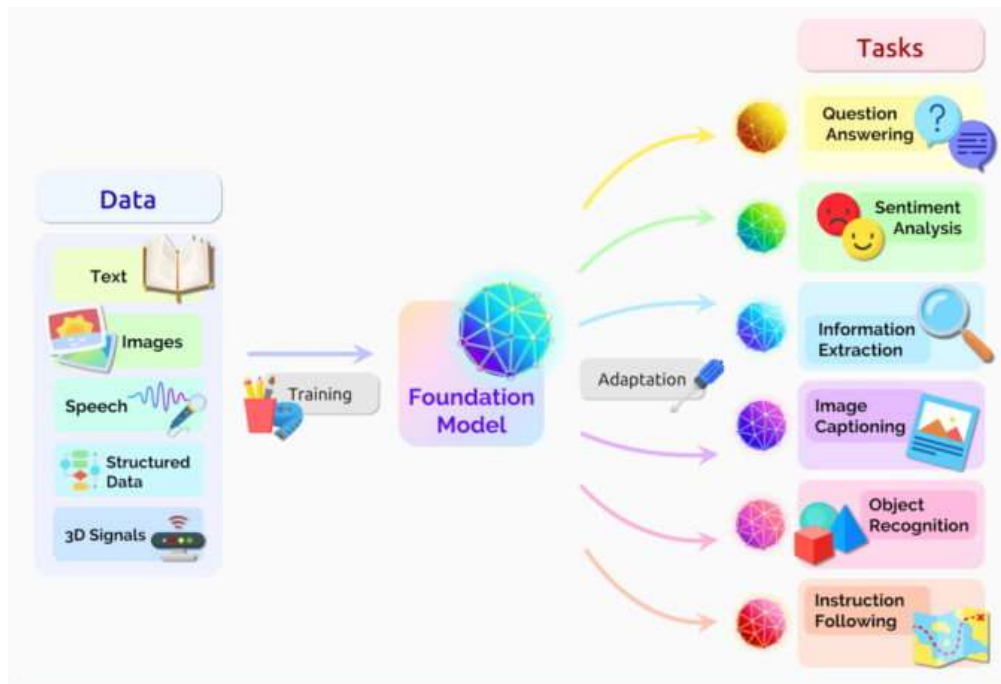


Figure 11 - Transformers, sometimes called foundation models, are already being used with many data sources for a host of applications (Source: Nvidia)

The transformer architecture

Originally devised for sequence transduction or neural machine translation, transformers excel in converting input sequences into output sequences.

The main core characteristic of the Transformers architecture is that they maintain the encoder-decoder model.

If we start considering a Transformer for language translation as a simple black box, it would take a sentence in one language, English for instance, as an input and output its translation in Spanish (Ferrer, 2024).

If we dive a little bit, we observe that this black box is composed of two main parts:

- The encoder takes in our input and outputs a matrix representation of that input. For instance, the English sentence “How are you?”
- The decoder takes in that encoded representation and iteratively generates an output. In our example, the translated sentence “¿Cómo estás?”

However, both the encoder and the decoder are actually a stack with multiple layers (same number for each). All encoders present the same structure, and the input gets into each of them and is passed to the next one. All decoders present the same structure as well and get the input from the last encoder and the previous decoder.

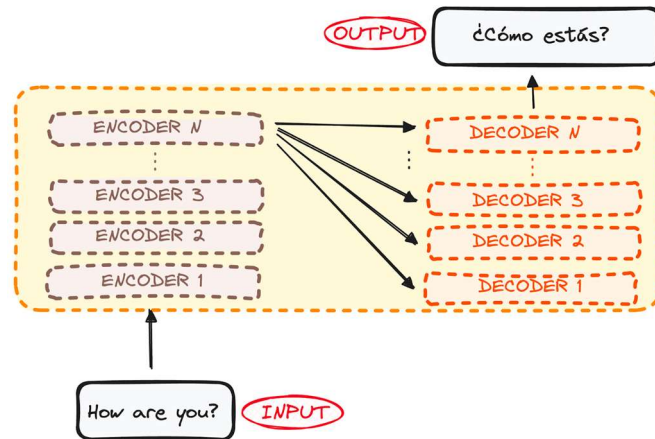


Figure 12 - Transformer Architecture (Source: datacamp)

There are two primary innovations that transformer models bring to the table. Consider these two innovations within the context of predicting text.

1. **Positional encoding:** Instead of looking at each word in the order that it appears in a sentence, a unique number is assigned to each word. This provides information about the position of each token (parts of the input such as words or subword pieces in NLP) in the sequence, allowing the model to consider the sequence's sequential information.

Sequence	Index of token	Positional Encoding Matrix			
I	0	P_{00}	P_{01}	...	P_{0d}
am	1	P_{10}	P_{11}	...	P_{1d}
a	2	P_{20}	P_{21}	...	P_{2d}
Robot	3	P_{30}	P_{31}	...	P_{3d}

Positional Encoding Matrix for the sequence 'I am a robot'

Figure 13 - Positional Encoding mechanism

2. **Self-attention:** Attention is a mechanism that calculates weights for every word in a sentence as they relate to every other word in the sentence, so the model can predict words which are likely to be used in sequence. This understanding is learned over time as a model is trained on lots of data. The self-attention mechanism allows each word to attend to every other word in the sequence in

parallel, weighing their importance for the current token. In this way, it can be said that machine learning models can “learn” the rules of grammar, based on statistical probabilities of how words are typically used in language.

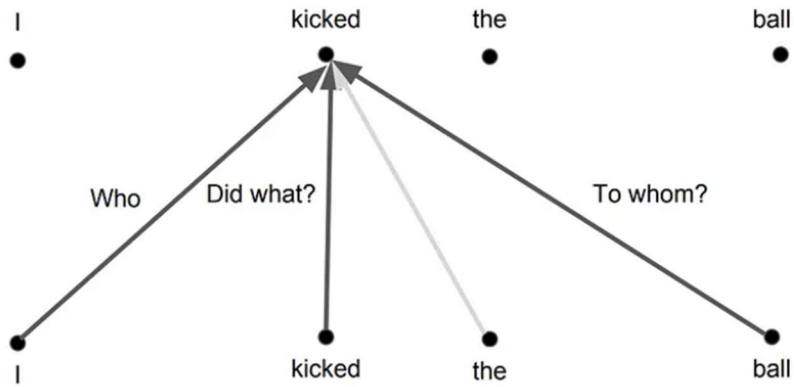


Figure 14 - Self-Attention mechanism

2 The Evolution of Generative AI: Milestones and Historical Perspectives

This chapter traces the historical development of generative AI, from its origins in early artificial intelligence research to its modern advancements. It outlines key milestones and breakthroughs that have shaped the field, offering context for how generative AI has evolved into a transformative technology. Additionally, the chapter introduces the Turing Test, a pivotal concept for evaluating machine intelligence, and discusses its relevance in understanding the capabilities and limitations of AI. This historical perspective provides a comprehensive background, enriching the reader's understanding of how generative AI has progressed over time.

2.1 The Evolution of Generative AI: A Historical Perspective

The Foundations of Artificial Intelligence: 1950s - 1970s

The journey of generative AI is deeply rooted in the broader development of artificial intelligence (AI), which began in the mid-20th century. Alan Turing, a pioneer in computing, introduced the concept of "intelligent machinery" in 1947 and later proposed the famous Turing Test in 1950, exploring whether machines could exhibit intelligent behavior indistinguishable from humans. This foundational work laid the groundwork for what would later become artificial intelligence as a discipline. (Toloka, 2023) (Foote, 2024)

The official birth of AI as a scientific field occurred in 1956 during the Dartmouth Conference, where researchers like John McCarthy coined the term "artificial intelligence." Soon after, significant advances followed. Arthur Samuel developed the first machine learning algorithm in 1952 for playing checkers, introducing the term "machine learning." Meanwhile, Frank Rosenblatt created the perceptron in 1957, an early neural network model designed to recognize patterns, which became a fundamental concept in machine learning.

In the 1960s, John McCarthy developed the LISP programming language specifically for AI research. During this period, early generative AI emerged with Joseph Weizenbaum's creation of ELIZA in 1961, a primitive chatbot capable of responding to human inputs in natural language. ELIZA's pattern-based responses, while limited, marked the first instance of a generative AI model, inspiring future advances in natural language processing (NLP) and the development of interactive AI applications.

Generative AI and Early Neural Networks: 1970s - 1980s

As AI research progressed, the 1970s introduced several breakthroughs and challenges. The term "backpropagation" emerged, an essential algorithm for training deep neural networks. Despite these advancements, AI experienced its first "winter" between 1973 and 1979, marked by reduced funding and disillusionment over unmet expectations.

However, machine learning continued to evolve as businesses recognized its utility in automating simple tasks.

In 1975, Kunihiko Fukushima developed the Cognitron, a multilayered neural network that could recognize visual patterns, paving the way for deep learning. Following this, his creation of the Neocognitron in 1979 enabled handwritten character recognition, setting the stage for more sophisticated pattern recognition systems. Meanwhile, the concept of neural networks gained further attention, as the introduction of models with hidden layers allowed for more complex data transformation processes, crucial for advancing both AI and generative AI.

During the 1980s, generative AI development slowed as AI encountered its second winter, a period of widespread skepticism. Nevertheless, innovations like the Hopfield network in 1982, which simulated associative memory, kept the field alive. In 1986, David Rumelhart and colleagues popularized backpropagation for training neural networks, leading to improvements in image and speech recognition.

Rise of Deep Learning and Modern Generative AI: 1990s - 2000s

In the 1990s, AI research rebounded, driven by advancements in machine learning and the rise of the internet, which provided vast amounts of data for training algorithms. Notably, the computer gaming industry contributed significantly to generative AI by developing GPUs (graphics processing units) to render 3D graphics. Researchers discovered that GPUs could also accelerate neural network training, making deep learning more feasible.

During this period, recurrent neural networks (RNNs) gained traction for tasks requiring sequential data processing, such as language modeling and speech recognition. The introduction of Long Short-Term Memory (LSTM) networks by Juergen Schmidhuber and Sepp Hochreiter in 1997 addressed issues with long-term dependency retention, which enhanced AI's ability to generate coherent text sequences and remember context across sentences.

In parallel, advancements in computer vision, particularly facial recognition, were accelerated by governmental initiatives like the U.S. Face Recognition Grand Challenge (2004-2006). The 2000s also saw the rise of practical AI applications, like Dragon Systems' voice recognition and IBM's DeepBlue defeating the world chess champion in 1997, illustrating AI's growing power. As deep learning became more sophisticated, it laid the foundation for modern generative AI models capable of creating complex and realistic outputs.

The Advent of Generative Adversarial Networks and Transformers: 2010s

The 2010s were a transformative decade for generative AI, largely due to the development of two revolutionary architectures: Generative Adversarial Networks (GANs) and transformers. In 2014, Ian Goodfellow introduced GANs, a breakthrough that fundamentally changed generative AI by enabling high-quality image, video, and audio generation. GANs consist of two neural networks—the generator, which creates

content, and the discriminator, which evaluates it. Through adversarial training, GANs produce outputs that are increasingly realistic, making them ideal for applications in media, entertainment, and synthetic data generation.

Another monumental leap came in 2017 with the introduction of transformer models. Transformers, designed for handling sequential data, enabled parallel processing of text, which drastically reduced training times and improved the scalability of NLP tasks. These models revolutionized the field by allowing for the creation of large language models (LLMs) such as BERT and GPT. Transformers became the backbone of many generative AI systems, extending applications from text generation to vision tasks with the development of vision transformers.

In 2018, OpenAI launched GPT, a generative pre-trained transformer that leveraged massive datasets to generate coherent and contextually relevant text. This model laid the groundwork for GPT-2 and GPT-3, which followed in 2019 and 2020, showcasing unprecedented abilities in text generation, summarization, and translation. Transformers unlocked new possibilities for generative AI, accelerating its integration across various fields and establishing it as a critical component in AI research and development.

Recent Breakthroughs: 2020s and the Rise of Large Language Models

The 2020s have continued to witness rapid advancements in generative AI, particularly with the release of large language models (LLMs) like GPT-4. In 2022, OpenAI introduced ChatGPT, an AI model that leveraged LLM capabilities to produce human-like conversational responses, setting a new benchmark in interactive AI. ChatGPT's success demonstrated generative AI's potential in customer service, content creation, and information retrieval, making AI-powered communication accessible to millions.

Alongside LLMs, text-to-image models like DALL-E, developed by OpenAI in 2021, enabled the generation of high-quality images from textual descriptions, showcasing generative AI's versatility beyond text. DALL-E's architecture combined the transformer with additional neural networks, like CLIP and GLIDE, to produce images that could be edited, expanded, or modified based on user inputs.

The year 2022 also saw the release of Stable Diffusion by Stability.ai, an open-source model that democratized access to text-to-image technology, and Midjourney, which applied diffusion models to create artistic images. Diffusion models work by iteratively adding noise to an image and then reversing the process, allowing for the generation of photorealistic visuals from scratch.

Today, generative AI models are increasingly used across industries for various applications, including media, healthcare, and finance. The ability of these models to create content indistinguishable from human-made work has opened discussions on ethics, copyright, and the impact of AI-generated media.

Timeline of Major Milestones in Generative AI Development

- **1956:** AI established as a scientific field during the Dartmouth Conference.
- **1961:** ELIZA chatbot created, an early example of generative AI in NLP.
- **1979:** Introduction of the Neocognitron, the first deep learning network.
- **1997:** Development of LSTM networks for improved sequence modeling.
- **2014:** Ian Goodfellow introduces GANs, enabling realistic content generation.
- **2017:** Introduction of transformer models, revolutionizing NLP and generative AI.
- **2018:** OpenAI launches GPT, pioneering large language models.
- **2021:** DALL-E launched by OpenAI, enabling text-to-image generation.
- **2022:** Release of Stable Diffusion and Midjourney, democratizing image generation.
- **2023:** GPT-4 released, capable of producing extensive, coherent text responses.

2.2 The Turing Test: Evaluating Machine Intelligence

What is the Turing Test in AI?

The Turing Test is a method of inquiry in artificial intelligence for determining whether or not a computer is capable of thinking like a human being. The test is named after Alan Turing, the founder of the Turing Test and an English computer scientist, cryptanalyst, mathematician and theoretical biologist.

Turing proposed that a computer can be said to possess artificial intelligence if it can mimic human responses under specific conditions. The original Turing Test requires three terminals, each of which is physically separated from the other two. One terminal is operated by a computer, while the other two are operated by humans.

During the test, one of the humans functions as the questioner, while the second human and the computer function as respondents. The questioner interrogates the respondents within a specific subject area, using a specified format and context. After a preset length of time or number of questions, the questioner is then asked to decide which respondent was human and which was a computer.

The test is repeated many times. If the questioner makes the correct determination in half of the test runs or less, the computer is considered to have artificial intelligence because the questioner regards it as "just as human" as the human respondent.

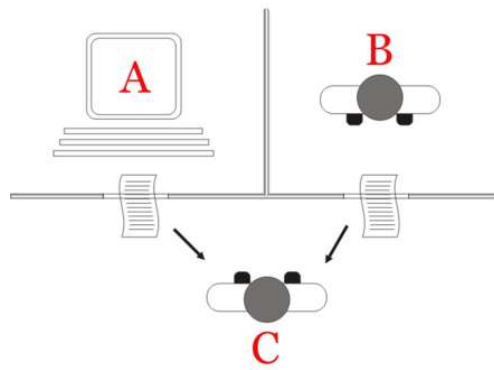


Figure 15 - Turing Test (Source: Google)

History of the Turing Test

The test is named after Alan Turing, who pioneered machine learning during the 1940s and 1950s. Turing introduced the test in his 1950 paper called "Computing Machinery and Intelligence" while at the University of Manchester. (George & Gillis, 2024)

In his paper, Turing proposed a twist on what is called "The Imitation Game." The Imitation Game involves no use of AI, but rather three human participants in three separate rooms. Each room is connected via a screen and keyboard, one containing a male, the other a female, and the other containing a male or female judge. The female tries to convince the judge that she is the male, and the judge tries to disseminate which is which.

Turing changes the concept of this game to include an AI, a human and a human questioner. The questioner's job is then to decide which is the AI and which is the human. Since the formation of the test, many AI have been able to pass; one of the first is a program created by Joseph Weizenbaum called ELIZA.

Limitations of the Turing Test

The Turing Test has been criticized over the years, in particular because historically, the nature of the questioning had to be limited in order for a computer to exhibit human-like intelligence. For many years, a computer might only score high if the questioner formulated the queries, so they had "Yes" or "No" answers or pertained to a narrow field of knowledge. When questions were open-ended and required conversational answers, it was less likely that the computer program could successfully fool the questioner.

In addition, a program such as ELIZA could pass the Turing Test by manipulating symbols it does not understand fully. John Searle argued that this does not determine intelligence comparable to humans.

To many researchers, the question of whether or not a computer can pass a Turing Test has become irrelevant. Instead of focusing on how to convince someone they are conversing with a human and not a computer program, the real focus should be on how to make a human-machine interaction more intuitive and efficient. For example, by using a conversational interface.

How is the Turing Test used today?

Although the variations of the Turing Test are often more applicable to our current understanding of AI, the original format of the test is still used to this day. For example, the Loebner Prize has been awarded annually since 1990 to the most human-like computer program as voted by a panel of judges. The competition follows the standard rules of the Turing Test. Critics of the award's relevance often downplay it as more about publicity than truly testing if machines can think.

At a competition organized by the University of Reading to mark the 60th anniversary of Turing's death in 2014, a chatbot called Eugene Goostman that simulates a 13-year-old boy passed the Turing Test, in the eyes of some, when it fooled 33% of the judges. This so-called first pass has been met with much criticism from those who argue that there weren't enough judges, that other machines have performed better at the test in the past and that the test is invalid for only lasting five minutes.

In 2018, Google Duplex successfully made an appointment with a hairdresser over the phone in front of a crowd of 7,000. The receptionist was completely unaware that they weren't conversing with a real human. This is considered by some to be a modern-day Turing Test pass, despite not relying on the true format of the test as Alan Turing designed it.

GPT-3, a natural language processing model created by OpenAI, is thought by some to have the best chance of beating the test in its true form of any technology that we have today. But, even with its advanced text-generation abilities, many have criticized the machine because it can be tricked into answering nonsensical questions and therefore would struggle under the conditions of the Turing Test.

Despite much debate about the relevance of the Turing Test today and the validity of the competitions that are based around it, the test still stands as a philosophical starting point for discussing and researching AI. As we continue to make advances in AI and better understand and map how the human brain functions, the Turing Test remains foundational for defining intelligence and is a baseline for the debate about what we should expect from technologies for them to be considered thinking machines.

3 Understanding Large Language Models: Foundations, Comparisons, and Evaluations

This chapter delves into the mechanics of Large Language Models (LLMs), explaining how they process and generate human-like text. It compares open-source and closed-source approaches, highlighting their advantages and limitations. Furthermore, the chapter explores how LLMs are evaluated through benchmarks that measure reasoning, coding, multilingual capabilities, and other critical metrics. This comprehensive analysis provides a foundation for understanding the technical and practical aspects of LLMs, as well as their performance and constraints in various applications.

3.1 Understanding Large Language Models

Large language models (LLMs) are advanced AI algorithms that leverage deep learning techniques and massive datasets to understand, generate, and predict content, primarily in natural language. As a specific category within generative AI, LLMs are designed to process and produce human-like text, making them invaluable across a range of applications. Their training involves vast amounts of data and billions of parameters, allowing these models to capture complex linguistic patterns and perform a variety of tasks, from answering questions to generating original content.

Language has been central to human communication for millennia, serving as a structured way to convey ideas, thoughts, and emotions. In the field of AI, language models serve a similar foundational purpose by enabling machines to interpret and generate human language. The origins of AI language models trace back to the 1960s with early projects like ELIZA, a pioneering chatbot created at MIT in 1966 that could respond to user inputs in natural language. Although primitive, ELIZA laid the groundwork for modern language models by demonstrating the potential of AI-driven communication. (Kerner, 2024)



Figure 16 - Eliza chatbot (Source: Google)

1. Expanding Capabilities with Large Language Models

The development of large language models (LLMs) has marked a significant leap in AI, enabling these models to handle extensive data and perform a vast range of tasks. Modern LLMs consist of billions of parameters, with some reaching hundreds of billions, allowing them to capture intricate language patterns and generate human-like responses. Parameters serve as adjustable variables during training, which help the model infer relationships within the data, forming a robust foundation for generating accurate, contextually appropriate content across diverse applications. This capability makes LLMs powerful tools in generating, predicting, and summarizing language-based content, distinguishing them within the broader scope of generative AI.

2. Foundation Models and the Transformer Revolution

A breakthrough in LLMs emerged in 2017 with transformer-based neural networks, which facilitate rapid, accurate processing of language sequences. Transformers have been central to the success of LLMs, enabling them to handle large-scale tasks efficiently. This innovation made LLMs highly adaptable and, as a result, foundational in areas ranging from business and research to everyday communication.

LLMs are a type of “foundation model,” a term popularized by the Stanford Institute for Human-Centered AI in 2021, which refers to models that provide a versatile base for multiple applications, saving organizations time and resources by allowing one model to support various functions.

Notable examples include OpenAI’s GPT-3 and GPT-4, Meta’s LLaMA models, Google’s BERT and PaLM models. These models are utilized widely in applications such as customer support, real-time interaction, and content creation, showcasing their adaptability across domains.

3. Transforming Human-Computer Interaction and Future Impact

At their core, LLMs are designed for a broad spectrum of language-related tasks beyond mere text generation. Their capabilities include understanding context, answering questions coherently, summarizing documents, translating languages, and assisting with creative tasks such as coding or writing. This versatility has made them invaluable in fields like virtual assistance, research, and multilingual communication.

As LLMs evolve, they are expected to redefine the way humans interact with technology, providing accessible, seamless experiences across digital platforms. Their continuous improvement positions them as a cornerstone in the digital landscape, with the potential to reshape information access, communication methods, and interaction with intelligent systems on a global scale.

3.2 How Large Language Models (LLMs) Work

Large language models (LLMs) are powerful AI systems designed to understand and generate human-like text. Their development relies on deep learning techniques and extensive data, often using billions of pages of text to build a strong foundation. This training enables them to handle complex language tasks, such as answering questions, generating new content, summarizing information, or performing sentiment analysis. (IBM, 2023)

The Training Process: Building Language Understanding

The foundation of an LLM begins with a large corpus of text data, which can measure up to petabytes in size. Training usually starts with an unsupervised approach, meaning that the data doesn't have labels or predefined categories. This approach allows models to recognize natural patterns within the language, as it learns relationships between words and phrases on its own. In certain stages, LLMs may also use self-supervised learning, where some data is labeled, enabling the model to refine its understanding of specific concepts.

A critical component of LLM architecture is the transformer model, which excels at processing language sequences. The transformer's self-attention mechanism allows the model to focus on relevant parts of a sentence or phrase, assigning weights (or scores) to specific words (or tokens) to capture their relationships. This weighting process helps the model understand context, such as recognizing that "bank" might mean a financial institution or a riverbank based on surrounding words.

Tokens, Embeddings, and Prediction

To interpret language, LLMs first break down text into smaller parts, known as tokens, which are sequences of characters or words. These tokens are transformed into embeddings, numerical values that represent the context of the text. Using embeddings, the model predicts the next word in a sentence by assigning a probability to each possible word based on the words that came before it. This process, repeated across billions of examples, enables LLMs to "learn" grammar, semantics, and conceptual relationships within language.

As a result, once the model is fully trained, it can generate coherent and contextually appropriate language. By inputting a prompt or question, the LLM can produce responses based on the patterns it learned, drawing from its extensive language database. This prediction ability allows the model to handle tasks such as generating summaries, completing sentences, translating text, and even creating new content.

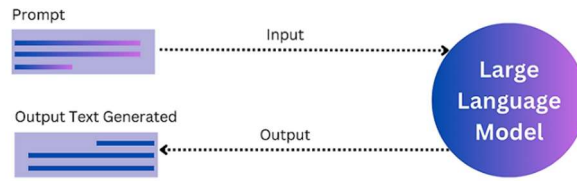


Figure 17 - LLM process representation (Source: Google)

Fine-Tuning and Reducing Errors

After the initial training, LLMs can be further improved through techniques like fine-tuning, where the model is adjusted to meet specific requirements. Another essential step is prompt engineering, which involves crafting prompts to help the model generate more accurate responses. Models can also undergo reinforcement learning with human feedback (RLHF) to reduce biases, prevent harmful language, and minimize errors known as "hallucinations," where the model may generate incorrect or misleading information.

By applying these methods, organizations ensure that LLMs are safe, reliable, and effective for professional and enterprise use, helping to prevent any unwanted outcomes or reputational risks. Through these enhancements, LLMs become robust tools capable of supporting a wide range of applications, from natural language understanding (NLU) to creative content generation, making them integral to modern AI-powered solutions.

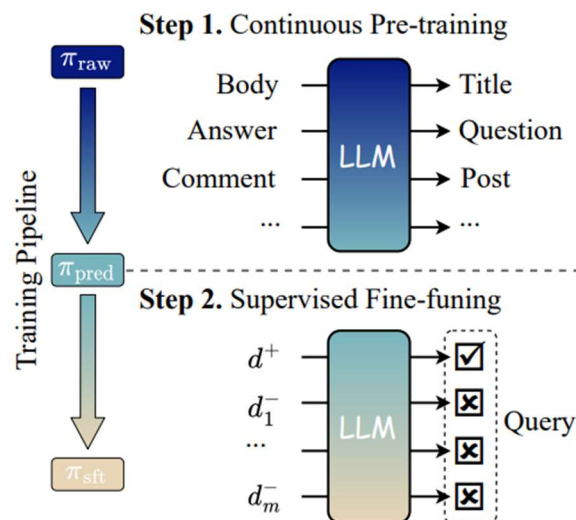


Figure 18 - Training pipeline (Source: Google)

Applications and Use Cases for Large Language Models (LLMs)

Large language models (LLMs) have become essential tools across industries, enabling a variety of language-related tasks that enhance efficiency, customer experience, and content creation. These AI models play a significant role in transforming business processes by converting text-based instructions into actions. Here are some of the most impactful applications:

1. **Conversational AI and Chatbots:** LLMs power advanced chatbots and virtual assistants like IBM Watsonx Assistant and Google's BARD, offering natural, context-aware responses to customer queries. These tools simulate human interaction, making customer support more responsive and autonomous by handling common inquiries in a self-service format.
2. **Text Generation:** LLMs are widely used for content creation, automating the writing of articles, marketing materials, and social media posts. They can produce text in response to prompts, ranging from simple emails to comprehensive reports, allowing companies to generate tailored content quickly and efficiently.
3. **Content Summarization:** Summarizing lengthy documents such as news articles, research reports, and corporate documentation is a key function of LLMs. This feature saves users time by condensing large volumes of text into concise summaries, making information easier to access and process.
4. **Language Translation:** Trained in multiple languages, LLMs can provide accurate, contextually relevant translations. This helps companies break down language barriers, expand their reach across regions, and communicate more effectively on a global scale.
5. **Code Generation and Review:** LLMs assist developers by generating code, identifying bugs, and suggesting fixes. They can even translate code between programming languages, making it easier for developers to adapt and optimize code across platforms and languages.
6. **Sentiment Analysis:** LLMs analyze text to detect the sentiment behind customer feedback, such as determining whether a review is positive or negative. This helps organizations better understand customer perceptions and manage brand reputation on a large scale.

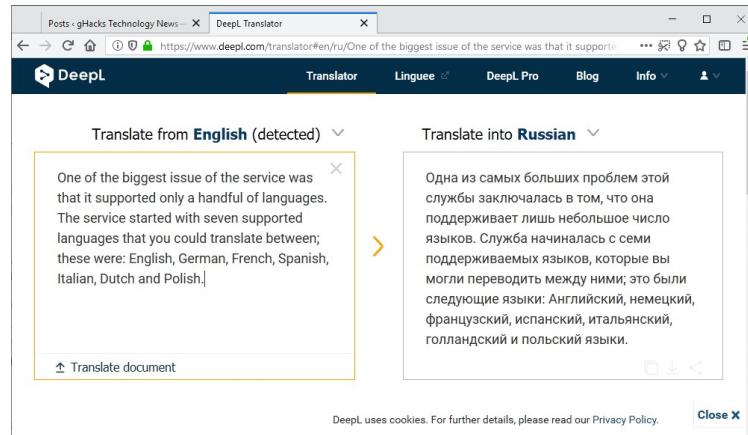


Figure 19 – Language Translation example

Advantages of Large Language Models

LLMs offer numerous benefits to organizations and users alike:

- **Extensibility and Adaptability:** LLMs serve as a foundation for custom applications. Additional training can tailor these models for specific tasks, making them adaptable to different organizational needs.
- **Flexibility:** A single LLM can perform multiple tasks, reducing the need for separate models for each application.
- **Performance:** Modern LLMs deliver high performance, providing low-latency, accurate responses across various tasks.
- **Ease of Training:** Many LLMs are trained on unlabeled data, which speeds up the training process.
- **Efficiency:** By automating repetitive tasks, LLMs help employees save time and focus on higher-value activities.

Challenges and Limitations of Large Language Models

Despite their many advantages, LLMs face certain challenges:

- **Development and Operational Costs:** LLMs require significant computational resources and data for training and operation, leading to high costs.
- **Bias and Ethical Concerns:** LLMs can perpetuate biases present in training data and occasionally generate harmful content, raising ethical issues.
- **Explainability:** The mechanisms behind LLM responses are often complex, making it difficult to explain how specific outputs were generated.
- **Hallucinations:** LLMs may occasionally produce inaccurate or irrelevant information not based on actual training data.
- **Security Risks:** LLMs can be exploited in phishing attacks, posing security concerns for organizations.

LLMs continue to evolve and reshape industries by improving processes, enhancing customer interactions, and supporting efficient, data-driven decision-making. As they advance, their transformative impact on business operations and digital communication will likely grow.

3.3 OPEN SOURCE VS CLOSED SOURCE

3.3.1 What is Open Source?

In general, *open source* refers to a “piece of knowledge,” often in the form of computer code, that is made publicly available and freely accessible. However, this concept is not restricted to software. For example, an algorithm can be described through natural language or mathematical formulas and then shared as public domain knowledge.

Open Source Software: What Can You Do with It?

Typically, open source code can be used in a wide variety of ways:

- **Freely Accessible:** Open source code can be used without cost.
- **Modifiable and Improvisable:** Users can adapt and improve the code.
- **Commercial Use:** Open source code can be utilized to build a business, though with some restrictions.

This code is usually hosted on public repositories like GitHub, where anyone can download and use it as a foundation for further work. However, with AI models, open source operates differently.

3.3.2 Open Weights vs. Open Source in AI Models

When an AI model is released as "open," such as Meta's Llama models, it often falls under the term “open weights,” rather than “open source.”

1. **Open Weights**

The company publicly releases the *weights* of the model, which are the numerical parameters that define it. In practice, these are contained in a single file storing these essential numerical values.

2. **Open Source**

Here, the entire source code used to develop and train the model is released, including the training data and the computational resources required for training.

Practical Implications: An Analogy

Consider baking a cake. With the "open weights" approach, you're provided with pre-measured and mixed ingredients (the model's weights), similar to a cake mix. You can use these ingredients to bake the cake, but you don't have insight into the exact

preparation process. In other words, you wouldn't be able to replicate the cake from scratch.

In contrast, with "open source," you have the full recipe: the ingredient list, step-by-step instructions, and the techniques required to bake the cake perfectly.

Current Practices Among AI Leaders

Notably, major companies like Meta and Mistral typically release models as open weights. Although these releases significantly contribute to AI development and offer a wide range of functionalities, they aren't fully "open source," as the models cannot be reproduced from scratch.

In this analogy, proprietary models like GPT, Gemini, and Claude are akin to ready-made cakes: the quality varies from budget options at a supermarket to premium options from high-end bakeries.

3.3.3 What is Closed Source?

In contrast to open source, *closed source* refers to software or technology whose underlying code is not publicly accessible. Instead, it remains proprietary, controlled solely by the organization or individual who created it.

Characteristics of Closed Source Software

Since the code is not made available to the public, closed source software usually has several key characteristics:

- **Restricted Access:** Only authorized individuals (typically the developers or licensed users) can view or modify the code.
- **Licensing Fees:** Users typically pay a licensing fee to access or use the software.
- **Proprietary Rights:** The owner maintains exclusive rights over the software, retaining control over updates, distribution, and customizations.

Why Keep Source Code Closed?

Organizations may choose to keep their software closed source for various reasons:

1. **Competitive Advantage:** Keeping the source code private preserves proprietary innovations and techniques.
2. **Revenue Generation:** Closed source licenses often provide a steady income stream through user subscriptions or purchase fees.
3. **Quality Control:** The company can ensure that any changes or updates maintain consistency and security across all versions.

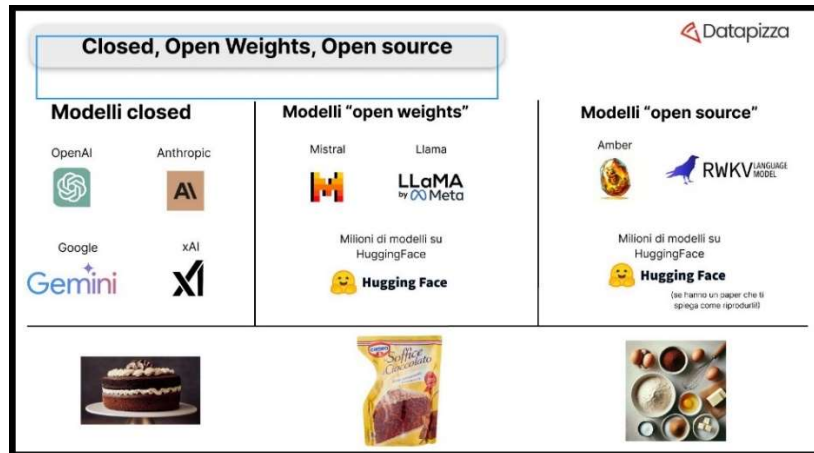


Figure 20 - differences between Closed, Open Weights and Open Source (Source: DataPizza)

Practical Example: Closed Source in AI Models

For AI models, closed source means that users receive access only to the *application*, not to the underlying code or parameters. This is common for proprietary AI models like GPT, Gemini, and Claude. In these cases, users can interact with the model through APIs or dedicated software but can't access, modify, or improve the model itself.

To continue with the cake analogy: closed source is like buying a slice of cake at a bakery. You can enjoy it, but you have no access to the recipe, ingredients, or process used to make it. The bakery retains full control over how the cake is made, and you only get the finished product, ready to consume.

Summary

Closed source keeps the "recipe" of the software or model private, allowing companies to protect their innovations, monetize access, and maintain consistent control over updates and usage.

3.4 LLM Evaluation

LLM evaluation refers to the discipline of ensuring a language model's outputs are consistent with the desired ethical, safety, and performance criteria — ultimately aligning with human values and intents. Some LLM evaluations look at the model's ability to perform specific tasks accurately and reliably, while others measure overall behavior, biases, and adherence to alignment objectives. (Dhinakaran, 2024)

This piece explores the essentials of LLM evaluation, LLM evaluation metrics, and a concrete exercise with everything you need to get started.

Approaches

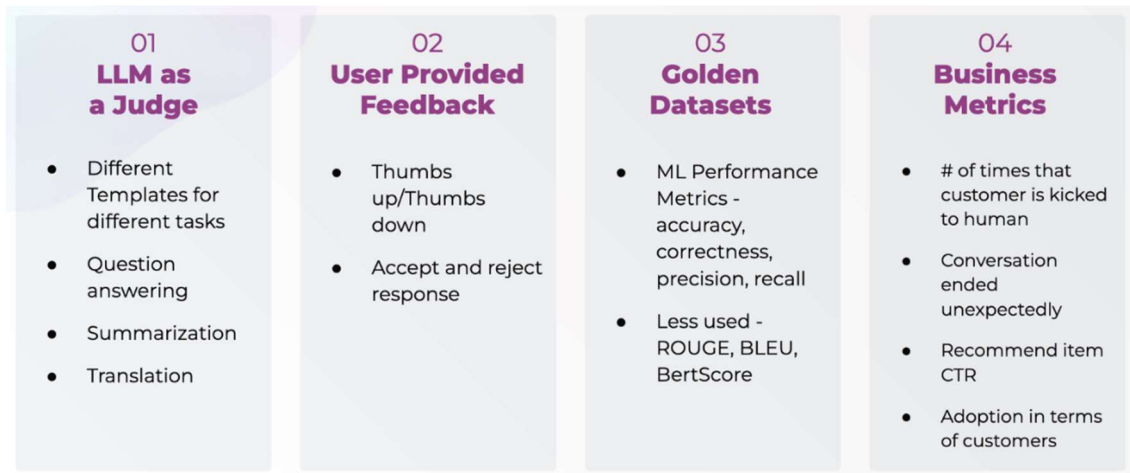


Figure 21 - LLM evaluation steps (Source: Arize)

There are many ways to quantify how your LLM application is doing, from user-provided feedback (i.e. thumbs-up/down, accept/reject response), golden datasets, and finally business metrics (i.e. recommended item purchases).

3.4.1 Exploring LLM Model Evaluations (Evals)

LLM model evals are focused on the overall performance of the foundational models. The companies launching the original customer-facing LLMs needed a way to quantify their effectiveness across an array of different tasks.

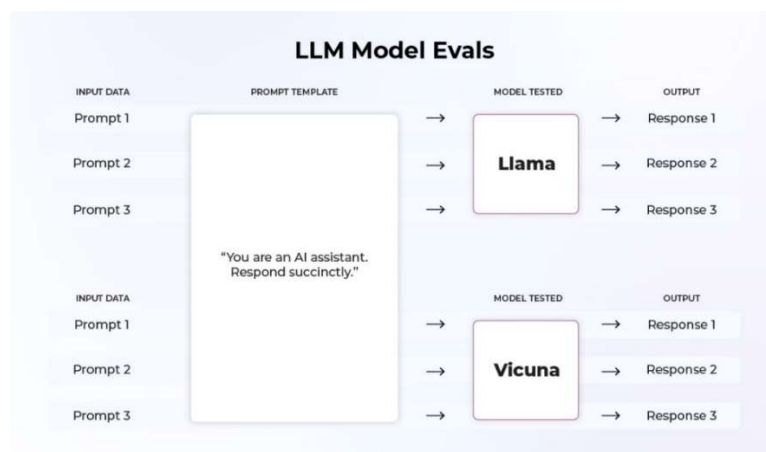


Figure 22 - Example of LLM evaluation (Source: Arize)

WHY DO WE NEED LLM BENCHMARKS?

They provide a standardized method to evaluate LLMs across tasks like coding, reasoning, math, truthfulness, and more.

By comparing different models, benchmarks highlight their strengths and weaknesses.

Model Performance Across Key LLM Benchmarks

These are the most commonly utilized LLM Benchmarks among models’ technical reports:

- 1. MMLU - Multitask accuracy
- 2. GPQA - Reasoning capabilities
- 3. HumanEval - Python coding tasks
- 4. MATH - Math problems with 7 difficulty levels
- 5. BFCL - The ability of the model to call functions/tools
- 6. MGSM - Multilingual capabilities

3.4.2 Benchmarking LLMs for Reasoning

HellaSwag - Measuring Commonsense Inference

This test measures the commonsense reasoning of LLM models. It tests if an LLM model could complete a sentence by choosing the correct option with common reasoning among 4 options.

For example:

How to catch dragonflies. Use a long-handled aerial net with a wide opening. Select an aerial net that is 18 inches (46 cm) in diameter or larger. Look for one with a nice long handle.

a) Loop 1 piece of ribbon over the handle. Place the hose or hose on your net and tie the string securely.	b) Reach up into the net with your feet. Move your body and head forward when you lift up your feet.	c) If possible, choose a dark-colored net over a light one. Darker nets are more difficult for dragonflies to see, making the net more difficult to avoid.	d) If it's not strong enough for you to handle, use a hand held net with one end shorter than the other. The net should have holes in the bottom of the net.
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Figure 23 - Responses by a LLM

Questions that seem simple to humans often posed challenges for state-of-the-art (SOTA) models released in 2019, as they had difficulty with commonsense inference, achieving only about 45% accuracy. By 2024, GPT-4 has achieved the highest benchmark score with 95.3% accuracy in this area, while among open-source models, Mixtral 8x7B leads with an accuracy of 84.4%.

DROP - A Reading Comprehension + Discrete Reasoning Benchmark

DROP evaluates models on their ability to pull important details from English-language paragraphs and then perform distinct reasoning actions, such as adding, sorting or counting items, to find the right answer. Here's an example:

Reasoning	Passage (some parts shortened)	Question	Answer	BiDAF
Subtraction (28.8%)	That year, his Untitled (1981) , a painting of a haloed, black-headed man with a bright red skeletal body, depicted amid the artists signature scrawls, was sold by Robert Lehrman for \$16.3 million, well above its \$12 million high estimate.	How many more dollars was the Untitled (1981) painting sold for than the 12 million dollar estimation?	4300000	\$16.3 million
Comparison (18.2%)	In 1517, the seventeen-year-old King sailed to Castile. There, his Flemish court In May 1518, Charles traveled to Barcelona in Aragon.	Where did Charles travel to first, Castile or Barcelona?	Castile	Aragon

Figure 24 - Example of DROP evaluation (Source: Google)

3.4.3 QA and Truthfulness Benchmarks

MMLU - Measuring Massive Multitask Language Understanding

This test measures model's multitask accuracy. It covers 57 tasks including elementary mathematics, US history, computer science, law, and more at varying depths, from elementary to advanced professional level. To get high accuracy on this test, models must have extensive world knowledge and problem solving ability.

TruthfulQA

This benchmark measures whether a language model is truthful in generating answers to questions. The benchmark comprises 817 questions that span 38 categories, including health, law, finance and politics. For this benchmark, GPT-4 seems to perform the best.

3.4.4 Math Benchmarks

MATH - Arithmetic Reasoning

MATH is a new benchmark, that has a dataset of 12,500 challenging competition mathematics problems. Each problem in MATH has a full step-by-step solution which can be used to teach models to generate answer derivations and explanations. The authors of this benchmark found out that increasing budgets and model parameter counts will be impractical for achieving strong mathematical reasoning, if scaling trends continues.

GSM8K - Arithmetic Reasoning

This dataset consists of 8.5K high quality linguistically diverse grade school math word problems. These problems take between 2 and 8 steps to solve, and solutions primarily involve performing a sequence of elementary calculations using basic arithmetic operations (+ - / *) to reach the final answer. A bright middle school student should be able to solve every problem, but some models find these tasks still challenging.

3.4.5 Coding Benchmarks

HumanEval - Coding Benchmark

This is the most used benchmark to evaluate the performance of LLMs in code generation tasks.

The HumanEval Dataset has a set of 164 handwritten programming problems that evaluate for language comprehension, algorithms, and simple mathematics, with some comparable to simple software interview questions. Each problem includes a function signature, docstring, body, and several unit tests, with an average of 7.7 tests per problem.

MBPP - Coding Benchmark for Python problems

The benchmark consists of around 1,000 crowd-sourced Python programming problems, designed to be solvable by entry level programmers, covering programming fundamentals, standard library functionality, and so on. Each problem consists of a task description, code solution and 3 automated test cases.

3.4.6 MultiModality Benchmarks

MMMU Benchmark

MMMU (Massive Multimodal Multidiscipline Understanding) is a benchmark for evaluating multimodal models on complex, college-level tasks requiring advanced knowledge and reasoning. It features 11.5K multimodal questions from six core disciplines, spanning 30 subjects and 183 subfields, with diverse image types like charts, diagrams, and maps. MMMU challenges models in both perception and reasoning. Testing on 14 open-source models and GPT-4V(ision) showed even GPT-4V achieved only 56% accuracy, highlighting significant room for improvement in multimodal AI models.

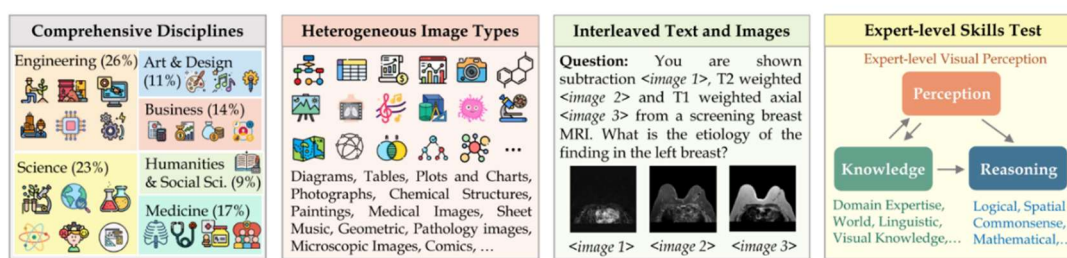


Figure 25 - Overview of the MMMU benchmark (Source: Google)

3.4.7 Multilingual Benchmarks

MGSM - Multilingual Benchmark

The Multilingual Grade School Math Benchmark (MGSM) is a collection of 250 grade-school math problems from the GSM8K dataset, translated into 10 languages by human annotators. GSM8K consists of 8.5K high-quality, linguistically diverse math word problems designed to evaluate models' ability to answer basic mathematical questions that involve multi-step reasoning.

3.4.8 Limitations of LLM Benchmarks

There are two major limitations of current LLM benchmarks:

1. Restricted scope

Many benchmarks have restricted scope, usually targeting capabilities on which LLMs have already proven some proficiency. Because they focus on areas where language models are known to be good, they're not great at finding new or unexpected skills that may emerge as language models get more advanced.

2. Short life span

Also, benchmarks for language modeling often don't last long in terms of usefulness. Once language models reach a level of performance that's as good as humans on these benchmarks, the benchmarks are typically either stopped and swapped out or updated by adding harder challenges.

This short lifespan is likely because these benchmarks don't cover tasks that are much harder than what current language models can do.

It's clear that as models continue to improve, they will achieve increasingly similar and higher scores on current benchmarks. So, we'll need to test models on future capabilities that are not possible now - with benchmarks like BBHard.

4 Closed Source Models: Innovations and Comparisons

This chapter focuses on the advancements and applications of leading closed-source AI models developed by organizations like Anthropic, OpenAI, and Google DeepMind. It explores the capabilities of models such as Claude, GPT-4 Turbo, and Gemini, emphasizing their technological breakthroughs and specialized use cases. The chapter also includes a comparative analysis of their features, performance, and subscription costs.

4.1 Anthropic: Pioneering AI Safety and Development

Anthropic is an artificial intelligence (AI) research and safety company established in 2021 by Dario and Daniela Amodei, both former OpenAI executives. The organization is committed to creating AI systems that are not only cutting-edge but also aligned with human values, ensuring they are interpretable, reliable, and steerable.

By prioritizing AI safety, Anthropic aims to mitigate potential risks while maximizing the benefits of transformative AI technologies for humanity. (Anthropic, n.d.)

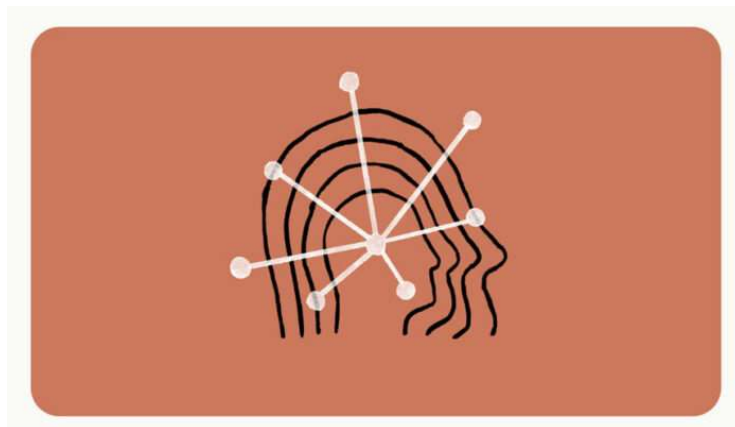


Figure 26 - Anthropic logo (Source: Anthropic.com)

Mission and Vision

Anthropic's mission revolves around the ethical advancement of AI technologies. The company seeks to:

- **Develop Aligned AI Systems:** By creating systems that can be guided and trusted to perform beneficial tasks, Anthropic is working to make AI systems that align with societal goals.
- **Advance AI Safety Research:** Anthropic approaches safety as a rigorous scientific discipline, conducting foundational and applied research to improve how AI systems behave in various scenarios.

- **Collaborate and Share Insights:** By engaging with global stakeholders and publishing research, Anthropic ensures that its findings contribute to broader conversations about AI ethics and safety.

The overarching vision is to deploy AI systems that support human flourishing, making significant contributions to fields such as education, healthcare, and business productivity.

Foundational Principles

Anthropic operates on a set of guiding principles that shape its work and internal culture:

1. **Mission-Driven Focus:** Every project and initiative prioritizes societal benefit, ensuring the AI tools developed are ultimately advantageous for humanity.
2. **High-Trust Environment:** The company fosters a culture of transparency, intellectual curiosity, and mutual respect among its team members.
3. **Empirical Pragmatism:** Anthropic adopts data-driven approaches, making decisions based on measurable outcomes rather than theoretical assumptions.
4. **Interdisciplinary Collaboration:** Teams work across domains, including machine learning, public policy, and product development, to address complex AI challenges holistically.

Research and Technological Innovations

Anthropic's research spans critical areas of machine learning, AI safety, and practical applications of AI. The organization has developed the Claude series of large language models (LLMs), which are designed to:

- Handle complex cognitive tasks such as content summarization, image transcription, and debugging code.
- Generate content that adheres to ethical standards, minimizing risks like bias or misinformation.
- Provide safe and efficient tools for industries ranging from customer support to software development.

These models incorporate advanced mechanisms to prevent common issues like hallucinations and biases, setting them apart as safer and more reliable tools for real-world applications.

Anthropic's Role in the AI Ecosystem

Anthropic is uniquely positioned in the AI landscape, bridging advanced technical research with ethical considerations. By focusing on safety and alignment, Anthropic ensures that its AI systems:

- Enhance human productivity and creativity.

- Address global challenges in sectors such as healthcare, education, and disaster management.
- Operate transparently, empowering users while protecting privacy and ethical standards.

As AI technologies evolve, Anthropic's pioneering work in AI safety and its collaborative ethos make it a critical player in shaping the future of ethical and impactful AI development.

4.1.1 Claude

Claude is an advanced AI assistant developed by Anthropic, designed to assist individuals and teams in various tasks. It excels in understanding and generating human-like text, making it useful for brainstorming, drafting content, and answering questions. (Anthropic, 2024)



Figure 27 - Claude: Large Language Model by Anthropic (Source: anthropic.com)

Key Features of Claude:

- *Advanced Reasoning:* Claude can perform complex cognitive tasks, such as analyzing data, solving problems, and providing detailed explanations.
- *Vision Analysis:* It can transcribe and analyze static images, including handwritten notes, graphs, and photographs, enhancing its ability to process visual information.
- *Code Generation:* Claude assists in creating websites using HTML and CSS, converting images into structured data formats like JSON, and debugging complex codebases.
- *Multilingual Processing:* It offers real-time translation between various languages, grammar practice, and the creation of multilingual content.

Why Choose Claude?

Anthropic emphasizes building AI models with integrity, focusing on:

- **Security:** Claude is accessible via AWS and GCP, holds SOC 2 Type II certification, and offers HIPAA compliance options, ensuring adherence to stringent security practices.
- **Trustworthiness:** It incorporates best-in-class jailbreak resistance and misuse prevention measures to mitigate risks for users.
- **Reliability:** Claude delivers accurate information with low rates of misinformation, even when processing long documents.

Accessing Claude:

Users can interact with Claude through:

- **Claude.ai:** A platform for direct engagement with Claude's capabilities.
- **API Access:** Developers can integrate Claude into their applications via the Anthropic API, enabling customized AI solutions.

Claude is designed to be a versatile AI assistant, supporting a wide range of tasks to enhance productivity and creativity for individuals and organizations alike.

New generation of Claude

Anthropic has introduced the Claude 3 model family, setting new benchmarks in artificial intelligence (AI) capabilities. This family comprises three models: Haiku, Sonnet, and Opus, each offering varying levels of intelligence, speed, and cost-effectiveness to cater to diverse user needs.

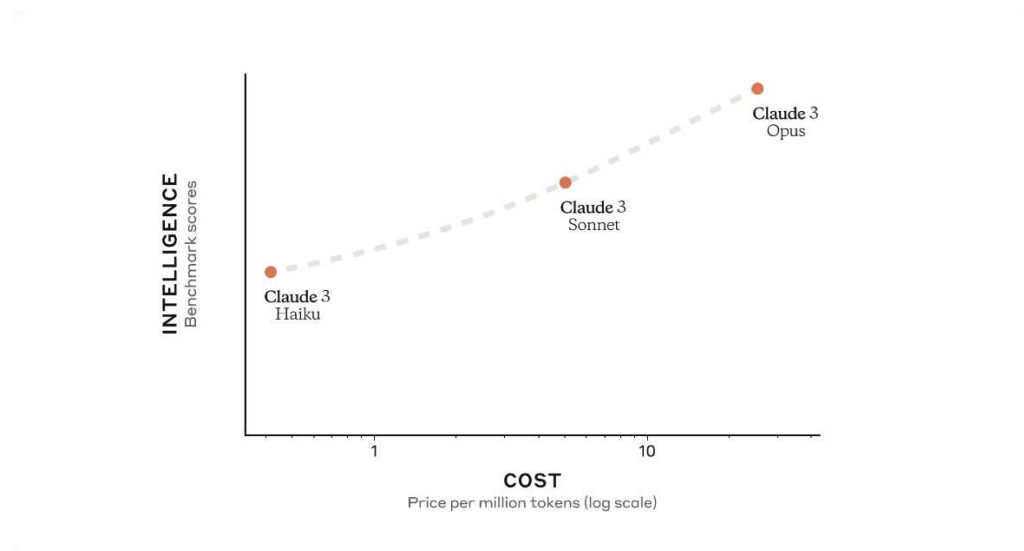


Figure 28 - Intelligence - Cost ratio graph (Source: Anthropic.com)

Key Features of Claude 3 Models:

- Enhanced Intelligence: Opus, the most advanced model, excels in tasks such as expert knowledge assessments, reasoning, and mathematics, demonstrating near-human comprehension and fluency.

	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge <i>MMLU</i>	86.8% 5-shot	79.0% 5-shot	75.2% 5-shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot
Graduate level reasoning <i>GPQA, Diamond</i>	50.4% 0-shot CoT	40.4% 0-shot CoT	33.3% 0-shot CoT	35.7% 0-shot CoT	28.1% 0-shot CoT	—	—
Grade school math <i>GSM8K</i>	95.0% 0-shot CoT	92.3% 0-shot CoT	88.9% 0-shot CoT	92.0% 5-shot CoT	57.1% 5-shot	94.4% Maj1@32	86.5% Maj1@32
Math problem-solving <i>MATH</i>	60.1% 0-shot CoT	43.1% 0-shot CoT	38.9% 0-shot CoT	52.9% 4-shot	34.1% 4-shot	53.2% 4-shot	32.6% 4-shot
Multilingual math <i>MGSM</i>	90.7% 0-shot	83.5% 0-shot	75.1% 0-shot	74.5% 8-shot	—	79.0% 8-shot	63.5% 8-shot
Code <i>HumanEval</i>	84.9% 0-shot	73.0% 0-shot	75.9% 0-shot	67.0% 0-shot	48.1% 0-shot	74.4% 0-shot	67.7% 0-shot
Reasoning over text <i>DROP, F1 score</i>	83.1 3-shot	78.9 3-shot	78.4 3-shot	80.9 3-shot	64.1 3-shot	82.4 Variable shots	74.1 Variable shots
Mixed evaluations <i>BIG-Bench-Hard</i>	86.8% 3-shot CoT	82.9% 3-shot CoT	73.7% 3-shot CoT	83.1% 3-shot CoT	66.6% 3-shot CoT	83.6% 3-shot CoT	75.0% 3-shot CoT
Knowledge Q&A <i>ARC-Challenge</i>	96.4% 25-shot	93.2% 25-shot	89.2% 25-shot	96.3% 25-shot	85.2% 25-shot	—	—
Common Knowledge <i>HellaSwag</i>	95.4% 10-shot	89.0% 10-shot	85.9% 10-shot	95.3% 10-shot	85.5% 10-shot	87.8% 10-shot	84.7% 10-shot

Figure 29 - Claude family capabilities (Source: anthropic.com)

- Multilingual Proficiency: All Claude 3 models are adept at understanding and generating content in multiple languages, including Spanish, Japanese, and French.
- Vision Capabilities: These models can process various visual formats like photos, charts, and technical diagrams, making them valuable for enterprises with diverse data types.

	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4V	Gemini 1.0 Ultra	Gemini 1.0 Pro
Math & reasoning <i>MMMU(val)</i>	59.4%	53.1%	50.2%	56.8%	59.4%	47.9%
Document visual Q&A <i>ANLS score, test</i>	89.3%	89.5%	88.8%	88.4%	90.9%	88.1%
Math <i>MathVista (testmini)</i>	50.5% CoT	47.9% CoT	46.4% CoT	49.9%	53.0%	45.2%
Science diagrams <i>AI2D, test</i>	88.1%	88.7%	86.7%	78.2%	79.5%	73.9%
Chart Q&A <i>Relaxed accuracy(test)</i>	80.8% 0-shot CoT	81.1% 0-shot CoT	81.7% 0-shot CoT	78.5% 4-shot CoT	80.8%	74.1%

Figure 30 - Claude family capabilities (Source: anthropic.com)

- **Reduced Refusals:** Compared to previous versions, Claude 3 models are less likely to decline responses, indicating improved contextual understanding.

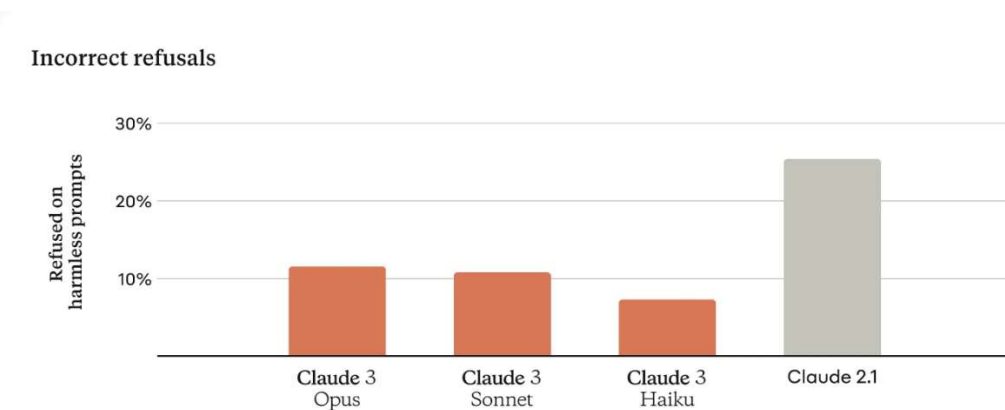


Figure 31 - Claude family incorrect refusal comparison (Source: anthropic.com)

- **Improved Accuracy:** Opus has shown a twofold improvement in accuracy over Claude 2.1 on complex, open-ended questions, providing more reliable information.
- **Extended Context Window:** Initially offering a 200,000-token context window, all three models can handle inputs exceeding one million tokens, accommodating extensive data processing needs.

Claude 3 Opus: Pushing the Boundaries of Intelligence

Claude 3 Opus is the most advanced model in the Claude 3 family, delivering industry-leading performance on highly complex and open-ended tasks. With unparalleled fluency and a human-like understanding of unseen scenarios, Opus exemplifies the potential of generative AI in redefining problem-solving and strategic analysis.

Potential Applications:

- *Task Automation:* Planning and executing complex actions across APIs and databases, interactive coding.
- *Research and Development:* Research reviews, brainstorming, hypothesis generation, and drug discovery.
- *Strategic Analysis:* Advanced chart and graph analysis, financial modeling, market trend forecasting.

Key Differentiator: Demonstrates superior intelligence and problem-solving capabilities compared to any other model currently available.

Claude 3 Sonnet: Balancing Intelligence and Scalability

Claude 3 Sonnet strikes a balance between high intelligence and operational efficiency, making it ideal for enterprise-level deployments. With robust performance and a significantly lower cost, Sonnet is engineered to handle large-scale workloads with endurance and speed.

Potential Applications:

- *Data Processing:* Retrieval-augmented generation (RAG), search, and knowledge retrieval across vast datasets.
- *Sales and Marketing:* Product recommendations, forecasting, and targeted marketing strategies.
- *Time-Saving Tasks:* Automated code generation, quality control, and text parsing from images.

Key Differentiator: Offers superior affordability and scalability compared to other models in its intelligence class, optimized for enterprise applications.

Claude 3 Haiku: Speed and Cost-Efficiency

Claude 3 Haiku is designed for rapid response and cost-effectiveness, making it the fastest and most compact model in the Claude 3 lineup. It is optimized for simple tasks and delivers near-instantaneous outputs, creating seamless AI experiences for real-time applications.

Potential Applications:

- *Customer Interactions:* Quick and accurate support for live interactions, including translations.
- *Content Moderation:* Identifying risky behaviors or inappropriate customer requests.

- *Cost-Effective Operations*: Logistics optimization, inventory management, and extracting knowledge from unstructured data.

Key Differentiator: Combines speed, intelligence, and affordability to outperform other models in its category.

4.1.2 Claude 3.5 Sonnet: A Leap Forward in AI Model Design

The release of Claude 3.5 Sonnet marks a significant advancement in the capabilities of AI models, focusing on a balance between enhanced performance, scalability, and cost efficiency. Designed to cater to enterprise-grade workloads, Claude 3.5 Sonnet exemplifies a model that adapts to high-demand environments while ensuring affordability and accuracy. (Anthropic, 2024)

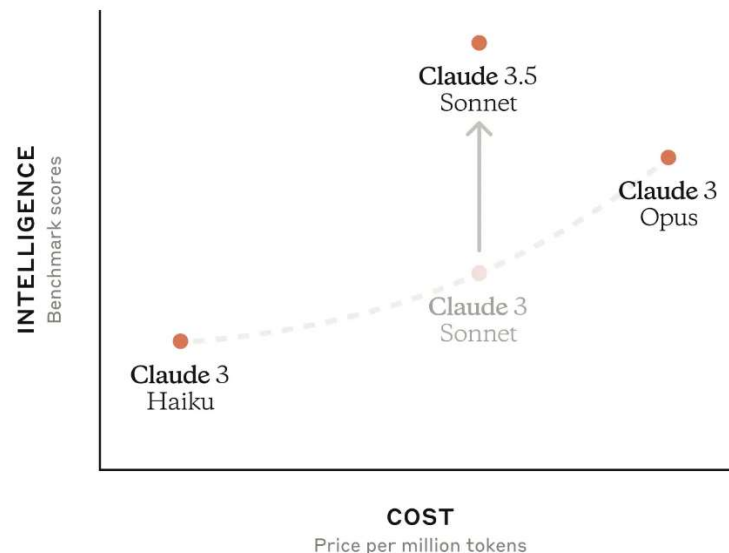


Figure 32 - Highlight of Claude 3.5 Sonnet potential (Source: anthropic.com)

Enhanced Capabilities of Claude 3.5 Sonnet

Claude 3.5 Sonnet builds on the strengths of its predecessors by incorporating advanced features that enhance both user interaction and operational efficiency. This model excels in tasks requiring extensive data processing, high-speed performance, and natural language understanding.

- **Scalability and Performance:**
Claude 3.5 Sonnet is optimized for large-scale deployments, making it ideal for enterprise environments. It handles complex workflows, including data retrieval, summarization, and categorization, without compromising speed or accuracy.

- Improved Context Handling:**
 With a context window of up to 200K tokens, Sonnet can process extensive datasets, ensuring more detailed and relevant outputs. This capability allows it to perform better in applications that require understanding intricate relationships within vast information.
- Efficiency and Cost Management:**
 By reducing operational costs compared to other models in its intelligence category, Claude 3.5 Sonnet enables businesses to leverage AI without exceeding budget constraints. It demonstrates how advanced AI can remain accessible to organizations of varying scales.

	Claude 3.5 Sonnet (new)	Claude 3.5 Haiku	Claude 3.5 Sonnet	GPT-4o*	GPT-4o mini*	Gemini 1.5 Pro	Gemini 1.5 Flash
Graduate level reasoning <i>GPQA (Diamond)</i>	65.0% 0-shot CoT	41.6% 0-shot CoT	59.4% 0-shot CoT	53.6% 0-shot CoT	40.2% 0-shot CoT	59.1% 0-shot CoT	51.0% 0-shot CoT
Undergraduate level knowledge <i>MMLU Pro</i>	78.0% 0-shot CoT	65.0% 0-shot CoT	75.1% 0-shot CoT	—	—	75.8% 0-shot CoT	67.3% 0-shot CoT
Code <i>HumanEval</i>	93.7% 0-shot	88.1% 0-shot	92.0% 0-shot	90.2% 0-shot	87.2% 0-shot	—	—
Math problem-solving <i>MATH</i>	78.3% 0-shot CoT	69.2% 0-shot CoT	71.1% 0-shot CoT	76.6% 0-shot CoT	70.2% 0-shot CoT	86.5% 4-shot CoT	77.9% 4-shot CoT
High school math competition <i>AIME 2024</i>	16.0% 0-shot CoT	5.3% 0-shot CoT	9.6% 0-shot CoT	9.3% 0-shot CoT	—	—	—
Visual Q/A <i>MMMU</i>	70.4% 0-shot CoT	—	68.3% 0-shot CoT	69.1% 0-shot CoT	59.4% 0-shot CoT	65.9% 0-shot CoT	62.3% 0-shot CoT
Agentic coding <i>SWE-bench Verified</i>	49.0%	40.6%	33.4%	—	—	—	—
Agentic tool use <i>TAU-bench</i>	Retail	Retail	Retail	—	—	—	—
	69.2%	51.0%	62.6%				
	Airline	Airline	Airline				
	46.0%	22.8%	36.0%				

* Our evaluation tables exclude OpenAI's o1 model family as they depend on extensive pre-response computation time, unlike typical models. This fundamental difference makes performance comparisons difficult.

Figure 33 - Claude 3.5 Sonnet capabilities (Source: anthropic.com)

	Claude 3.5 Sonnet	Claude 3 Opus	GPT-4o	Gemini 1.5 Pro
Visual math reasoning <i>MathVista (testmini)</i>	67.7% 0-shot CoT	50.5% 0-shot CoT	63.8% 0-shot CoT	63.9% 0-shot CoT
Science diagrams <i>AI2D, test</i>	94.7% 0-shot	88.1% 0-shot	94.2% 0-shot	94.4% 0-shot
Visual question answering <i>MMU (val)</i>	68.3% 0-shot CoT	59.4% 0-shot CoT	69.1% 0-shot CoT	62.2% 0-shot CoT
Chart Q&A <i>Relaxed accuracy (test)</i>	90.8% 0-shot CoT	80.8% 0-shot CoT	85.7% 0-shot CoT	87.2% 0-shot CoT
Document visual Q&A <i>ANLS score, test</i>	95.2% 0-shot	89.3% 0-shot	92.8% 0-shot	93.1% 0-shot

Figure 34 - Claude 3.5 Sonnet capabilities (Source: anthropic.com)

4.2 About OpenAI: Advancing AI for the Benefit of Humanity

OpenAI is a pioneering research and deployment company whose mission is to ensure that artificial general intelligence (AGI) benefits all of humanity. Founded in 2015 by a group of visionaries including Elon Musk, Sam Altman, Greg Brockman, Ilya Sutskever, Wojciech Zaremba, and John Schulman, OpenAI has consistently pushed the boundaries of artificial intelligence, promoting innovation and responsibility in the AI field. (OpenAI, n.d.)

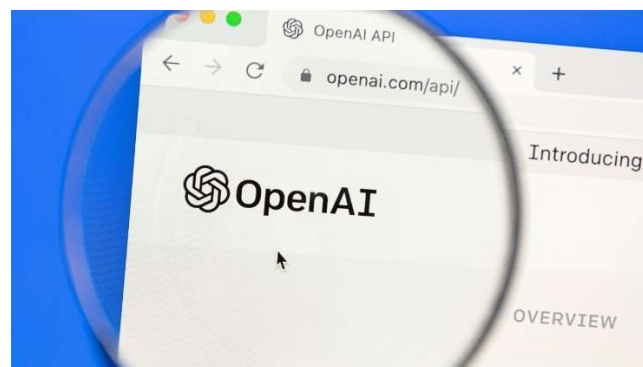


Figure 35 - OpenAI logo

Mission and Values

OpenAI operates under a clear mission: to ensure that AGI—highly autonomous systems capable of outperforming humans at most economically valuable work—is aligned with human values and distributed equitably across society. Key principles guiding OpenAI’s operations include:

- **Safety First:** Developing AI technologies in a manner that prioritizes safety, ensuring alignment with ethical standards and societal goals.
- **Collaborative Approach:** OpenAI values collaboration with global institutions, academia, and industries to accelerate progress in AI responsibly.
- **Broad Distribution:** Striving to ensure that the benefits of AI technology are accessible to everyone, preventing concentrated power or misuse.
- **Iterative Deployment:** Releasing technology incrementally to understand its impact, enabling robust feedback loops and course correction.

Key Milestones

OpenAI has made significant contributions to the field of AI, including:

1. **GPT Series:**
OpenAI developed the Generative Pre-trained Transformer (GPT) models, which have become benchmarks for natural language understanding and generation. GPT-4, the latest model, demonstrates capabilities in complex reasoning, creativity, and problem-solving.
2. **DALL·E:**
OpenAI's image-generation model, DALL·E, blends art and AI by creating photorealistic images from textual descriptions, showcasing the potential of multimodal AI systems.
3. **OpenAI API:**
A comprehensive platform for developers to integrate OpenAI's powerful models into diverse applications, fostering innovation across industries.

4.2.1 GPT-4 Turbo: Redefining AI Efficiency and Accessibility

OpenAI's release of GPT-4 Turbo represents a significant advancement in generative AI, offering enhanced performance and efficiency for developers and enterprises. Positioned as the successor to GPT-4, GPT-4 Turbo is both faster and more cost-effective, making cutting-edge AI more accessible and versatile for real-world applications. (OpenAI, 2023)

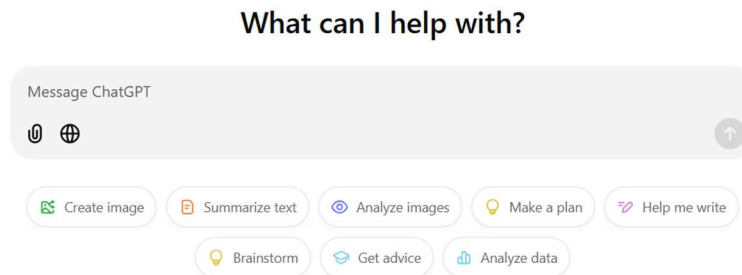


Figure 36 - ChatGPT user interface

Key Features of GPT-4 Turbo

1. **Improved Efficiency:**
GPT-4 Turbo is designed to be faster and more resource-efficient compared to its predecessor. This optimization ensures reduced latency, enabling seamless integration into applications requiring real-time interaction.
2. **Cost-Effectiveness:**
A significant reduction in costs for API usage makes GPT-4 Turbo an attractive option for businesses. Developers can now leverage advanced language generation capabilities without incurring prohibitive expenses, facilitating widespread adoption across industries.
3. **Scalability for Diverse Applications:**
GPT-4 Turbo supports extensive context windows, allowing it to handle large and complex datasets. This scalability is ideal for tasks such as document summarization, extended conversations, and intricate problem-solving.

Applications of GPT-4 Turbo

GPT-4 Turbo has been tailored for a range of applications, including:

- **Customer Interaction:** Real-time chatbot systems powered by GPT-4 Turbo offer natural, fluid conversations, enhancing customer satisfaction and support efficiency.
- **Content Creation:** The model's improved generation capabilities facilitate the creation of high-quality text for marketing, education, and entertainment.
- **Research and Development:** Its ability to process extensive context windows supports data analysis, hypothesis generation, and scientific exploration.

- Coding Assistance: GPT-4 Turbo excels in translating natural language into code, identifying errors, and offering solutions, making it a valuable tool for developers.

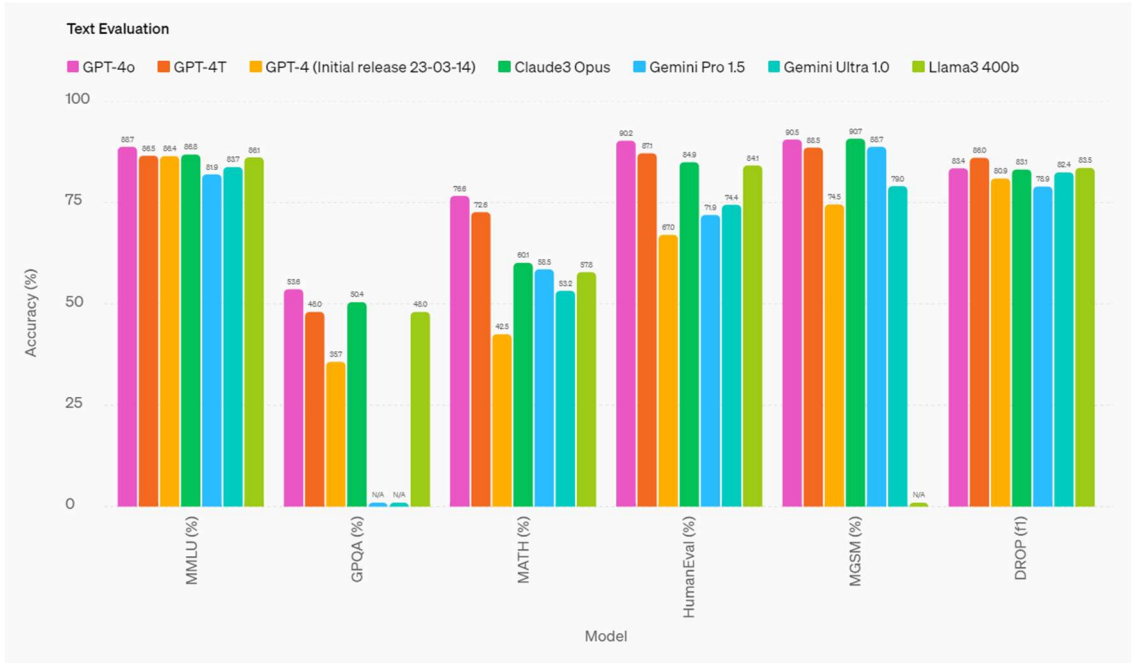


Figure 37 - GPT Benchmarks (Source: OpenAI.com)

Eval Sets	GPT-4o	GPT-4T 2024-04-09	Gemini 1.0 Ultra	Gemini 1.5 Pro	Claude Opus
MMMU (%) (val)	69.1	63.1	59.4	58.5	59.4
MathVista (%) (testmini)	63.8	58.1	53.0	52.1	50.5
AI2D (%) (test)	94.2	89.4	79.5	80.3	88.1
ChartQA (%) (test)	85.7	78.1	80.8	81.3	80.8
DocVQA (%) (test)	92.8	87.2	90.9	86.5	89.3
ActivityNet (%) (test)	61.9	59.5	52.2	56.7	
EgoSchema (%) (test)	72.2	63.9	61.5	63.2	

Figure 38 - GPT Benchmarks (Source: OpenAI.com)

4.2.2 Comparing Free and Premium Versions of OpenAI's ChatGPT

OpenAI's ChatGPT service offers two distinct tiers of access: the free version (typically based on GPT-3.5) and the premium version (featuring GPT-4 or GPT-4 Turbo). Each option is designed to cater to different user needs, with notable differences in performance, capabilities, and cost.

Free Version: Powered by GPT-3.5

The free version provides access to GPT-3.5, a robust and versatile language model. While it serves as an excellent entry point to generative AI, it has some limitations when compared to the more advanced GPT-4 series.

1. **Model:**
Users interact with GPT-3.5, a powerful model capable of text generation, question answering, and task handling. However, it is less advanced than GPT-4 in terms of handling complex or nuanced inputs.
2. **Performance:**
GPT-3.5 performs well for general inquiries and simpler tasks but may struggle with highly detailed or intricate queries that require a deep understanding of context or creativity.
3. **Speed:**
Response times can be slower, especially during peak usage periods, as GPT-3.5 is the default choice for the majority of free users.
4. **Features:**
While the free version supports basic text generation and conversational capabilities, it has limitations in handling long prompts, maintaining coherence over extended dialogues, and producing highly detailed outputs.
5. **Cost:**
The primary advantage of the free version is its accessibility—it is completely free to use. However, users must accept trade-offs in speed, reliability, and advanced functionality.

Premium Version: Powered by GPT-4 or GPT-4 Turbo

The premium version, available through a ChatGPT Plus subscription, offers access to GPT-4 or GPT-4 Turbo. These models provide enhanced capabilities for users with more demanding requirements.

1. **Model:**
Premium users gain access to GPT-4, a state-of-the-art language model, or GPT-4 Turbo, a variant optimized for faster and more cost-efficient operation. Both models surpass GPT-3.5 in complexity, accuracy, and scalability.
2. **Performance:**
GPT-4 shines in addressing complex, nuanced, or highly technical queries. It excels at maintaining coherence in long conversations, generating creative and

context-aware content, and offering more reliable results across diverse use cases.

3. Speed:

The inclusion of GPT-4 Turbo ensures faster response times, making the premium version ideal for high-demand scenarios where efficiency is critical.

4. Advanced Capabilities:

- Handles larger text inputs with greater ease.
- Delivers highly detailed and structured responses.
- Supports advanced tasks such as coding, multilingual translations, problem-solving, and creative writing.
- Demonstrates improved accuracy and adaptability across a wide range of tasks and domains.

5. Cost:

The premium service is available via a subscription priced at \$20 per month.

This fee grants users enhanced reliability during peak times, priority access, and faster performance alongside GPT-4's superior capabilities.

Choosing the Right Option

For casual users or those exploring generative AI for the first time, the free version offers a cost-free entry point with solid capabilities. On the other hand, the premium version is tailored for users who require advanced performance, faster response times, and greater reliability—making it an invaluable tool for professionals, researchers, and businesses leveraging AI for critical applications.

This tiered structure allows OpenAI to cater to diverse user needs, democratizing access to AI while offering cutting-edge solutions for those with more demanding requirements.

4.3 Google DeepMind: Advancing AI to Benefit Humanity

DeepMind, a pioneering artificial intelligence (AI) research lab founded in 2010 and acquired by Google in 2014, is dedicated to advancing AI technologies to improve the world. With a mission to "solve intelligence, and use it to solve everything else," DeepMind is at the forefront of creating groundbreaking technologies that address complex challenges across multiple disciplines. From healthcare to climate science, the organization's work aims to revolutionize industries and contribute to human knowledge. (Google DeepMind, n.d.)



Figure 39 - Google DeepMind logo

DeepMind's Vision and Approach

DeepMind's vision centers on building artificial general intelligence (AGI) — systems capable of performing any intellectual task that a human can do. This ambition goes beyond individual applications to encompass a future where AI can act as a general-purpose tool for solving global challenges.

Key Pillars of DeepMind's Approach:

1. **Interdisciplinary Research:** DeepMind emphasizes collaboration across diverse fields, including neuroscience, biology, and computer science, to better understand intelligence. By integrating these disciplines, the organization aims to create AI systems that mirror the problem-solving flexibility of the human brain.
2. **Ethics and Safety:** Ethical considerations are at the heart of DeepMind's mission. The organization works closely with ethicists, policymakers, and researchers to ensure its AI systems are transparent, safe, and aligned with societal values.
3. **Partnerships and Collaboration:** DeepMind partners with academic institutions, government agencies, and non-profits to apply its technologies to pressing global issues, such as advancing medical research and optimizing energy systems.

4.3.1 Gemini: A Leap Forward in AI Technology

DeepMind's Gemini represents a groundbreaking step in the evolution of artificial intelligence (AI). Designed to merge cutting-edge research with practical application, Gemini is poised to redefine the possibilities of AI systems by integrating advanced reasoning, memory, and tool-use capabilities. As part of DeepMind's commitment to creating impactful and ethical AI, Gemini sets a new benchmark for intelligent systems. (Google DeepMind, 2024)



Figure 40 - Gemini logo

What Is Gemini?

Gemini is DeepMind's latest AI model, developed to operate as a general-purpose system with enhanced problem-solving abilities. Unlike earlier AI models, Gemini builds on the principles of reinforcement learning, natural language processing, and tool-use integration, making it more versatile and capable of understanding complex tasks.

Key Features of Gemini

1. Advanced Reasoning Capabilities

Gemini incorporates state-of-the-art reasoning abilities, allowing it to tackle complex problems across various domains. By leveraging vast datasets and applying logical inference, Gemini excels in tasks that require strategic thinking and nuanced decision-making.

2. Memory Integration

A distinguishing feature of Gemini is its enhanced memory architecture, which enables it to store and retrieve information dynamically. This allows the model to draw on prior interactions, making it particularly effective in applications requiring continuity, such as customer support or multi-step workflows.

3. Tool Use

Gemini stands out for its ability to integrate with external tools and APIs. This feature allows the model to expand its functionality by accessing specialized systems, enabling applications in fields like data analysis, software development, and real-time diagnostics.

4. Human-Like Interactions

Gemini's advanced natural language understanding (NLU) enables it to engage in conversations that mimic human interactions. It adapts its tone, context, and content to suit different users, making it ideal for applications in education, healthcare, and customer engagement.

Gemini Model Variants

DeepMind has developed Gemini in different configurations to cater to various use cases and scalability needs. Each model offers unique strengths designed for specific applications:

1. Gemini Ultra

- **Description:** The most advanced model in the Gemini family, designed for handling highly complex tasks with precision.
- **Key Features:**
 - Capable of advanced reasoning.
 - Best for research, advanced diagnostics, and strategic forecasting.

2. Gemini Pro

- **Description:** A balanced model offering a mix of speed and intelligence, ideal for enterprise-level workloads.
- **Key Features:**
 - Handles mid-complexity tasks efficiently.
 - Optimized for cost and scalability.

3. Gemini Nano

- **Description:** A lightweight, high-speed model for simple, real-time interactions.
- **Key Features:**
 - Prioritizes speed and cost-efficiency.
 - Ideal for tasks requiring instant responses.

Longer context

1.5 Pro and 1.5 Flash both have a default context window of up to one million tokens — the longest context window of any large scale foundation model. They achieve near-perfect recall on long-context retrieval tasks across modalities, unlocking the ability to process long documents, thousands of lines of code, hours of audio, video, and more. For 1.5 Pro, developers and enterprise customers can also sign up to try a two-million-token context window.

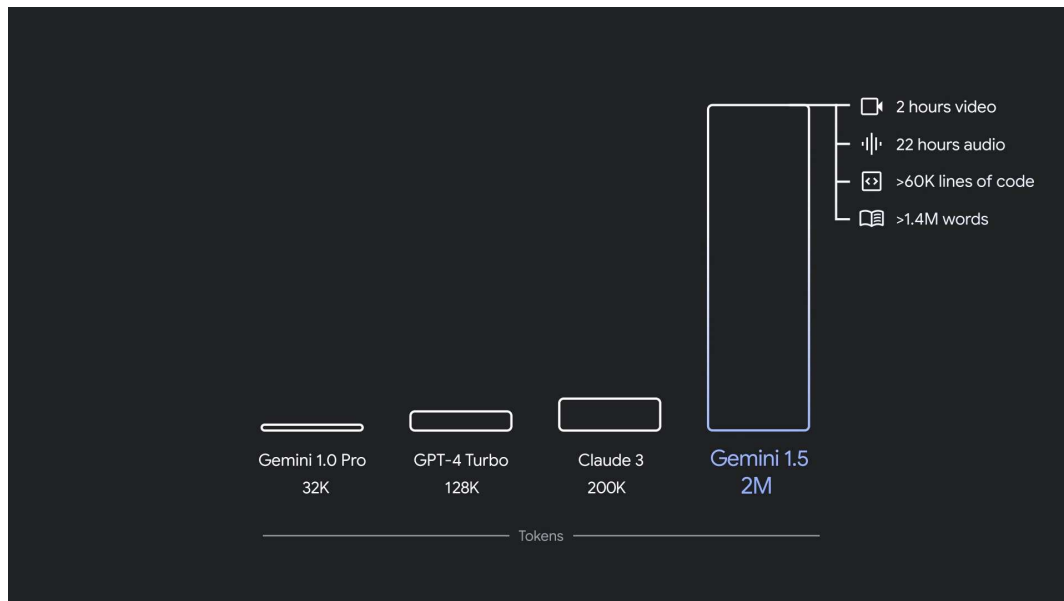


Figure 41 - Gemini 1.5 number of tokens (Source: deepmind.google)

4.3.2 Gemini Ultra: The Pinnacle of AI Innovation

DeepMind's Gemini Ultra is the flagship model in the Gemini family, designed to tackle the most complex and demanding AI tasks. With its cutting-edge architecture and unparalleled reasoning capabilities, Gemini Ultra represents the forefront of artificial intelligence, offering solutions for advanced use cases that require precision, adaptability, and scalability. (Google DeepMind, n.d.)

What Sets Gemini Ultra Apart?

1. Unmatched Complexity Handling

Gemini Ultra is built to navigate the most intricate problems, whether they involve advanced data analysis, strategic forecasting, or scientific research. Its ability to process vast amounts of data and draw meaningful insights ensures it stands out as a leader in AI performance.

2. Enhanced Reasoning and Memory

The model excels in multi-step reasoning and integrates an advanced memory system that enables it to recall and synthesize information dynamically. This feature makes Gemini Ultra especially effective in applications requiring continuity and high-level decision-making.

3. Seamless Tool Integration

Gemini Ultra supports the integration of external tools and APIs, allowing it to extend its functionality across diverse domains. This adaptability makes it suitable for industries such as healthcare, finance, and R&D, where custom workflows are crucial.

4. High-Performance Infrastructure

The model operates on state-of-the-art hardware optimized for speed, reliability, and efficiency. This infrastructure ensures that Gemini Ultra delivers rapid responses without compromising accuracy or depth.

The first model to outperform human experts on MMLU (Massive Multitask Language Understanding), a benchmark that uses 57 subjects including math, physics, history, law, medicine and ethics to test both world knowledge and problem-solving ability.

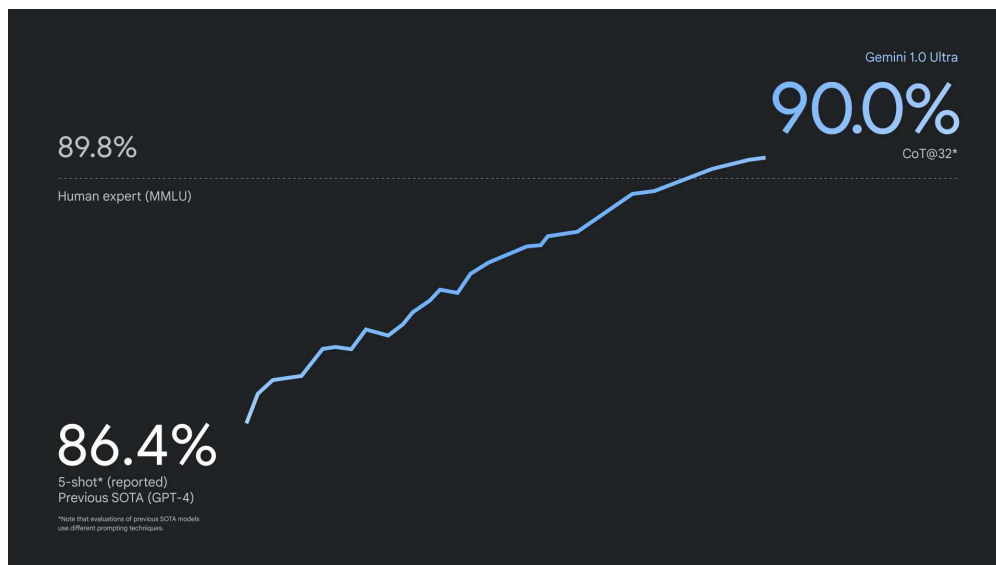


Figure 42 - First model to outperform human experts on MMLU (Source. Deepmind.google)

Gemini: Serving Billions of Google Users Across Diverse Platforms

The Gemini AI models are deeply integrated into Google's expansive suite of products and services, enhancing functionality and delivering transformative experiences to billions of users globally. By embedding these advanced models across a range of applications, Gemini redefines productivity, creativity, and innovation.

Key Applications of Gemini Across Google

Gemini App

Designed as a tool to supercharge creativity, the Gemini App provides users with an AI-powered conversational experience, fostering ideation and problem-solving with real-time feedback.

Google Workspace

Gemini elevates productivity in Workspace applications, such as:

- Gmail: Craft smarter and more concise emails with AI-assisted drafting.
- Docs: Automatically generate summaries or suggestions for content improvement.
- Sheets: Enable data-driven insights and forecasting with advanced analysis tools.
- Slides: Suggest creative layouts or generate visuals based on textual content.

Pixel Devices

The Pixel 8 Pro introduces Gemini Nano, Google's most efficient AI model tailored for on-device tasks. This enables functionalities like:

- Real-time photo enhancement.
- AI-driven personal assistant features.
- Seamless offline AI capabilities.

Chrome Browser

Gemini Nano powers AI-driven web experiences in Chrome, enabling developers to:

- Build interactive, intelligent web applications.
- Integrate real-time user personalization features.
- Explore new dimensions of AI-assisted browsing.

Google Ads

Gemini models optimize advertising strategies by:

- Delivering personalized ad recommendations.
- Enhancing campaign performance and ROI with data-driven insights.
- Automating audience segmentation and targeting for maximum impact.

Google Cloud

Enterprises leverage Gemini through Google Cloud to:

- Innovate faster with scalable, enterprise-ready AI solutions.
- Automate customer interactions with advanced chatbots.
- Gain deeper insights through predictive analytics and data modeling.

Android Platform

Gemini empowers developers in Android Studio by:

- Accelerating app development with AI-driven coding tools.
- Offering real-time error detection and optimization suggestions.
- Simplifying integration of AI features into Android applications.

Transformative Impact of Gemini

By embedding Gemini into its ecosystem, Google achieves unparalleled integration of cutting-edge AI technologies into everyday user experiences. Whether through personal productivity tools, enterprise solutions, or developer platforms, Gemini ensures that AI becomes a seamless part of life and work for billions of users worldwide.

	Gemini Ultra	Gemini Pro
MMLU Multiple-choice questions in 57 subjects (professional & academic) (Hendrycks et al., 2021a)	90.04% CoT@32*	79.13% CoT@8*
GSM8K Grade-school math (Cobbe et al., 2021)	94.4% Maj1@32	86.5% Maj1@32
MATH Math problems across 5 difficulty levels & 7 subdisciplines (Hendrycks et al., 2021b)	53.2% 4-shot	32.6% 4-shot
BIG-Bench-Hard Subset of hard BIG-bench tasks written as CoT problems (Srivastava et al., 2022)	83.6% 3-shot	75.0% 3-shot
HumanEval Python coding tasks (Chen et al., 2021)	74.4% 0-shot (PT****)	67.7% 0-shot (PT****)
Natural2Code Python code generation. (New held-out set with no leakage on web)	74.9% 0-shot	69.6% 0-shot
DROP Reading comprehension & arithmetic. (metric: F1-score) (Dua et al., 2019)	82.4 Variable shots	74.1 Variable shots
HellaSwag (validation set) Common-sense multiple choice questions (Zellers et al., 2019)	87.8% 10-shot	84.7% 10-shot
WMT23 Machine translation (metric: BLEURT) (Tom et al., 2023)	74.4 1-shot (PT****)	71.7 1-shot

Figure 43 - Gemini Ultra benchmarks (Source: deepmind.google)

	Gemini Ultra (pixel only)	Gemini Pro (pixel only)
MMMU (val) Multi-discipline college-level problems (Yue et al., 2023)	59.4% pass@1 62.4% Maj1@32	47.9%
TextVQA (val) Text reading on natural images (Singh et al., 2019)	82.3%	74.6%
DocVQA (test) Document understanding (Mathew et al., 2021)	90.9%	88.1%
ChartQA (test) Chart understanding (Masry et al., 2022)	80.8%	74.1%
InfographicVQA (test) Infographic understanding (Mathew et al., 2022)	80.3%	75.2%
MathVista (testmini) Mathematical reasoning (Lu et al., 2023)	53.0%	45.2%
AI2D (test) Science diagrams (Kembhavi et al., 2016)	79.5%	73.9%
VQAv2 (test-dev) Natural image understanding (Goyal et al., 2017)	77.8%	71.2%

Figure 44 - Others Gemini Ultra benchmarks (Source: deepmind.google)

MMMU (val)	Gemini Ultra (0-shot)	
	Maj@32	pass@1
Art & Design	74.2	70.0
Business	62.7	56.7
Science	49.3	48.0
Health & Medicine	71.3	67.3
Humanities & Social Science	78.3	78.3
Technology & Engineering	53.0	47.1
Overall	62.4	59.4

Figure 45 - Gemini Ultra benchmarks on different subjects (Source: Deepmind.google)

4.4 Comparative Overview of Advanced AI Chatbot Models

The evolution of AI-powered chatbots has introduced a variety of specialized tools, each designed to cater to specific user needs. From coding to image generation and data analysis, these tools have revolutionized how individuals and businesses interact with technology. The following sections provide an in-depth look at the most notable AI chatbot services, highlighting their unique features and optimal use cases. (Artificial Analysis, n.d.)

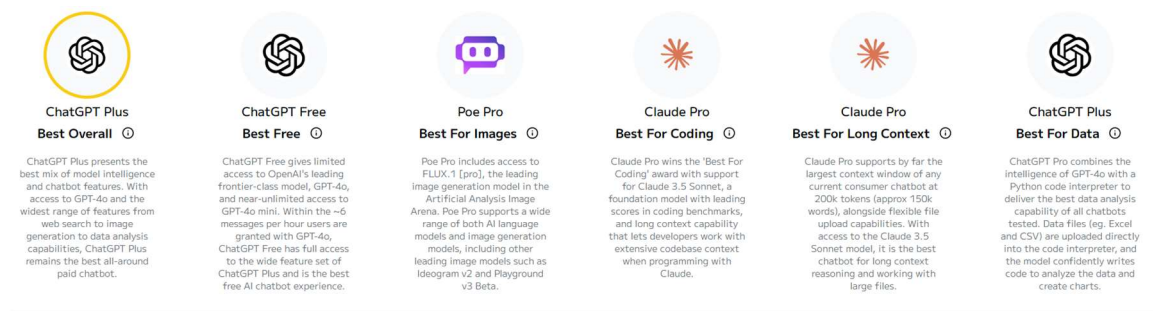


Figure 46 - AI chatbots comparison (Source: Artificial Analysis)

ChatGPT Plus: The All-Rounder

ChatGPT Plus represents the most versatile option for users seeking a comprehensive chatbot experience. Powered by GPT-4 Turbo, this premium model offers unparalleled intelligence and flexibility. Its wide range of applications includes web search, image generation, and advanced data analysis, making it the best choice for general users. This service is subscription-based, offering reliable access to state-of-the-art AI capabilities.

ChatGPT Free: Accessible AI for Everyone

As the free-tier version of OpenAI's flagship chatbot, ChatGPT Free provides access to GPT-4 in a limited capacity. While it may not support complex or resource-intensive tasks like its premium counterpart, it is well-suited for basic interactions and general inquiries. The service ensures accessibility, enabling users to explore AI's potential without financial commitment.

Claude Pro: Ideal for Coding and Long-Context Tasks

The Claude Pro series introduces two distinct capabilities:

1. **Best for Coding:** Equipped with the Claude 3.5 Sonnet model, this service is optimized for software developers. It excels in handling extensive codebases, debugging, and generating high-quality programming solutions.
2. **Best for Long Context:** Offering the largest context window of any AI chatbot (up to 200k tokens), Claude Pro is ideal for processing large datasets and

handling complex reasoning tasks. This feature benefits researchers and professionals managing detailed reports or intricate projects.

ChatGPT Pro: Leading in Data Analysis

ChatGPT Pro integrates GPT-4 Turbo with a robust Python code interpreter, making it the go-to tool for data-driven professionals. Users can directly upload data files such as Excel or CSV and rely on the model to write code for analysis or create visual charts. This functionality is crucial for analysts and data scientists seeking streamlined, accurate results.

Key Takeaways

This comparative analysis highlights the unique strengths of modern AI chatbots, demonstrating how tailored functionalities can meet diverse user requirements. Whether for creative, technical, or analytical purposes, these tools showcase the dynamic potential of AI to simplify complex tasks and enhance productivity across industries. Each model provides distinct advantages, enabling users to select the most appropriate solution for their specific needs.

Intelligence Comparison of Language Models

The Artificial Analysis Quality Index provides a synthesis metric to evaluate the overall intelligence and reasoning capabilities of various foundation models. This index assesses model performance using datasets like MMLU, GPQA, Math, and HumanEval, reflecting a model's ability to handle diverse and complex tasks.

The comparison chart highlights the following:

- ChatGPT Plus, ChatGPT Free, and Microsoft Copilot Free all achieve the highest intelligence score of 77.2, showcasing their robust analytical capabilities.
- Claude Pro and Claude Free follow closely with a score of 76.9, indicating their competence in advanced reasoning tasks.
- Gemini Advanced scores slightly lower at 71.5, but it balances intelligence with other features like integration within Google's ecosystem.
- Gemini Free achieves a score of 60, representing a simpler model more suitable for basic tasks.

This comparison underscores that while high intelligence scores are critical for complex scenarios, models like Gemini emphasize complementary strengths like real-time integration and cost-effectiveness.

Quality Index: Foundation Model Intelligence and Reasoning

Artificial Analysis Quality Index; Higher is better

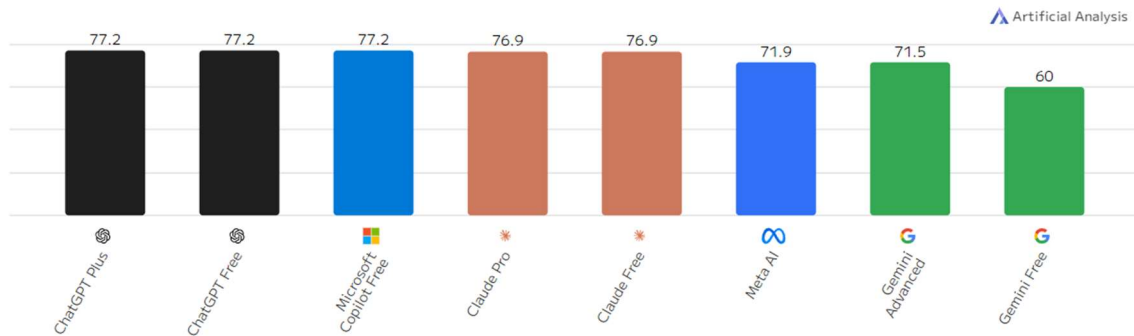


Figure 47 - Intelligence comparison (Source: Artificial Analysis)

Context Window Capabilities

The Effective Context Window measures the maximum number of tokens (units of text) a model can process simultaneously. This capability directly influences a model's ability to handle long and complex inputs like lengthy documents or intricate conversational threads.

Key observations include:

- Claude Pro leads with a remarkable context window of 160,000 tokens, making it the best choice for analyzing extensive datasets or conducting detailed research.
- ChatGPT Plus supports 25,000 tokens, offering a strong balance between length and cost efficiency.
- Models like Gemini Advanced (17,000) and Claude Free (13,000) provide respectable performance, catering to mid-range input lengths.
- Simpler models, including ChatGPT Free (5,000) and Gemini Free (2,000), are designed for shorter interactions, making them suitable for quick tasks or beginner-level applications.

The effective context window is a critical factor when selecting a model for applications like research reviews, legal documentation, or extensive knowledge synthesis.

Effective Context Window

Effective Context Window (Tokens); Higher is better

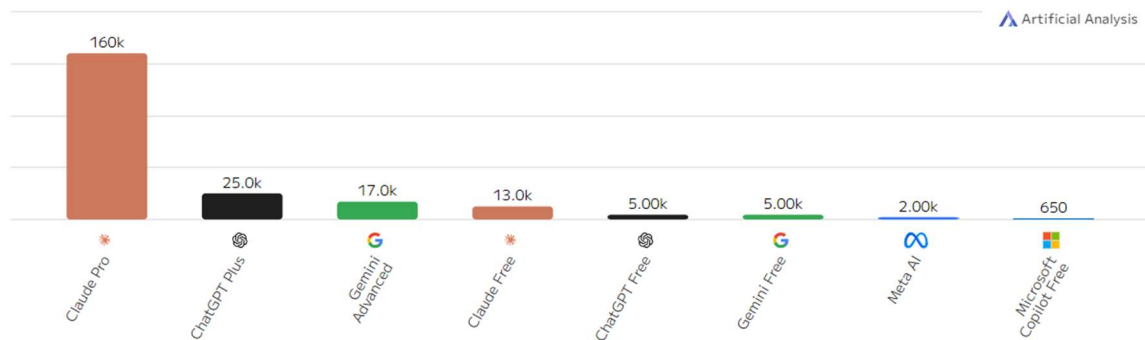


Figure 48 - Context window capabilities comparison (Source: Artificial Analysis)

Speed Comparison: Tokens per Second

The Output Speed Metric quantifies the number of tokens a model generates per second, directly reflecting its responsiveness and efficiency in real-time applications.

The data reveals:

- Gemini Free outpaces all competitors with a speed of 150 tokens per second, making it ideal for applications requiring rapid interaction, such as live customer support.
- Claude Pro and Claude Free deliver 70 tokens per second, balancing speed with high-context processing capabilities.
- ChatGPT Plus and ChatGPT Free, alongside Gemini Advanced, generate 50 tokens per second, aligning with their robust processing power and versatile applications.
- Models like Meta AI (40 tokens/second) and Microsoft Copilot Free (30 tokens/second) focus on specialized use cases where speed is secondary to task complexity.

For time-sensitive tasks like real-time translations or interactive customer queries, speed is a pivotal metric, and Gemini Free stands out as the top performer in this category.

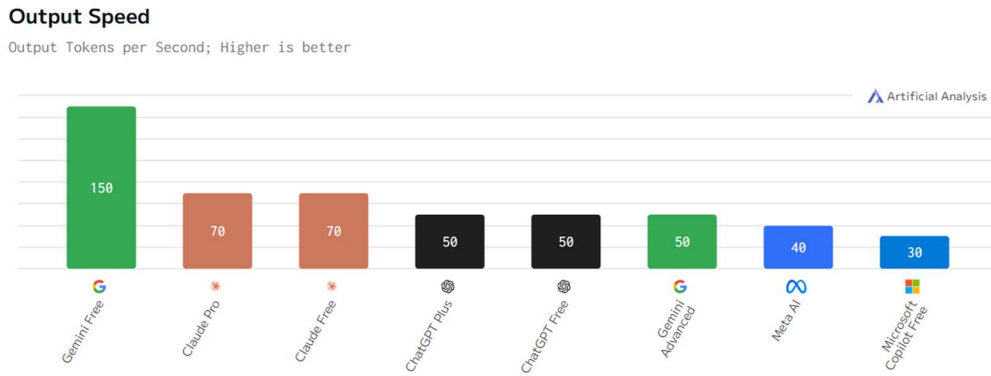


Figure 49 - Output speed comparison (Source: Artificial Analysis)

4.5 Cost Comparison and Feature Analysis of Large Language Model Platforms

The rapid development of large language models (LLMs) has given rise to various platforms, each offering a unique combination of features and pricing models. This section provides an in-depth analysis of six prominent LLM platforms—ChatGPT Plus, ChatGPT Free, Claude Pro, Claude Free, Gemini Advanced, and Gemini Free—focusing on cost structures, tools, input capabilities, and privacy policies.

4.5.1 Cost and Performance: Comparing Plan Options

The cost and performance of LLM platforms vary based on the model used, rate limits, and overall speed. The premium options, including ChatGPT Plus, Claude Pro, and Gemini Advanced, are priced at approximately \$20/month and cater to advanced users requiring enhanced capabilities.

- ChatGPT Plus utilizes the GPT-4o model (Aug '24) with an intelligence score of 77.2, offering high accuracy and rapid performance with unlimited rate limits.
- Claude Pro, also priced at \$20/month, is based on Claude 3.5 Sonnet with an intelligence score of 76.9. It provides users with up to 46 messages per hour, balancing performance and affordability.
- Gemini Advanced is slightly cheaper at \$19.99/month, using Gemini 1.5 Pro, with an intelligence score of 71.5 and optimized for high-speed responses.

Conversely, the free plans (ChatGPT Free, Claude Free, and Gemini Free) are more accessible but come with limitations:

- ChatGPT Free has a lower intelligence score of 60, capped at 6 messages per hour.
- Claude Free and Gemini Free similarly have restricted capabilities but still offer basic functionality for casual users.

The distinctions in performance and cost reflect the platforms’ target demographics, ranging from enterprise-level users to individual learners exploring generative AI capabilities.

	 ChatGPT Plus	 ChatGPT Free	 Claude Pro	 Claude Free	 Gemini Advanced	 Gemini Free
Price ⓘ	\$20.00	Free	\$20.00	Free	\$19.99	Free
Foundation Model ⓘ	GPT-4o (Aug '24)	GPT-4o (Aug '24) ⓘ	Claude 3.5 Sonnet	Claude 3.5 Sonnet ⓘ	Gemini 1.5 Pro (May '24) ⓘ	Gemini 1.5 Flash (May '24)
Intelligence ⓘ	77.2	77.2	76.9	76.9	71.5	60.0
Rate Limit ⓘ	High	6 messages per hour ⓘ	46 messages per hour	7 messages per hour ⓘ	High	High
Speed ⓘ ▼	Fairly fast	Fairly fast	Fairly fast	Fairly fast	Fast	Fast

Figure 50 - Comparing different plan options (Source: Artificial Analysis)

4.5.2 Tools and Functionalities: Enhancing User Experience

Each platform integrates tools to enhance user experience, but the availability of these tools varies across the tiers.

- ChatGPT Plus offers comprehensive tool support, including web search capabilities for real-time information retrieval, a code interpreter for programming tasks, and data analysis tools. These features make it an all-encompassing platform for professionals handling diverse tasks.
- Gemini Advanced mirrors ChatGPT Plus in terms of tools, providing similar functionalities that cater to enterprises and developers.
- Claude Pro, on the other hand, is more limited, focusing primarily on HTML output capabilities, which are ideal for structured content generation but less suited for advanced data analysis or web search.

Free plans (ChatGPT Free, Claude Free, Gemini Free) provide minimal tool support, restricting advanced functionalities. For example, Claude Free lacks the tools required for real-time web searches or advanced data processing. This limitation makes free tiers suitable only for basic tasks such as simple content generation and conversational interactions.

Tools  ^	3/4	3/4	1/4	1/4	3/4	1/4
Web Search 	✓	✓ 	-	-	✓	✓
Code Interpreter 	✓	✓ 	- 	- 	✓	-
Data Analysis 	✓	✓ 	- 	- 	✓	- 
HTML Output 	-	-	✓ 	✓ 	-	-

Figure 51 - tools and functionalities comparison

4.5.3 Input Capabilities: File Handling and Data Processing

The platforms differ significantly in their ability to process various input formats, a critical factor for professional and enterprise users.

- ChatGPT Plus and Gemini Advanced stand out with full input capabilities, including support for image uploads, PDF processing, and handling Excel/CSV files. These features allow users to analyze complex datasets and extract information from structured and unstructured files.
- Claude Pro and Claude Free are slightly limited in input capabilities, lacking certain features such as file source connection, which restricts their usability for data-intensive tasks.

The free versions of ChatGPT and Gemini also lack advanced input features, making them less effective for handling complex file types.

Input Capabilities  ^	4/4	4/4	3/4	3/4	4/4	1/4
Image Upload 	✓	✓	✓	✓	✓	✓
PDF Upload 	✓ 	✓ 	✓ 	✓ 	✓	-
Excel/CSV Upload 	✓	✓	✓	✓	✓	-
File Source Connect 	✓	✓	-	-	✓	-

Figure 52 - Input capabilities

4.5.4 Privacy and Data Sharing Policies

Privacy remains a critical concern when selecting an LLM platform, particularly for enterprise users handling sensitive data. The privacy policies of the platforms reveal stark contrasts:

- Claude Pro and Claude Free prioritize privacy by not sharing user data by default, ensuring a secure environment for sensitive information.
- ChatGPT Plus, ChatGPT Free, and Gemini platforms allow data sharing by default but provide an opt-out option for users who wish to enhance their privacy settings.

This difference underscores the importance of aligning platform selection with organizational privacy requirements, especially in regulated industries such as healthcare and finance.

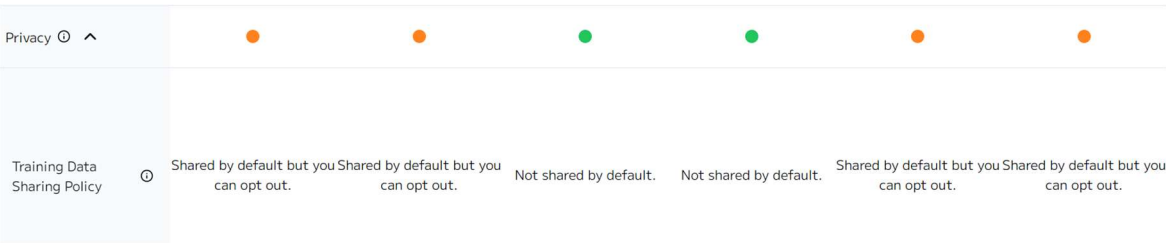


Figure 53 - Privacy and data sharing comparison

5 Advancing Open Source AI

This chapter delves into the realm of open-source AI, highlighting its significance in fostering collaboration, transparency, and innovation. It begins with an exploration of the evolution of open-source AI, featuring Llama 3.1 as a pivotal milestone. The chapter also showcases groundbreaking contributions from Google's Gemma models, emphasizing the 27B model's advanced capabilities. Additionally, Mistral's cutting-edge advancements, including the Mixtral 8x22B mixture-of-experts model, are analyzed to underline the power of open-source AI in driving accessible and high-performance solutions. Together, these developments demonstrate how open-source AI is shaping the future of generative technologies.

5.1 The Evolution of Open Source AI: Llama 3.1's Arrival

The release of Llama 3.1 405B represents a landmark in the realm of open-source large language models (LLMs), pushing the boundaries of what openly accessible AI can achieve. This model, the largest and most capable open-source AI to date, delivers state-of-the-art performance in various tasks, from multilingual translation and tool use to math reasoning and general knowledge. By providing capabilities on par with leading closed models like GPT-4 and Claude 3.5, Meta underscores its commitment to innovation through transparency and community engagement. (Meta, 2023)



Figure 54 - Meta AI logo

This development underscores Meta's commitment to democratizing AI technology, providing the global community with access to cutting-edge AI tools.

Key Features of Llama 3.1:

- **Expanded Context Length:** Llama 3.1 supports a context window of up to 128,000 tokens, enabling it to handle extensive inputs effectively. This feature is particularly beneficial for applications requiring long-form text summarization and complex coding assistance.
- **Multilingual Support:** The model is proficient in eight languages, enhancing its applicability across diverse linguistic contexts and promoting inclusivity in AI applications.

- **Synthetic Data Generation and Model Distillation:** Llama 3.1 introduces workflows that facilitate the generation of synthetic data and the distillation of models, processes that are instrumental in refining smaller models and fostering innovative AI applications.

In addition to the flagship 405B model, Meta has released upgraded versions of the 8B and 70B models. These models inherit the multilingual capabilities and extended context length of Llama 3.1, making them suitable for a wide range of applications, including multilingual conversational agents and coding assistants. Notably, Meta has revised its licensing to permit developers to utilize outputs from Llama models, including the 405B, to enhance other models, thereby fostering a collaborative and open AI development environment.

Performance Evaluation:

Llama 3.1 underwent rigorous evaluation across over 150 benchmark datasets encompassing various languages and tasks. Extensive human evaluations were also conducted to compare its performance with competing models in real-world scenarios. The results indicate that Llama 3.1 is competitive with leading foundation models, including GPT-4 and Claude 3.5 Sonnet, across a spectrum of tasks. This performance underscores the model's robustness and versatility, positioning it as a formidable tool in the AI landscape. (Meta , 2024)

Category Benchmark	Llama 3.1 405B	Nemotron 4 340B Instruct	GPT-4 (0125)	GPT-4 Omni	Claude 3.5 Sonnet
General					
MMLU (0-shot, CoT)	88.6	78.7 (non-CoT)	85.4	88.7	88.3
MMLU PRO (5-shot, CoT)	73.3	62.7	64.8	74.0	77.0
IFEval	88.6	85.1	84.3	85.6	88.0
Code					
HumanEval (0-shot)	89.0	73.2	86.6	90.2	92.0
MBPP EvalPlus (base) (0-shot)	88.6	72.8	83.6	87.8	90.5
Math					
GSM8K (8-shot, CoT)	96.8	92.3 (0-shot)	94.2	96.1	96.4 (0-shot)
MATH (0-shot, CoT)	73.8	41.1	64.5	76.6	71.1
Reasoning					
ARC Challenge (0-shot)	96.9	94.6	96.4	96.7	96.7
GPQA (0-shot, CoT)	51.1	-	41.4	53.6	59.4
Tool use					
BFCL	88.5	86.5	88.3	80.5	90.2
Nexus	58.7	-	50.3	56.1	45.7
Long context					
ZeroSCROLLS/QuALITY	95.2	-	95.2	90.5	90.5
InfiniteBench/En.MC	83.4	-	72.1	82.5	-
NIH/Multi-needle	98.1	-	100.0	100.0	90.8
Multilingual					
Multilingual MGSM (0-shot)	91.6	-	85.9	90.5	91.6

Figure 55 - Performance evaluations (Source: llama.com)

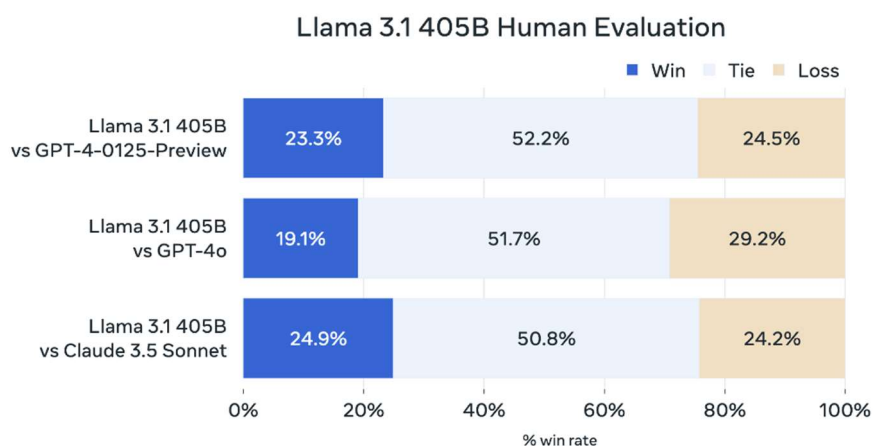


Figure 56 - Comparison between Llama 3.1 405B and the bests closed source models (Source: llama.com)

Human Evaluations for Enhanced Model Performance

In addition to the traditional evaluations performed on benchmark datasets, Meta conducts human evaluations to refine and optimize the performance of Llama models

further. These evaluations are particularly valuable for assessing aspects of the model that are not easily quantifiable through standard benchmarks. Human reviewers analyze how well the model performs in terms of tone, verbosity, and its understanding of nuances and cultural contexts. This process ensures that the models not only perform technically well but also align with human expectations for natural and contextually appropriate interactions. For instance, these evaluations allow Meta to identify areas where the model could be improved to better capture the subtleties of multilingual conversations or maintain a balanced level of detail in responses.

The visual representation in the chart highlights these human evaluation outcomes, showing how the model fares across various categories such as English comprehension, reasoning, coding, multilingual tasks, and multiturn dialogues. By incorporating this human-centric feedback loop, Meta ensures that its models remain effective and culturally adaptable, making them suitable for a broad range of real-world applications.

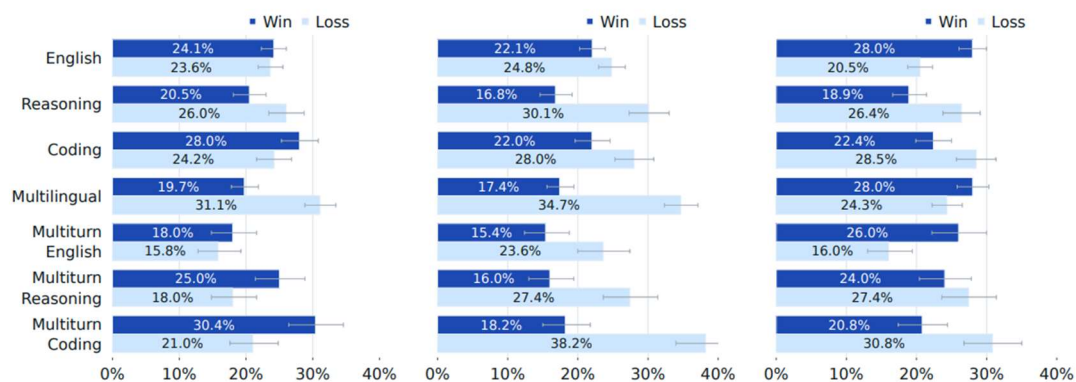


Figure 57 - Evaluations in different subjects (Source: llama.com)

The Llama System:

Meta envisions Llama as an integral component of a broader system capable of orchestrating various elements, including external tools. This vision aims to provide developers with the flexibility to design and create custom solutions aligned with their specific objectives. To support this, Meta has released a comprehensive reference system featuring sample applications and new components such as Llama Guard 3, a multilingual safety model, and Prompt Guard, a prompt injection filter. These resources are open-source, encouraging community engagement and collaborative development.

To further enhance interoperability and standardization, Meta has introduced the "Llama Stack," a set of standardized interfaces designed to streamline the development of toolchain components and agentic applications. By collaborating with industry partners, startups, and the broader community, Meta aims to establish these interfaces as ecosystem-wide standards, thereby reducing development barriers and fostering innovation.

Openness and Innovation:

A distinguishing feature of Llama models is their open-source nature. Developers have the freedom to download model weights, customize them for specific applications, train on new datasets, and perform additional fine-tuning. This openness enables deployment across various environments, including on-premises, cloud, or local machines, without necessitating data sharing with Meta. Such flexibility democratizes access to advanced AI tools, empowering a broader spectrum of developers and organizations to leverage AI capabilities.

Contrary to the perception that closed models are more cost-effective, Llama models offer some of the lowest costs per token in the industry, as evidenced by testing from Artificial Analysis. This cost efficiency, combined with the model's advanced capabilities, makes Llama an attractive option for developers and organizations seeking powerful yet affordable AI solutions.

The Remarkable Growth and Adoption of Llama Models: A Case for Open Source AI

The rapid growth of Llama models highlights their emergence as a leading force in the AI landscape. Since their introduction, Llama models have become the cornerstone of open-source AI innovation, with their popularity accelerating dramatically in just over a year. With 350 million downloads to date, including 20 million downloads in the last month alone, Llama models are now the most downloaded open-source model family globally. This trajectory underscores the immense value of open source in democratizing AI and fostering innovation.

Exponential Growth and Market Leadership

The adoption rate of Llama models has seen unprecedented growth. In 2023, Meta reported steady usage, but since the release of Llama 3.1 in May 2024, the numbers have skyrocketed:

- 10x increase in downloads compared to the same time last year, reflecting exponential adoption across industries.
- Doubling of token volume usage from May to July 2024 among major cloud service providers.
- A 10x surge in monthly token usage from January to July 2024 for some of Meta's largest cloud provider partners.

In August 2024, the 405B variant of Llama 3.1 emerged as the most utilized foundation model across cloud platforms, showcasing the traction gained by this frontier-level AI model.

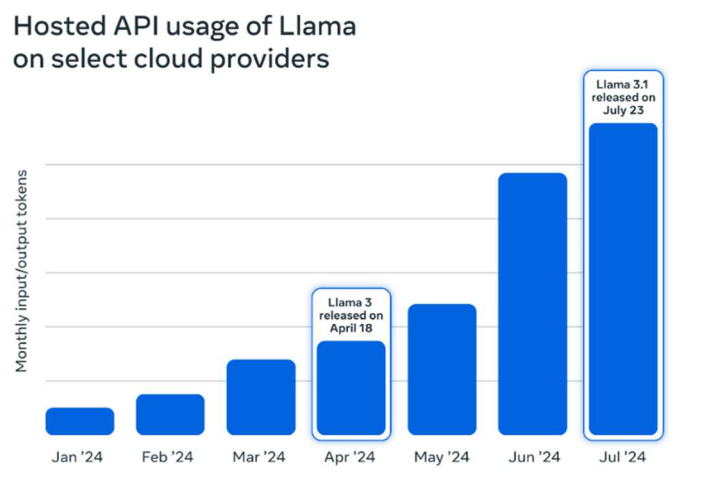


Figure 58 - Growth of the use of Llama (Source: llama.com)

The Power of Open Source: A Vibrant Ecosystem

Llama's success is a testament to the power of open source AI. Meta's decision to make Llama models openly accessible has catalyzed a thriving and diverse AI ecosystem. This approach has empowered developers and enterprises to innovate without the constraints of proprietary systems, enabling:

- Startups to experiment and push the boundaries of AI capabilities.
- Enterprises to build on-premise AI systems or deploy Llama through major cloud service providers such as AWS, Microsoft Azure, and Google Cloud.

Meta's partnerships with industry leaders, including Databricks, Dell, Groq, NVIDIA, IBM Watsonx, and Snowflake, have further amplified Llama's accessibility and utility. These collaborations ensure that developers can seamlessly integrate Llama models into existing workflows, unlocking their full potential.

5.2 Gemma: Google's Open Source AI Innovation

In the ever-evolving landscape of artificial intelligence, Google has consistently stood at the forefront of innovation, advancing technologies that redefine possibilities. Among its groundbreaking contributions is Gemma, a family of lightweight, state-of-the-art open models designed to foster responsible AI development. Rooted in Google's long-standing commitment to open-source innovation, Gemma builds on the company's history of releasing transformative technologies such as Transformers, TensorFlow, and BERT. By openly sharing its advancements, Google not only empowers developers and researchers but also ensures that AI's benefits are accessible globally.



Figure 59 - Gemma logo

Developed by Google DeepMind and other teams across Google, Gemma leverages the same research and technology behind Gemini models, Google’s largest AI offering. However, Gemma takes a distinct approach by emphasizing accessibility and responsibility, making it ideal for developers seeking high performance without the barriers often associated with proprietary systems.

From its inception, Gemma has been synonymous with scalability and adaptability. Available in multiple sizes, it is designed to cater to diverse computational needs, running seamlessly across laptops, workstations, and cloud environments. Its compatibility with major AI frameworks like PyTorch, TensorFlow, and JAX, alongside integrations with platforms such as Hugging Face and Google Cloud Vertex AI, makes Gemma a versatile choice for developers aiming to innovate responsibly.

5.2.1 Gemma 2: Highlighting the 27B Model as Google’s Best Open-Source Offering

Google continues to pave the way for open-source AI innovation with the release of the Gemma 2 family, a new generation of lightweight and state-of-the-art open models. Among these, the Gemma 2 27B stands out as the most powerful and capable model in the series, solidifying its position as Google’s premier open-source large language model. Designed for both accessibility and performance, Gemma 2 27B combines cutting-edge research with responsible AI principles, offering unparalleled capabilities for developers and researchers alike. (Google, 2024)

The Gemma 2 27B: A Breakthrough in Open AI Models

The Gemma 2 27B represents a significant milestone in Google’s AI journey. With its 27 billion parameters, this model achieves a remarkable balance between power and efficiency, delivering state-of-the-art results on key benchmarks. Unlike many larger, closed-source models, Gemma 2 27B is openly accessible, providing developers with the opportunity to integrate its capabilities into their projects while maintaining flexibility and transparency.

Unmatched Performance at Scale

Gemma 2 27B is specifically engineered to deliver superior performance across a wide range of tasks, including:

- *Multilingual conversational agents*: Its enhanced multilingual capabilities ensure accurate and contextually appropriate outputs across various languages.
- *Long-form content summarization*: The model excels at processing and summarizing lengthy documents, making it invaluable for research and data-intensive industries.
- *Retrieval-augmented generation (RAG)*: With its advanced reasoning abilities, the 27B model can generate detailed and accurate outputs based on retrieved information.

Thanks to its lightweight design, Gemma 2 27B outperforms many models with significantly larger parameter sizes, proving that size isn’t the sole determinant of AI excellence. Its ability to achieve top-tier results while maintaining computational efficiency ensures that it can be deployed seamlessly across diverse hardware and software environments.

			Gemma 2		Llama 3		Grok-1
	BENCHMARK	METRIC	9B	27B	8B	70B	314B
General	MMLU	5-shot, top-1	71.3	75.2	66.6	79.5	73.0
Reasoning	BBH	3-shot, CoT	68.2	74.9	61.1	81.3	–
	HellaSwag	10-shot	81.9	86.4	82	–	–
Math	GSM8K	5-shot, maj@1	68.6	74.0	45.7	–	62.9 (8-shot)
	MATH	4-shot	36.6	42.3	–	–	23.9
Code	HumanEval	pass@1	40.2	51.8	–	–	63.2 (0-shot)

Figure 60 - Gemma 2 benchmarks evaluations (Source: Google AI for Developers)

Developer-Friendly and Scalable

Gemma 2 27B is built to cater to developers of all levels, offering seamless integration with popular tools and platforms:

- *Multi-framework compatibility*: Developers can work with their preferred frameworks, including PyTorch, JAX, and TensorFlow.
- *Cross-platform deployment*: The model can run on various devices, from local desktops to cloud environments, ensuring accessibility across diverse use cases.

- Optimization for cutting-edge hardware: The 27B model is optimized for NVIDIA GPUs and Google Cloud TPUs, delivering industry-leading performance for both training and inference.

For enterprise users, Gemma 2 27B integrates effortlessly with Google Cloud Vertex AI, offering advanced MLOps tools for deployment and customization. Its scalability ensures that organizations can leverage its capabilities across a range of applications, from small-scale experiments to large-scale deployments.

The Benchmark for Open-Source Excellence

Google's introduction of Gemma 2 27B redefines the standards for open-source AI models. By combining state-of-the-art performance, responsible design, and developer-friendly tools, this model empowers innovation across industries. Whether used for advanced research, language processing, or data generation, Gemma 2 27B stands out as a transformative tool for developers and organizations.

As the largest and most capable model in the Gemma 2 lineup, the 27B version highlights Google's dedication to advancing open-source AI. By providing access to such a powerful tool, Google ensures that the benefits of cutting-edge AI technology are shared widely, fostering a more innovative and inclusive AI ecosystem.

Comparative Analysis of Gemma 27B's Performance

The Gemma 27B model, developed by Google, stands out as a state-of-the-art open-source AI model. Despite having fewer parameters compared to larger models like Llama 3.1 (405B), the Gemma 27B consistently demonstrates competitive, and in some cases superior, performance across a range of benchmarks. This achievement underscores the efficiency and optimization underlying the Gemma architecture, setting a high standard for open-source language models.

Artificial Analysis and Reasoning Capabilities

The first chart evaluates multiple dimensions of language model performance, including general reasoning (MMLU), scientific reasoning (GPQA), and multilingual capabilities.

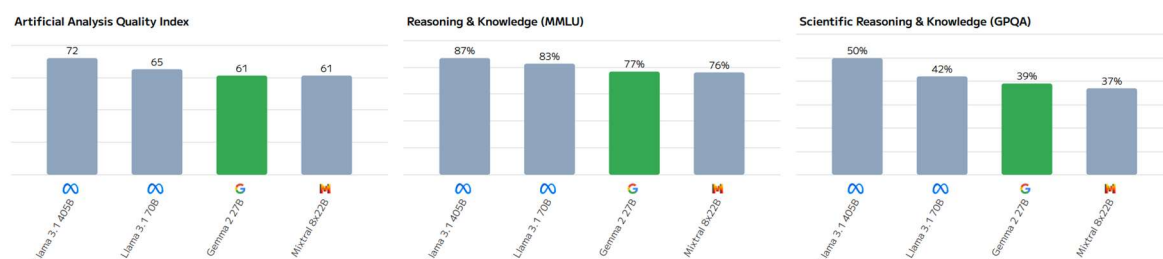


Figure 61 – Reasoning capabilities comparison (Source: Artificial Analysis)

This demonstrates that size alone does not determine effectiveness, as Gemma’s design enables it to excel in practical applications requiring diverse linguistic and analytical skills.

Coding and Communication Benchmarks

The second chart highlights Gemma 27B’s strength in coding and communication tasks.

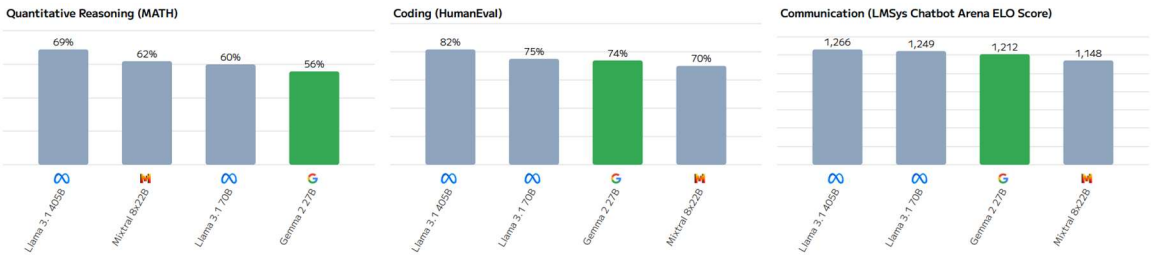


Figure 62 – Coding and communication benchmarks comparison (Source: Artificial Analysis)

Superior Performance with Optimized Design

What makes these results particularly noteworthy is that Gemma 27B achieves these scores with significantly fewer parameters than its competitors. This compact size allows for greater deployment flexibility, enabling the model to run effectively on devices with lower computational power while maintaining high performance. The design and optimization of the Gemma 27B model exemplify Google’s commitment to making advanced AI capabilities more accessible and efficient.

These charts visually highlight how Gemma 27B competes closely with significantly larger models, affirming its place as a leading solution in the open-source AI landscape.

5.3 Mistral: frontier AI in your hands

Mistral AI is a French company dedicated to developing state-of-the-art generative AI models with a strong emphasis on openness and innovation. Founded in April 2023 by a team of former engineers from Meta Platforms and Google DeepMind, Mistral AI quickly gained prominence in the artificial intelligence landscape. Its mission is to make advanced AI tools accessible to everyone, fostering transparency and collaboration through the release of open-source models. (Mistral AI, n.d.)



Figure 63 – Mistral AI logo

Commitment to Openness

One of Mistral AI's defining characteristics is its commitment to open-source technologies. The company believes that sharing AI advancements with the global community promotes responsible innovation and accelerates the development of impactful solutions. Unlike proprietary systems, Mistral AI's models are designed to be flexible and modifiable, allowing developers to adapt them to specific use cases. This open approach enables organizations to deploy AI tools in their preferred environments, whether on-premises, in virtual private clouds (VPC), or through public cloud platforms like Amazon Bedrock and Azure AI.

Technological Leadership

Mistral AI has introduced several innovative models that set new benchmarks in efficiency and usability. A notable example is the Mistral 7B, a highly optimized model released in September 2023. Despite its relatively small size, Mistral 7B delivers outstanding performance across a variety of tasks, outperforming larger models in several key benchmarks. This efficiency allows it to be deployed on standard hardware, making advanced AI capabilities more accessible.

Ecosystem and Industry Impact

Mistral AI is not just about developing models; it aims to build an ecosystem that supports developers and enterprises in integrating AI into their operations. The company's models are compatible with popular frameworks such as PyTorch and TensorFlow, and they can be fine-tuned using widely available tools like Hugging Face. Mistral also provides extensive documentation and pre-trained models, ensuring developers have the resources they need to deploy AI solutions effectively.

The adoption of Mistral AI's technologies spans multiple industries, including finance, healthcare, and technology. For instance, financial institutions use Mistral's models to optimize data analysis and decision-making processes, while technology companies integrate these tools into applications for content generation, virtual assistance, and coding support.

Future Vision

Mistral AI's vision extends beyond simply creating models. The company aims to drive the global AI community toward more collaborative and responsible innovation. By focusing on open-source principles, Mistral AI challenges the dominance of closed

systems, advocating for a more equitable distribution of AI technologies. This approach not only democratizes access to powerful tools but also ensures that organizations retain control over their data and implementations.

5.3.1 Introduction of Mixtral 8x22B: A Breakthrough in Mixture-of-Experts Models

Mistral AI, a pioneering company in the generative AI space, has unveiled its latest innovation: Mixtral 8x22B, a state-of-the-art mixture-of-experts (MoE) model. This cutting-edge AI model demonstrates Mistral AI's commitment to creating advanced, efficient, and open-source AI technologies. Mixtral 8x22B is not just another AI release—it represents a significant leap forward in model design, performance, and adaptability, solidifying Mistral's position as a leader in the AI ecosystem. (Mistral AI, n.d.)

Key Features of Mixtral 8x22B

The Mixtral 8x22B model is an MoE architecture comprising eight active experts, each equipped with 22 billion parameters. What makes Mixtral unique is its ability to activate only the specific expert components needed for a task, significantly optimizing both performance and computational efficiency. While other models rely on a single neural network for every input, Mixtral dynamically selects the most relevant expert, reducing redundancy and enhancing precision.

Efficiency Without Compromise

Mixtral is designed to deliver exceptional performance without requiring excessive computational resources. Each query activates only two out of the eight available experts, effectively reducing the overall computational load by 75%. This selective activation mechanism allows Mixtral 8x22B to achieve the performance of models with significantly higher parameter counts, while maintaining the scalability and cost-efficiency that enterprises demand.

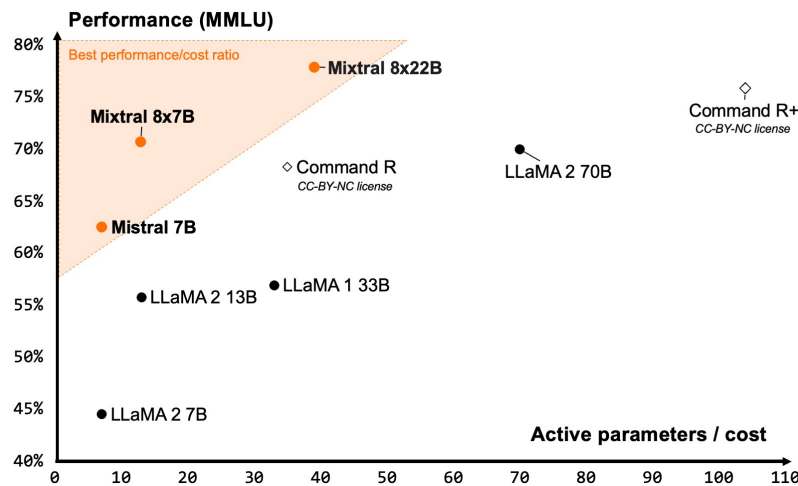


Figure 64 – Measure of the performance (MMLU) versus inference budget tradeoff (number of active parameters). Mistral 7B, Mistral 8x7B and Mistral 8x22B all belong to a family of highly efficient models compared to the other open models. (Source: Mistral.ai)

State-of-the-Art Performance

Despite its leaner activation strategy, Mixtral excels across a wide array of tasks. The model has achieved top-tier results on standard benchmarks such as MMLU (Massive Multitask Language Understanding), HumanEval for coding tasks, and BIG-bench for general reasoning. Its superior performance in tasks requiring nuanced understanding, multilingual proficiency, and technical problem-solving places Mixtral 8x22B among the most capable AI models currently available.

Reasoning and knowledge

Model	Active parameters	Common sense and reasoning					Knowledge	
		MMLU	HellaS	WinoG	Arc C (5)	Arc C (25)	TriQA	NaturalQS
LLaMA 2 70B	70B	69.9%	87.1%	83.2%	86.0%	85.1%	77.57%	35.5%
Command R	35B	68.2%	87.0%	81.5%	-	66.5%	-	-
Command R+	104B	75.7%	88.6%	85.4%	-	71.0%	-	-
Mistral 7B	7B	62.47%	83.1%	78.0%	77.2%	78.1%	68.8%	28.1%
Mixtral 8x7B	12.9B	70.63%	86.6%	81.2%	85.8%	85.9%	78.4%	36.5%
Mixtral 8x22B	39B	77.75%	88.5%	84.7%	91.3%	91.3%	82.2%	40.1%

Figure 65 – Reasoning and knowledge comparison (Source: Mistral.ai)

Multilingual capabilities

Model												
	Arc-C	HellaS	MMLU	Arc-C	HellaS	MMLU	Arc-C	HellaS	MMLU	Arc-C	HellaS	MMLU
Mistral 7B	44.2%	63.9%	50.6%	39.7%	58.4%	49.6%	43.9%	64.8%	51.4%	41.2%	60.8%	51.3%
Mixtral 8x7B	54.3%	76.0%	66.1%	52.7%	71.0%	64.9%	53.7%	76.3%	67.5%	51.1%	72.9%	65.9%
Mixtral 8x22B	59.1%	79.2%	75.1%	57.1%	75.9%	74.1%	58.4%	79.9%	75.7%	55.3%	77.1%	75.8%
LLaMA 2 70B	49.9%	72.5%	64.3%	47.3%	68.7%	64.2%	50.5%	74.5%	66.0%	49.4%	70.9%	65.1%

Figure 66 - Multilingual capabilities comparison (Source: Mistral.ai)

Maths & Coding

Model	Active parameters	Coding		Maths		
		HumanE	MBPP	GSM8K maj@1 (5-shot)	GSM8K maj@8 (8-shot)	Math maj@4
LLaMA 2 70B	70B	29.3%	49.8%	53.6%	69.6%	13.8%
Command R	35B	-	-	56.6%	-	-
Command R+	104B	-	-	70.7%	-	-
Mistral 7B	7B	26.22%	50.2%	36.5%	50.0%	12.7%
Mixtral 8x7B	12.9B	40.2%	60.7%	58.4%	74.4%	28.4%
Mixtral 8x22B	39B	45.1%	71.2%	78.6%	88.4%	41.8%

Figure 67 - Maths and coding comparison (Mistral.ai)

Applications Across Industries

Mixtral 8x22B’s versatility opens doors for applications across multiple domains. Some of its most prominent use cases include:

- **Multilingual Assistance:** The model’s proficiency in handling over 100 languages makes it an excellent tool for global businesses looking to enhance customer support or streamline international operations.
- **Code Generation and Debugging:** Mixtral delivers exceptional performance in software development tasks, assisting engineers in generating clean, error-free code and optimizing existing codebases.
- **Complex Decision-Making:** Its advanced reasoning capabilities make Mixtral suitable for industries such as finance and healthcare, where precision and accuracy are paramount.

Mistral AI has positioned Mixtral as a transformative tool for enterprises seeking to leverage AI for competitive advantage while maintaining control over computational costs.

Open-Source Availability and Accessibility

In line with Mistral AI's commitment to openness, Mixtral 8x22B is available as an open-source model, allowing developers to integrate it into their workflows without restrictions. The model can be deployed on various platforms, including on-premises systems, virtual private clouds, and public cloud services like Amazon Web Services (AWS) and Google Cloud. By offering Mixtral 8x22B as an open-source tool, Mistral ensures that organizations of all sizes can access cutting-edge AI capabilities without the barriers imposed by proprietary systems.

Additionally, Mixtral is fully compatible with major machine learning frameworks such as PyTorch and Hugging Face Transformers, making it easy for developers to fine-tune the model for specific applications. This flexibility empowers organizations to deploy AI solutions tailored to their unique needs while retaining complete control over their data and operations.

Performance Comparisons

Mixtral 8x22B stands out not only for its efficiency but also for its performance on par with—or even surpassing—larger, more resource-intensive models. Comparative studies have shown that Mixtral performs exceptionally well in areas such as reasoning, multilingual translation, and technical task execution. For example:

- **Reasoning Tasks:** Mixtral achieves results comparable to models with up to 175 billion parameters, showcasing its ability to compete with leading closed-source systems.
- **Multilingual Translation:** Its extensive language capabilities place it among the top models for real-world applications requiring cross-lingual understanding.
- **Coding and Problem-Solving:** The model consistently outperforms many alternatives in benchmarks like HumanEval, making it a valuable asset for software development teams.

6 Introduction to Generative AI Beyond Text

This chapter explores the transformative potential of generative AI in domains beyond text. It begins with an introduction to DALL·E, a groundbreaking tool revolutionizing visual content creation, followed by a detailed look at DALL·E 3, which sets new standards in image generation. The chapter concludes with OpenAI's Sora, showcasing the evolution of AI-powered assistance in various applications. Together, these advancements illustrate how generative AI is expanding into visual creativity and intelligent assistance, pushing the boundaries of what AI can achieve across different mediums.

6.1 Introducing DALL·E: Revolutionizing Visual Creativity Through Generative AI

DALL·E, developed by OpenAI, is a cutting-edge generative AI model designed to create visually compelling and imaginative images based on textual prompts. This transformative technology represents a significant leap forward in the intersection of artificial intelligence and visual creativity, enabling users to translate words into intricate, high-quality visuals. As an integral part of OpenAI's ecosystem, DALL·E stands as a testament to the potential of AI in fostering artistic innovation and accessibility. (OpenAI, n.d.)

Overview of DALL·E

DALL·E is an AI model specifically built for generating images that are not only photorealistic but also conceptually imaginative. Unlike traditional graphic design tools, DALL·E uses natural language processing (NLP) and advanced deep learning algorithms to interpret user input and produce highly detailed visuals. Whether prompted to generate surreal landscapes, realistic objects, or abstract art, DALL·E excels at producing creative and unique outputs that align closely with user intent.

With the ability to synthesize visual content across a vast array of styles and themes, DALL·E is redefining how individuals and organizations approach image creation. Its application spans industries, from marketing and entertainment to education and design, offering a versatile tool for creators at all levels.

Core Features of DALL·E

1. **Text-to-Image Generation:** At its core, DALL·E transforms textual descriptions into highly detailed and visually rich images. This capability allows users to articulate their ideas in words and receive customized visuals that align with their vision.
2. **Inpainting:** DALL·E includes advanced inpainting features, enabling users to edit existing images by replacing or modifying specific areas. This feature is

particularly valuable for refining creative projects or adapting visuals to new contexts.

3. **Style Transfer:** Users can instruct DALL·E to generate images in specific artistic styles, from Renaissance paintings to modern graphic design. This flexibility empowers artists and designers to experiment with aesthetics without the need for extensive manual effort.
4. **Resolution Scaling:** The latest iterations of DALL·E prioritize generating high-resolution images, ensuring outputs are suitable for professional applications such as advertising campaigns, presentations, and publications.
5. **User Customization:** DALL·E is designed with user interactivity in mind. By enabling iterative feedback and prompt adjustments, the model ensures that the generated outputs are refined and meet user expectations.



Figure 68 - Example of DALL·E work (Source: OpenAI.com)

Applications and Use Cases

DALL·E's versatility has unlocked a broad range of applications across multiple sectors:

- **Creative Industries:** Artists, graphic designers, and content creators can use DALL·E to conceptualize and generate unique visuals, saving time and inspiring innovation.
- **Marketing and Advertising:** DALL·E enables brands to create tailored visual content for campaigns, adapting images to align with brand identity and market trends.
- **Education and Academia:** Educators and researchers leverage DALL·E to produce engaging visual aids, helping to simplify complex concepts and make learning more interactive.
- **Entertainment and Media:** Game developers, filmmakers, and writers utilize DALL·E to visualize settings, characters, and story elements, enhancing the creative process.
- **E-commerce:** Businesses can design product imagery, promotional banners, and personalized customer experiences using DALL·E's advanced image generation capabilities.

Ethical Considerations and Safeguards

OpenAI has integrated robust ethical frameworks into DALL·E's development to ensure responsible use. The model employs advanced filters and moderation tools to prevent the generation of harmful or inappropriate content. By leveraging reinforcement learning with human feedback (RLHF), DALL·E aligns its outputs with ethical guidelines while prioritizing user safety and inclusivity.

Additionally, OpenAI has implemented a watermarking system for DALL·E outputs, distinguishing AI-generated images from human-created visuals. This initiative promotes transparency and addresses concerns surrounding the misuse of AI-generated content.

The Role of DALL·E in OpenAI's Ecosystem

DALL·E complements OpenAI's existing models, such as GPT, by bridging the gap between text and visual content. Its integration with OpenAI's broader ecosystem allows users to generate holistic multimedia outputs, combining text, images, and even code for comprehensive creative projects.

DALL·E also supports API integration, enabling businesses and developers to embed its capabilities directly into their applications. This makes it a scalable solution for organizations looking to incorporate AI-driven image generation into their workflows.

Future Potential of DALL·E

As DALL·E continues to evolve, its potential applications are expanding. OpenAI is exploring ways to enhance DALL·E's capabilities, such as enabling multi-modal generation (combining text, visuals, and audio) and improving its adaptability for real-time user interaction.

The future of DALL·E lies in its ability to democratize creativity, making high-quality image generation accessible to users of all skill levels. Its impact is expected to resonate across industries, empowering individuals and organizations to push the boundaries of what is visually possible.

6.1.1 Introducing DALL·E 3: A New Benchmark in Image Generation

DALL·E 3, the latest iteration in OpenAI's renowned generative AI series, redefines the capabilities of text-to-image models. Building upon the successes of its predecessors, DALL·E 3 integrates enhanced natural language understanding and improved image generation technologies. This leap forward empowers users to craft highly detailed, creative, and visually striking images from simple textual prompts, setting a new standard for AI-driven creativity. (OpenAI, 2023)



Figure 69 - PROMPT: An illustration of an avocado sitting in a therapist's chair, saying 'I just feel so empty inside' with a pit-sized hole in its center. The therapist, a spoon, scribbles notes (Source: OpenAI.com)

Key Advancements in DALL·E 3

1. **Enhanced Comprehension of Complex Prompts:** One of the standout features of DALL·E 3 is its superior ability to interpret intricate and nuanced prompts. Unlike earlier models, DALL·E 3 can understand longer and more detailed descriptions, enabling users to generate visuals that closely match their specific requirements.
2. **Seamless Integration with GPT-4:** DALL·E 3 integrates directly with OpenAI's GPT-4, allowing for a unique synergy between text-based and image-based creativity. Users can iteratively refine prompts through natural language interactions with GPT-4, enabling precise adjustments and yielding outputs that align more closely with their creative vision.
3. **Inpainting and Editing Features:** The model includes advanced inpainting capabilities, allowing users to modify existing images by focusing on specific sections. Whether refining details or incorporating new elements, this feature makes DALL·E 3 an invaluable tool for both iterative design and creative exploration.
4. **Safety and Ethical Guardrails:** OpenAI has implemented robust safety measures to mitigate the generation of harmful or inappropriate content. These include advanced filtering systems that detect and block problematic prompts, ensuring that DALL·E 3 is used responsibly.

DALL·E 3: Pioneering New Applications

DALL·E 3's enhanced functionality broadens its applications across multiple domains:

- **Creative Industries:** Artists and designers can use DALL·E 3 to conceptualize and produce unique visuals, from abstract art to photorealistic renderings. Its ability to translate imaginative prompts into tangible outputs accelerates the creative process.

- **Marketing and Advertising:** Brands leverage DALL·E 3 for tailored visual content in campaigns, creating unique imagery that resonates with target audiences.
- **Education and Research:** Educators and researchers utilize DALL·E 3 for visualizing concepts, creating teaching aids, and simplifying complex ideas.
- **Film and Gaming:** Filmmakers and game developers can visualize characters, settings, and story elements, streamlining the pre-production and design phases.
- **Publishing and Media:** Writers and journalists use DALL·E 3 to generate illustrations that complement their narratives, making content more engaging.

How DALL·E 3 Stands Out

1. **Image Quality:** DALL·E 3 generates high-resolution images with unparalleled detail and clarity. It excels at producing outputs that are both realistic and creative, ensuring its utility in professional environments.
2. **Creative Flexibility:** The model supports a wide range of styles, including photorealism, digital art, and surrealism. This versatility makes it suitable for a broad spectrum of creative projects.
3. **Ease of Use:** With its integration into the ChatGPT ecosystem, DALL·E 3 provides a user-friendly interface for prompt crafting and image generation. This accessibility democratizes image creation, enabling individuals without technical expertise to utilize the model effectively.
4. **Responsible AI: Use By** embedding ethical guidelines into its framework, DALL·E 3 prioritizes user safety. Features such as watermarking ensure transparency, distinguishing AI-generated content from human-created visuals.

Impact of DALL·E 3 on the Creative Landscape

DALL·E 3's ability to interpret nuanced prompts and generate high-quality visuals marks a significant milestone in the evolution of generative AI. By enabling the creation of complex and visually stunning images, it empowers individuals and organizations to push the boundaries of their creative endeavors. Whether for commercial purposes or personal projects, DALL·E 3 democratizes access to advanced image generation, ensuring that creativity is no longer limited by technical expertise.

Future Potential of DALL·E 3

As DALL·E 3 continues to evolve, OpenAI is exploring opportunities to enhance its features further. Future iterations may include even higher resolution outputs, real-time generation capabilities, and deeper integration with other AI models. These advancements will ensure that DALL·E remains at the forefront of generative AI, driving innovation and expanding the possibilities of AI-assisted creativity.

6.2 Introducing OpenAI's Sora: The New Frontier in AI-Powered Assistance

OpenAI's latest innovation, Sora, represents a groundbreaking leap in AI-powered assistance, offering users a seamlessly integrated platform designed to enhance productivity, accessibility, and collaboration. By combining cutting-edge artificial intelligence with practical tools, Sora is positioned as a transformative solution for individuals and organizations across various domains. This model is a testament to OpenAI's commitment to making advanced technology both accessible and user-centric. (OpenAI, n.d.)

Overview of Sora

Sora is more than a typical AI model—it is a comprehensive ecosystem designed to augment user productivity. Its features include instantaneous problem-solving, creative brainstorming capabilities, and robust support for highly specialized tasks such as programming and content generation. What sets Sora apart is its capacity to integrate seamlessly into user workflows, ensuring that its capabilities are both practical and easy to adopt.

The core features of Sora reflect OpenAI's dedication to innovation. Built on an advanced foundation model, Sora has been optimized for multitasking, contextual understanding, and user adaptability. OpenAI's focus is not only on raw computational power but also on aligning the model to user needs through iterative fine-tuning and feedback mechanisms.

Key Features of Sora

1. *Enhanced Contextual Awareness:* Sora excels at processing and responding to complex user queries. Its context-aware algorithms enable it to handle lengthy conversations or intricate prompts while maintaining coherence. This positions it as an ideal tool for tasks like drafting long-form content or summarizing extensive documents.
2. *Tool Integration:* OpenAI has embedded Sora with tools that allow users to interact with their data dynamically. This includes support for file analysis, code debugging, and project management, enabling seamless interactions between the AI and real-world workflows.
3. *Creative Applications:* Creativity lies at the heart of Sora's design. Whether generating innovative ideas for marketing campaigns, drafting creative writing, or solving novel problems, the AI serves as an effective collaborator that adapts to a range of tasks.
4. *Multimodal Capabilities Reflecting:* OpenAI's advances in multimodal AI, Sora is capable of processing and generating text, images, and structured data. This versatility enables users to input diverse types of data and receive tailored outputs that align with their goals.

Use Cases and Benefits

Sora has been crafted to support a broad spectrum of applications, ensuring relevance for diverse professional and personal settings. Key use cases include:

- **Business and Productivity:** Sora can draft emails, create reports, and automate mundane tasks, empowering professionals to focus on higher-value activities.
- **Research and Development:** With its ability to analyze large datasets and generate actionable insights, Sora aids researchers and engineers in solving complex problems more efficiently.
- **Education and Learning:** Students and educators can use Sora as a learning companion to simplify concepts, explore creative ideas, and improve academic outcomes.
- **Creative Industries:** Artists, writers, and content creators can leverage Sora to push the boundaries of creativity, enabling new forms of expression and artistic exploration.

Beyond these specific applications, Sora is designed to democratize access to AI, ensuring that individuals and organizations of all sizes can harness the power of generative AI.

Ethical Considerations

Aligned with OpenAI's broader principles, Sora emphasizes responsible AI usage. Its development has prioritized safety, fairness, and transparency. Mechanisms like reinforcement learning with human feedback (RLHF) and continuous monitoring ensure that Sora provides accurate, ethical, and inclusive outputs. OpenAI's commitment to safety is reflected in its detailed user guidelines and the implementation of safeguards against misuse.

Positioning Sora in the OpenAI Ecosystem

Sora is a pivotal addition to OpenAI's product lineup, complementing other offerings like GPT-4 and Codex. By addressing user needs that extend beyond text generation into interactive and dynamic problem-solving, Sora bridges the gap between traditional AI models and real-world functionality.

The model also integrates with OpenAI's existing tools, creating a cohesive ecosystem where users can leverage the strengths of multiple AI systems. Sora's compatibility with APIs and support for cloud-based and local applications further enhances its versatility.

7 Introduction to Platforms for Accessing LLMs

This chapter delves into two prominent platforms that provide streamlined access to Large Language Models (LLMs): Amazon Bedrock and Hugging Face. Amazon Bedrock offers a comprehensive solution for businesses to integrate generative AI seamlessly into their workflows, emphasizing scalability and enterprise-grade capabilities. Hugging Face, on the other hand, fosters a vibrant ecosystem for collaboration, innovation, and open-source machine learning. Together, these platforms demonstrate the diverse ways organizations and developers can leverage LLMs for tailored applications, showcasing the growing accessibility and versatility of AI technologies.

7.1 Amazon Bedrock: Revolutionizing Access to Generative AI

Amazon Bedrock, a cutting-edge service introduced by AWS, represents a transformative step in how organizations access and deploy generative AI models. Designed with enterprise users in mind, Bedrock eliminates the complexities traditionally associated with AI model deployment by offering seamless integration of foundation models directly into applications, without the need to manage infrastructure or host servers. This innovation aligns with AWS's commitment to democratizing AI technology and enabling businesses to focus on scaling and innovation rather than technical overhead. (Amazon Web Service, n.d.)

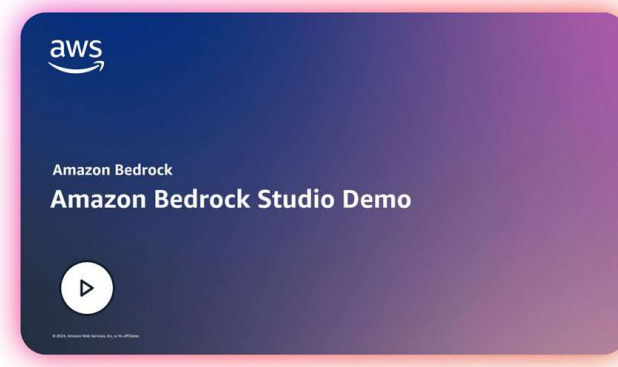


Figure 70 - Amazon Bedrock

Key Features of Amazon Bedrock

1. **Access to Foundation Models (FMs)**
Bedrock provides access to a variety of pre-trained foundation models from leading AI providers such as Anthropic, Cohere, Stability AI, and Amazon Titan. This diversity allows users to select models optimized for specific tasks, such as text generation, summarization, image creation, and chatbot functionality, ensuring tailored solutions for unique business needs.

2. **No Infrastructure Management**
Unlike traditional AI model deployment, which often requires significant investment in server infrastructure and GPU resources, Bedrock offers a fully managed environment. Users can access and deploy FMs without worrying about infrastructure scaling or maintenance, dramatically simplifying the adoption process and reducing operational costs.
3. **Customizable Outputs with Fine-Tuning**
Bedrock empowers users to fine-tune foundation models with their proprietary datasets, ensuring the outputs align closely with specific organizational needs. This customization capability allows businesses to maintain a competitive edge by generating highly relevant and context-specific results.

Use Cases and Security Framework

Amazon Bedrock stands as a comprehensive platform designed to unlock the potential of generative AI for enterprises. Beyond offering access to cutting-edge foundational models, Bedrock empowers organizations to implement AI-driven solutions with a strong emphasis on security, privacy, and governance. This section explores its diverse use cases and robust security framework.

Key Use Cases

1. **Text Generation**
Amazon Bedrock enables the creation of original written content, such as blog posts, social media updates, and website copy. By leveraging pre-trained foundational models, users can generate high-quality and contextually relevant text efficiently.
2. **Virtual Assistants**
The platform supports the development of virtual assistants that understand user requests, streamline task management, and execute actions autonomously. These assistants enhance customer service, internal workflows, and user engagement.
3. **Text and Image Search**
Bedrock allows users to search and synthesize relevant information from vast datasets, encompassing both text and images. This capability is particularly useful for answering complex queries and providing data-driven recommendations.
4. **Text Summarization**
By employing Bedrock's models, users can condense lengthy documents—such as reports, articles, or research papers—into concise summaries. This feature helps extract key insights and supports faster decision-making.
5. **Image Generation**
Bedrock facilitates the creation of visually compelling and realistic images. From marketing campaigns to web design, this feature provides businesses with a powerful tool for enhancing visual communication.

Security and Privacy

Amazon Bedrock prioritizes data security and privacy, ensuring organizations can safely adopt generative AI technologies without compromising sensitive information. Its security features include:

1. **Secure Customization**
Users retain complete control over the data used to customize foundational models. Data is encrypted both in transit and at rest, safeguarding it from unauthorized access.
2. **Data Protection and Privacy**
Bedrock guarantees that customer data remains under their control. When a foundation model is fine-tuned, it operates on a private copy of the data, ensuring it is not shared with external model providers.
3. **Governance and Auditability**
The platform integrates robust monitoring and logging tools to support compliance and governance requirements. These features enable organizations to maintain transparency and ensure adherence to regulatory standards.

Bridging Innovation and Security

Amazon Bedrock exemplifies a balanced approach to generative AI adoption, combining advanced functionality with comprehensive safeguards. By addressing key business needs—such as content creation, intelligent assistants, and secure data handling—Bedrock emerges as a trusted platform for organizations seeking to leverage AI responsibly and effectively.

7.2 Hugging Face: A Hub for Collaboration and Innovation in Machine Learning

Hugging Face has become one of the most prominent platforms in the world of Machine Learning (ML), especially in the field of Natural Language Processing (NLP). At its core, Hugging Face represents more than just a library or repository—it is a community-driven hub where researchers, data scientists, and developers converge to share resources, collaborate on projects, and contribute to the advancement of ML technologies. (Hugging Face, n.d.)

The Hugging Face Hub: A Community-Centric Platform

The Hugging Face Hub serves as a central repository and collaborative platform designed to simplify the workflows of ML practitioners. It hosts an impressive array of resources, including:

- Over 900,000 models for tasks like NLP, computer vision, and audio processing.
- 200,000 datasets tailored for various domains and modalities.

- 300,000 demos, enabling users to interactively experiment with state-of-the-art ML models.

The Hub fosters a community-centric approach to AI development, emphasizing the importance of knowledge sharing and collective innovation. Hugging Face operates under the belief that no single company, regardless of its resources, can "solve AI" alone. By enabling seamless collaboration, the Hub ensures that advancements in AI are accessible to all.

One of the key advantages of Hugging Face is the availability of pre-trained models, which eliminates the need for developers to build ML models from scratch. Instead, they can simply load a pre-trained model, fine-tune it for their specific task, and deploy it with minimal effort. This ease of use has significantly lowered the barrier to entry for working with sophisticated ML technologies.

As the platform describes itself, Hugging Face is “The AI community for building the future,” a statement that encapsulates its mission to democratize AI development and empower its users through shared resources and collaboration.

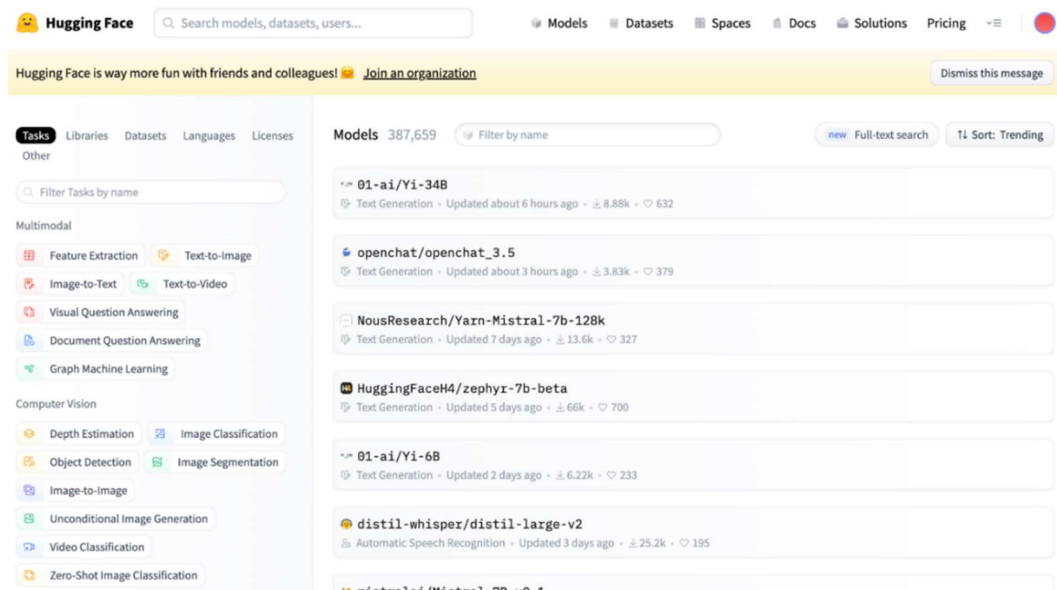


Figure 71 - Hugging Face interface

Core Components and Features

Hugging Face’s success lies in its rich ecosystem of tools and features, each designed to streamline and enhance ML workflows. Below are the most notable components:

1. Transformers Library

The Transformers library is at the heart of Hugging Face’s offerings. It is a comprehensive collection of state-of-the-art pre-trained models optimized for various NLP tasks, such as:

- Text classification
- Language generation
- Translation
- Summarization

The library abstracts much of the complexity involved in working with ML models, offering a simple `pipeline()` method that allows users to easily implement advanced functionalities without delving into the technical details of model initialization, tokenization, or preprocessing.



Figure 72 - Key features of transformer library (Source: huggingface.co)

Key features of the Transformers library include:

- *Abstraction of complexity*: Users can quickly set up workflows without needing deep technical expertise.
- *Pre-trained models*: The extensive collection of pre-trained models accelerates development and reduces resource requirements.
- *Flexibility and modularity*: The library is designed for customization, enabling users to adapt models for their specific needs.
- *Community-driven support*: Tutorials, documentation, and forums foster an active user base, ensuring newcomers and experts alike have access to guidance.
- *Continuous updates*: Hugging Face keeps the library up-to-date with the latest developments in ML, ensuring users can leverage cutting-edge advancements.

2. The Hugging Face Hub: Models, Datasets, and Spaces

The Hugging Face Hub hosts a variety of resources essential for ML development:

- **Models**:
The Hub offers access to thousands of open-source models for NLP, vision, and audio tasks. Each model is accompanied by a Model Card, which provides important information about its capabilities, limitations, biases, and evaluation metrics.

Users can experiment with models directly in their browsers via inference widgets or access them programmatically through serverless APIs.

- **Datasets:**

With over 200,000 datasets in more than 8,000 languages, the Hub supports a wide range of tasks across NLP, computer vision, and audio. Datasets are accompanied by Dataset Cards, offering detailed documentation about their content, purpose, and structure.

For large datasets, Hugging Face supports streaming, allowing users to access data efficiently without needing to download it in its entirety.

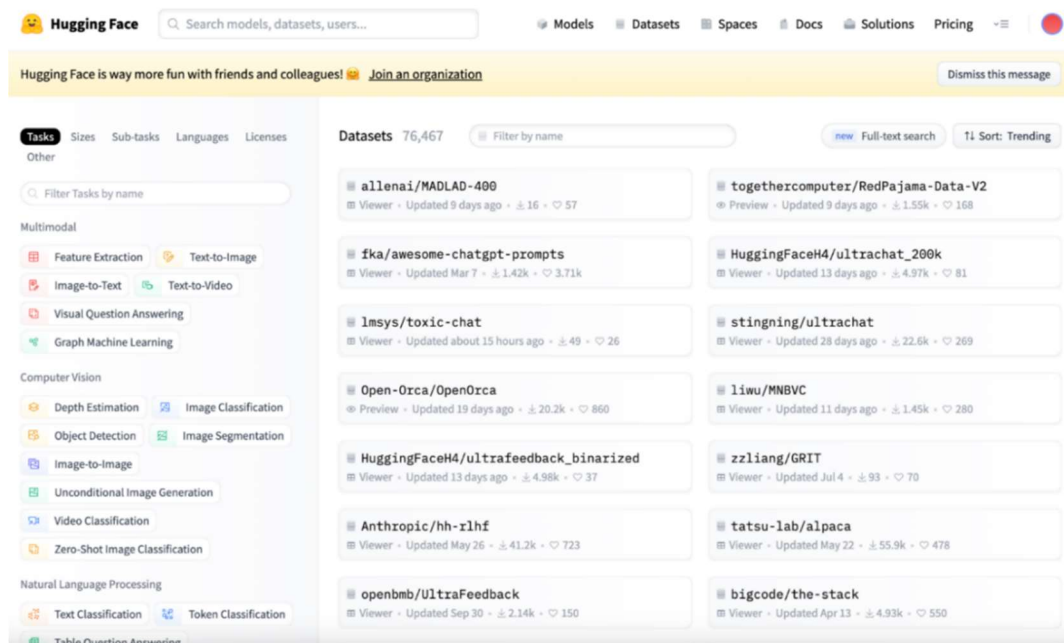


Figure 73 - Hugging Face datasets

- **Spaces:**

Spaces enable users to create and host interactive ML applications directly on the Hub. Using tools like Gradio and Streamlit, developers can build demo apps to showcase their projects, present ideas at conferences, or collaborate with others. Spaces also support static content, such as HTML/CSS/JavaScript pages, making them a versatile tool for ML practitioners.

Why Hugging Face Stands Out

Hugging Face is more than just a repository of tools—it is a thriving ecosystem built on the principles of open-source collaboration and community support. By providing pre-trained models, accessible APIs, and a robust community, Hugging Face has revolutionized how developers approach ML tasks.

Whether through its Transformers library, datasets, or Spaces, Hugging Face has successfully bridged the gap between cutting-edge research and practical application, empowering organizations and individuals to innovate in ways that were previously out of reach.

In summary, Hugging Face represents a paradigm shift in the field of ML, making advanced AI technologies more accessible, collaborative, and impactful than ever before.

8 Case Studies

This chapter presents real-world applications of generative AI, demonstrating its transformative potential across industries. By examining two detailed case studies, we explore how organizations are leveraging generative AI to address complex challenges. The first case focuses on how Google uses generative AI to accelerate incident response, enhancing efficiency and communication during security events. The second case highlights the rapid development capabilities of generative AI, showcasing its ability to streamline workflows and innovate solutions in dynamic environments. These examples illustrate the practical impact of generative AI in addressing real-world problems while optimizing resources and outcomes.

8.1 Accelerating incident response using generative AI

Today Google is highlighting another way they use generative AI to help the defenders gain the advantage: Leveraging LLMs (Large Language Model) to speed-up their security and privacy incidents workflows. (Google)

Introduction

Incident management is a team sport. They have to summarize security and privacy incidents for different audiences including executives, leads, and partner teams. This can be a tedious and time-consuming process that heavily depends on the target group and the complexity of the incident. Google estimates that writing a thorough summary can take nearly an hour and more complex communications can take multiple hours. But they hypothesized that they could use generative AI to digest information much faster, freeing up their incident responders to focus on other more critical tasks - and it proved true. Using generative AI we could write summaries 51% faster while also improving the quality of them.

Google incident response approach

At Google, when an incident is reported, Detection & Response Team works to restore normal service as quickly as possible, while meeting both regulatory and contractual compliance requirements. They do this by following the five main steps in the Google incident response program:

1. Identification: Monitoring security events to detect and report on potential data incidents using advanced detection tools, signals, and alert mechanisms to provide early indication of potential incidents.
2. Coordination: Triaging the reports by gathering facts and assessing the severity of the incident based on factors such as potential harm to customers, nature of the incident, type of data that might be affected, and the impact of the incident on customers. A communication plan with appropriate leads is then determined.

3. Resolution: Gathering key facts about the incident such as root cause and impact, and integrating additional resources as needed to implement necessary fixes as part of remediation.
4. Closure: After the remediation efforts conclude, and after a data incident is resolved, reviewing the incident and response to identify key areas for improvement.
5. Continuous improvement: Is crucial for the development and maintenance of incident response programs. Teams work to improve the program based on lessons learned, ensuring that necessary teams, training, processes, resources, and tools are maintained.

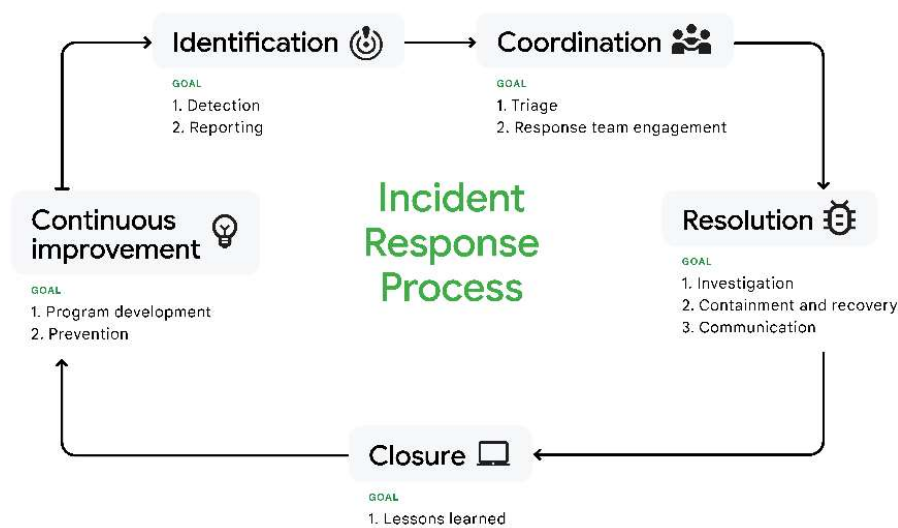


Figure 74 - Google incident response process (Source: Google Security Blog)

Leveraging generative AI

Google detection and response processes are critical in protecting their billions of global users from the growing threat landscape, which is why they're continuously looking for ways to improve them with the latest technologies and techniques. The growth of generative AI has brought with it incredible potential in this area, and they were eager to explore how it could help them improve parts of the incident response process. Google teams started by leveraging LLMs to not only pioneer modern approaches to incident response, but also to ensure that their processes are efficient and effective at scale.

Managing incidents can be a complex process and an additional factor is effective internal communication to leads, executives and stakeholders on the threats and status of incidents. Effective communication is critical as it properly informs executives so that they can take any necessary actions, as well as to meet regulatory requirements. Leveraging LLMs for this type of communication can save significant time for the incident commanders while improving quality at the same time.

Human vs LLMs

Given that LLMs have summarization capabilities, Google wanted to explore if they are able to generate summaries on par, or as well as humans can. They ran an experiment that took 50 human-written summaries from native and non-native English speakers, and 50 LLM-written ones with our finest (and final) prompt, and presented them to security teams without revealing the author.

The teams learned that the LLM-written summaries covered all of the key points, they were rated 10% higher than their human-written equivalents, and cut the time necessary to draft a summary in half.

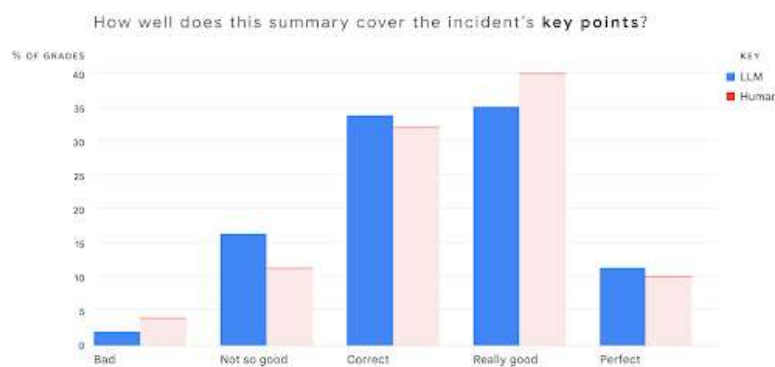


Figure 75 - Comparison of human vs LLM content completeness (Source: Google Security Blog)

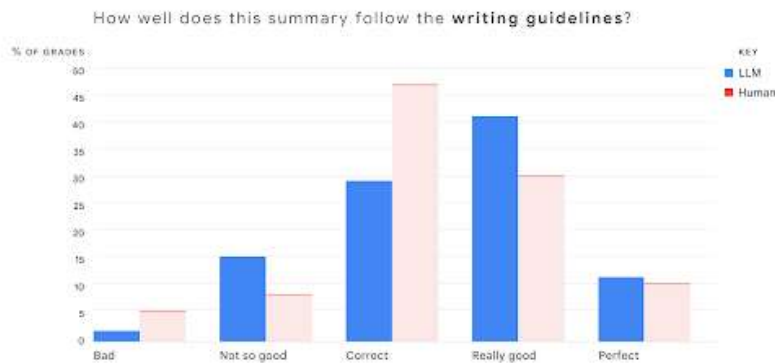


Figure 76 - Comparison of human vs LLM writing styles (Source: Google Security Blog)

Managing risks and protecting privacy

Leveraging generative AI is not without risks. In order to mitigate the risks around potential hallucinations and errors, any LLM generated draft must be reviewed by a human. But not all risks are from the LLM - human misinterpretation of a fact or statement generated by the LLM can also happen. That is why it's important to ensure there is human accountability, as well as to monitor quality and feedback over time.

Given that their incidents can contain a mixture of confidential, sensitive, and privileged data, they had to ensure they built an infrastructure that does not store any data. Every component of this pipeline - from the user interface to the LLM to output processing - has logging turned off. And, the LLM itself does not use any input or output for re-training. Instead, they use metrics and indicators to ensure it is working properly.

Input processing

The type of data they process during incidents can be messy and often unstructured: Free-form text, logs, images, links, impact stats, timelines, and code snippets. They needed to structure all of that data so the LLM “knew” which part of the information serves what purpose. For that, they first replaced long and noisy sections of codes/logs by self-closing tags (<Code Section/> and <Logs/>) both to keep the structure while saving tokens for more important facts and to reduce risk of hallucinations.

During prompt engineering, they refined this approach and added additional tags such as <Title>, <Actions Taken>, <Impact>, <Mitigation History>, <Comment> so the input’s structure becomes closely mirrored to their incident communication templates. The use of self-explanatory tags allowed them to convey implicit information to the model and provide them with aliases in the prompt for the guidelines or tasks, for example by stating “Summarize the <Security Incident>”.

```
<Security Incident>
<Title> [tool_name.verdict] Abuse verdict for project id: xyz.</Title>
<Metadata> This ticket was filled and submitted on the 2023-10-01. It was marked with the labels:
"Investigation" and "AB".</Metadata>
<Description> Counter-Abuse has issued an abuse verdict against a GCP project.</Description>
<Additional Information> The incident was reported through the xyz pipeline with a policy violation
of "COIN_MINING".
The infraction can be found in the project xyz.</Additional Information>
<Data Incident> 2023-10-01 11:50:19</Date Incident><Incident Causes> The identified causes are:
MISCONFIGURATION, WEAK_OR_NO_PASSWORD</Incident Causes><Actions Taken> The following actions were
taken:
1) Action1
2) Action2</Actions Taken>
<Software Involved> Software1</Software Involved>
<Sensitive Data> - NONE, TEST</Sensitive Data>
<Mitigation History><Comment index="1" author="user1@domain.com"> Looks like there was a CPU spike:
URL around 05:00. Running application1 now.</Comment>
<Comment index="2" author="user3@domain.com"> Instance compromised, shutting it down</Comment>
<Comment index="3" author="user4@domain.com"> InstanceMetadata</Comment>
<Comment index="4" author="user8@domain.com"> Get additional information on InstanceMetadata:
URL"<Code Section/>"</Comment>
<Comment index="5" author="user3@domain.com"> Looks like it was compromised through successfully
authentication as root account using SSH with password authentication: "<Code Section/>"</Comment>
<Comment index="6" author="user3@domain.com"> A malicious cron job was created on the machine
"<Code Section/>". The cron job downloaded a bash script from IP and executed it. The script was
not present under "<Code Section/>" at the time of the investigation "<Code Section/>"</Comment>
<Comment index="10" author="user3@domain.com"> Exec update sent.</Comment>
</Mitigation History>
```

Figure 77 - tags

Prompt engineering

Once they added structure to the input, it was time to engineer the prompt. They started simple by exploring how LLMs can view and summarize all of the current incident facts with a short task:

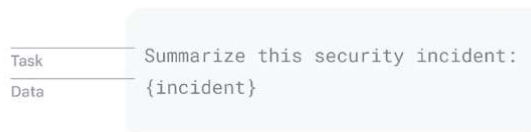


Figure 78 - short task for the model

Limits of this prompt:

- The summary was too long, especially for executives trying to understand the risk and impact of the incident
- Some important facts were not covered, such as the incident’s impact and its mitigation
- The writing was inconsistent and not following our best practices such as “passive voice”, “tense”, “terminology” or “format”
- Some irrelevant incident data was being integrated into the summary from email threads
- The model struggled to understand what the most relevant and up-to-date information was

Workflow integration

In integrating the prompt into their workflow, they wanted to ensure it was complementing the work of their teams, vs. solely writing communications. They designed the tooling in a way that the UI had a ‘Generate Summary’ button, which would pre-populate a text field with the summary that the LLM proposed. A human user can then either accept the summary and have it added to the incident, do manual changes to the summary and accept it, or discard the draft and start again.

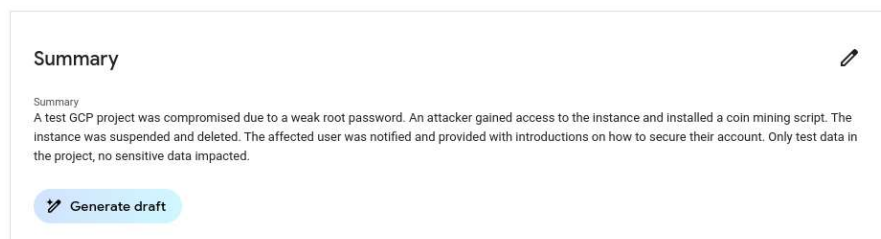


Figure 79 - UI showing the ‘generate draft’ button and LLM proposed summary around a fake incident

Quantitative wins

Their newly-built tool produced well-written and accurate summaries, resulting in 51% time saved, per incident summary drafted by an LLM, versus a human.

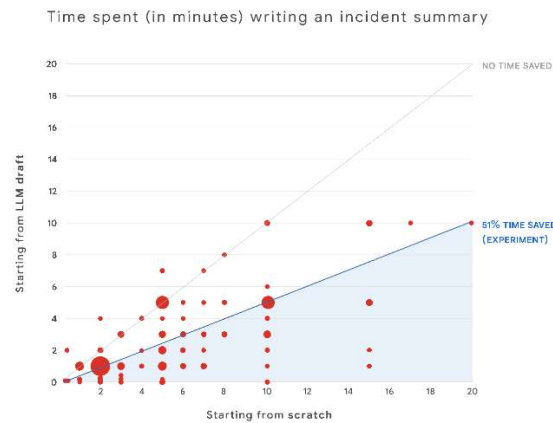


Figure 80 - Time saved (Source: Google Security Blog)

The only edge cases they have seen were around hallucinations when the input size was small in relation to the prompt size. In these cases, the LLM made up most of the summary and key points were incorrect. They fixed this programmatically: If the input size is smaller than 200 tokens, they won't call the LLM for a summary and let the humans write it.

Evolving to more complex use cases: Executive updates

Given these results, they explored other ways to apply and build upon the summarization success and apply it to more complex communications. They improved upon the initial summary prompt and ran an experiment to draft executive communications on behalf of the Incident Commander (IC). The goal of this experiment was to ensure executives and stakeholders quickly understand the incident facts, as well as allow ICs to relay important information around incidents. These communications are complex because they go beyond just a summary - they include different sections (such as summary, root cause, impact, and mitigation), follow a specific structure and format, as well as adhere to writing best practices (such as neutral tone, active voice instead of passive voice, minimize acronyms).

This experiment showed that generative AI can evolve beyond high level summarization and help draft complex communications. Moreover, LLM-generated drafts, reduced time ICs spent writing executive summaries by 53% of time, while delivering at least on-par content quality in terms of factual accuracy and adherence to writing best practices.

8.2 Developing rapidly with Generative AI

Let's see the approach of Discord for solving problems with generative AI, along with insights on rapidly launching new features leveraging this technology.

They break down the process of building with LLMs into a few stages. Starting with product ideation and defining requirements, they first need to figure out what they're building and how it can benefit users. Next, they develop a prototype of our idea, learn

from small-scale experiments, and repeat that process until our feature is in a good state. Finally, they fully launch and deploy their product at scale. In this section, we will dive deeper into each stage of this process. (Discord)

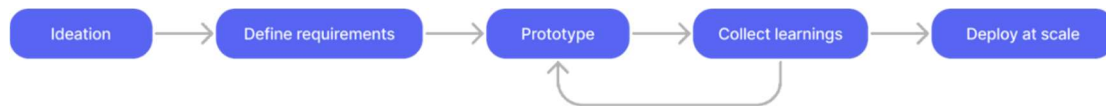


Figure 81 - The different stages of building an LLM-powered feature (Source: discord.com)

How Discord identifies use cases for generative AI

They start by having empathy for their users and for their staff - what are the opportunities that generative AI can help address? Like machine learning in general, generative AI is a tool — and one that shouldn't be applied when other tools are a better fit. When it comes to identifying where generative AI can make an impact, they dig into challenges that commonly:

- Involve analysis, interpretation, or review of unstructured content (e.g. text) at scale
- Require massive scaling that may be otherwise prohibitive due to limited resources
- Would be challenging for rules-based or traditional ML approaches

Defining product requirements

Once they've identified a potential use case for a generative AI application, the next step involves defining the product requirements. This phase requires a thoughtful analysis to select the best-suited LLM and to frame their problem as a prompt to an LLM.

They consider these aspects of their problem:

- Latency: How fast does the system need to respond to user input?
- Task Complexity: What level of understanding is required from the LLM? Is the input context and prompt super domain-specific?
- Prompt Length: How much context needs to be provided for the LLM to do its task?
- Quality: What is the acceptable level of accuracy for the generated content?
- Safety: How important is it to sanitize user input or prevent the generation of harmful content and prompt hacking?
- Language Support: Which languages does the application need to support?
- Estimated QPS: What throughput does their system eventually need to handle?

Several factors, such as complexity, prompt length, and quality, often conflict with the need for low latency, primarily because a bigger, more capable LLM usually delivers better outcomes but operates more slowly during inference owing to the model's larger size. Consequently, if minimizing response time is critical, they can consider either incurring higher costs (e.g. by having more available compute) or accepting a drop in quality by using smaller models.

Prototyping AI applications: From Idea to MVP

The product requirements the workers define then play into your selection of which off-the-shelf LLM you'll use for your prototype. You generally lean towards picking more advanced commercial LLMs to quickly validate your ideas and obtain early feedback from users. Although they may be expensive, the general idea is that if problems can't be adequately solved with state-of-the-art foundational models like GPT-4, then more often than not, those problems may not be addressable using current generative AI tech. If an off-the-shelf LLM can address the problem, then you can step into the learning stage and concentrate on iterating on your product rather than diverting engineering resources towards building and maintaining machine learning infrastructure.

Evaluating Prompts

The key step at this stage is to create the right prompt. You start with a basic prompt that tells ChatGPT (or whatever LLM you selected for our prototype) what you want it to do. Then, you make adjustments to this prompt, changing the wording to make the task clearer. However, after a lot of adjustments, it's often difficult to tell if these changes are actually improving our results. That's where evaluating the prompts becomes crucial. By using metrics to guide your changes, you know you are moving the needle on the quality of your results.

To do this, Discord employs a technique known as AI-assisted evaluation, alongside traditional metrics for measuring performance. This helps them pick the prompts that lead to better quality outputs, making the end product more appealing to users. AI-assisted evaluation uses best-in-class LLMs (like GPT-4) to automatically critique how well the AI's outputs match what they expected or how they score against a set of criteria. This method uses GPT-4 in a way that's similar to the critic model found in the actor-critic algorithm in reinforcement learning where a separate model is used to evaluate how well the model used for inference performed. Automating evaluation allows them to quickly see what's working well and what needs to be tweaked in their prompts, without having to manually check everything. When evaluating, they design prompts that ask for simple yes or no answers or rate the outputs on a scale, making the evaluation process straightforward.

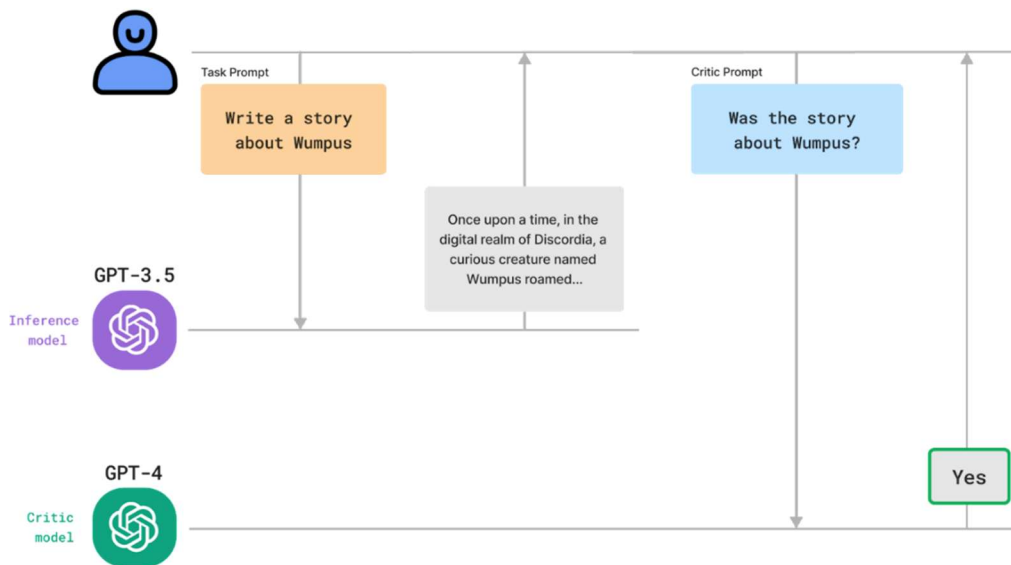


Figure 82 - AI-assisted evaluation consists of 2 separate prompts: one for your task and another to evaluate your results. The task prompt is passed to the inference model whereas the critic prompt is passed to the more advanced critic model. (Source: discord.com)

Launch and Learn

Once they are sufficiently confident in the quality of the results their prompt generates, they roll out a limited release (e.g. A/B test) of their product and observe the system's performance in situ. The exact metrics they use depend on the application — their main goal is to understand how users use the feature and quickly make improvements to better meet their needs. For internal applications, this might mean measuring efficiency and sentiment. For consumer-facing applications, they similarly focus on measures of user satisfaction - direct user feedback, user engagement measures, etc. This feedback is critical to identify areas for improvement, including highlighting incorrect answers or instances where LLM hallucinations might be causing a strange user experience.

Beyond user satisfaction, they also pay attention to system health metrics, such as response speed (latency), throughput (tokens per second), and error rates. LLMs sometimes have trouble generating output in a consistently structured format, which is crucial for minimizing data parsing errors and ensuring the output is robustly usable in our services. Insights here can inform how much post-hoc processing might be needed to fully productionize this capability at scale.

Keeping an eye on costs is equally important for understanding how much they will spend when they fully scale up the feature. They look at how many tokens per second they're using in their initial limited release to predict the costs of a complete launch if they were to use the same technology that's powering their prototype.

All of the above information is critical to understanding if their product is working as intended and providing value to users. If it is, then they can proceed to the next step: deploying at scale. If not, then they look to take their learnings, iterate on the system, and try again.

Deploying at Scale

LLM Application Architecture

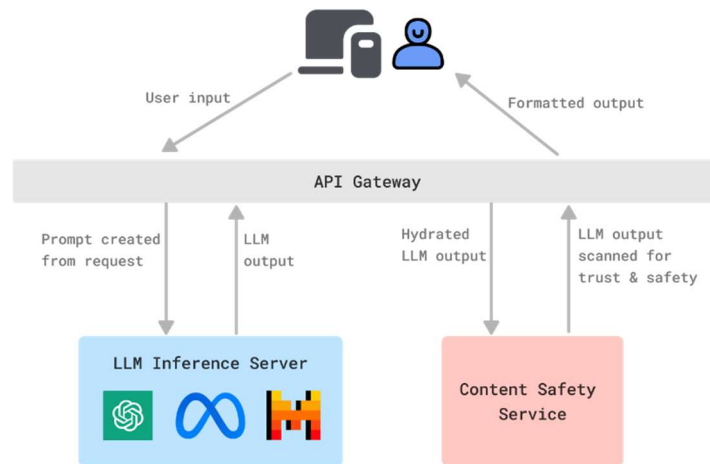


Figure 83 - A high-level architecture for an LLM application (Source: discord.com)

The basic setup for apps using LLMs consists of several essential parts. Inputs to the inference server are prepared into a prompt that they've tested and evaluated on a robust set of examples. At the heart of the architecture lies the LLM inference server, tasked with the job of operating the LLM to produce answers from the inputs it gets. Examples of such servers commercially include ChatGPT or other OpenAI GPT APIs, which are specialized in generating content with low latency.

Because they care deeply about the user experience, privacy, and safety, they work with cross-functional partners like Legal and other Safety teams to ensure they've implemented thoughtful mitigations, while adhering to privacy principles such as data minimization. For example, they chose to incorporate content safety filters to the output of the inference server to identify undesired material before it reaches the user. They can leverage in-house or third-party trust and safety ML models to detect inappropriate content.

All these elements put together form a system that taps into the capabilities of LLMs efficiently while monitoring the content's quality and safety to ensure that they're delivering a quality end product.

Self-hosted LLMs

When they're thinking about adding a feature that uses LLMs, they consider many tradeoffs when designing their LLM inference server such as balancing the costs and the amount of engineering effort. Using commercial LLMs is great because it gives us access to top-notch models and they don't have to worry about setting up the tech ourselves, but the expenses can add up quickly. For privacy reasons, they may also prefer to process full-scale data completely in-house. A solution is to self-host an open-sourced or custom fine-tuned LLM. Opting for a self-hosted model can reduce costs

dramatically - but with additional development time, maintenance overhead, and possible performance implications. Considering self-hosted solutions requires weighing these different trade-offs carefully.

Recent open-source models, like Llama and Mistral, are making high-quality results possible right out of the gate, even for complex tasks that traditionally required a model to be trained specifically for them. However, for domain-specific or complex tasks, the team might still need to fine-tune the model to achieve excellent performance. They've found it's best to start with smaller models and only move up to bigger ones if needed for quality reasons.

Setting up the necessary machine learning infrastructure to run these big models is another challenge. They need a dedicated model server for running model inference (using frameworks like Triton or vLLM), powerful GPUs to run everything robustly, and configurability in our servers to make sure they're high throughput and low latency. Tuning the inference servers for optimal performance is task-specific - the best configuration depends on the models they're using, as well as the input and output token lengths, and ultimately impacts how efficiently the server can batch input requests to maximize throughput.

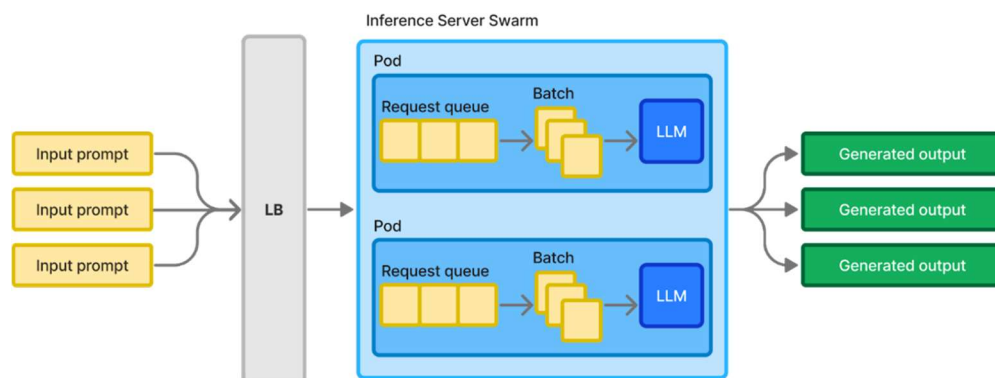


Figure 84 - Self-hosted inference server (Source: discord.com)

Closing Thoughts

Looking ahead, there's little doubt that generative AI will only grow more important as a means of solving massive-scale business-critical problems. Balancing cost, engineering effort, and performance will remain challenging, and Discord's team is excited to see (and contribute to) the rapid development of novel technology and tools to more effectively do so in the coming years!

CONCLUSIONS

The rapid advancements in Generative AI have ushered in a transformative era, redefining the boundaries of creativity, efficiency, and innovation across multiple industries. This thesis has explored the evolution, foundational technologies, applications, and ethical considerations of Generative AI, providing a comprehensive understanding of its potential and limitations. From early neural networks to state-of-the-art large language models, Generative AI has proven to be a versatile and powerful tool capable of generating text, images, and even assisting in critical workflows.

A significant focus was placed on the dichotomy between open-source and closed-source models, highlighting how community-driven initiatives like Llama 3.1 and Gemma democratize AI access while proprietary models like GPT-4 Turbo and Gemini Ultra push the boundaries of performance and innovation. Both approaches offer unique advantages and challenges, emphasizing the importance of collaboration and ethical development in shaping AI's future.

The case studies included in this research demonstrate the tangible impact of Generative AI in real-world scenarios, such as accelerating incident response and enhancing productivity. These examples underscore the potential of AI to solve complex problems efficiently while also raising critical concerns about accountability, data privacy, and the risk of misuse.

In conclusion, Generative AI represents a monumental leap in technological progress, but its integration into society must be approached with caution and responsibility. Developers, policymakers, and businesses must work together to ensure that this technology is not only accessible but also aligned with ethical principles and societal values. The insights provided in this thesis aim to contribute to this dialogue, offering a roadmap for harnessing the power of Generative AI to drive innovation while mitigating its risks.

Bibliography

- Amazon Web Service. (n.d.). *Aamazon Bedrock*. Retrieved from https://aws.amazon.com/bedrock/?nc1=h_ls
- Amazon. (n.d.). *What are GANs?* Retrieved from Amazon Web Service.
- anshumanm2fja. (2024, 05 01). *What is Forward Propagation in Neural Networks?* Retrieved from Geeksforgeeks: <https://www.geeksforgeeks.org/what-is-forward-propagation-in-neural-networks/>
- Anthropic. (n.d.). Retrieved from Anthropic: anthropic.com
- Anthropic. (2024). Claude 3.5 Sonnet Model Card Addendum. Retrieved from Anthropic: anthropic.com
- Anthropic. (2024). The Claude 3 Model Family: Opus, Sonnet, Haiku. *Anthropic*. Retrieved from Anthropic: anthropic.com
- Artificial Analysis. (n.d.). *Artificial Analysis*. Retrieved from <https://artificialanalysis.ai/models>
- Bergmann, D., & Stryker, C. (2023, 11 23). *What is an autoencoder?* Retrieved from IBM: <https://www.ibm.com/topics/autoencoder#:~:text=An%20autoencoder%20is%20a%20type,input%20from%20this%20compressed%20representation.>
- Dhinakaran, A. (2024, 11). *LLM Evaluation: Everything You Need To Run, Benchmark LLM Evals*. Retrieved from arize: <https://arize.com/blog-course/llm-evaluation-the-definitive-guide/#why-precision-recall>
- Discord. (n.d.). *Developing Rapidly with Generative AI*. Retrieved from Discord Blog: <https://discord.com/blog/developing-rapidly-with-generative-ai>
- Ferrer, J. (2024, 01 09). *How Transformers Work: A Detailed Exploration of Transformer Architecture*. Retrieved from datacamp: <https://www.datacamp.com/tutorial/how-transformers-work>
- Foote, K. (2024, 03 05). *A Brief History of Generative AI*. Retrieved from Dataversity: <https://www.dataversity.net/a-brief-history-of-generative-ai/>

George, B., & Gillis, A. (2024, 08). *What is the Turing Test?* Retrieved from Techtarget: <https://www.techtarget.com/searchenterpriseai/definition/Turing-test>

Google. (2024). Gemma 2: Improving Open Language Models.

Google. (n.d.). *Accelerating incident response using generative AI*. Retrieved from Google Security Blog: <https://security.googleblog.com/2024/04/accelerating-incident-response-using.html>

Google DeepMind. (n.d.). Retrieved from deepmind.google

Google DeepMind. (2024). *Gemini: A Family of Highly Capable*. Retrieved from arxiv.org

Google DeepMind. (n.d.). *Google DeepMind*. Retrieved from <https://deepmind.google/technologies/gemini/ultra/>

Hugging Face. (n.d.). *Hugging Face*. Retrieved from <https://huggingface.co/>

IBM. (2023, 07 06). *AI vs. machine learning vs. deep learning vs. neural networks: What's the difference?* Retrieved from IBM: <https://www.ibm.com/think/topics/ai-vs-machine-learning-vs-deep-learning-vs-neural-networks>

IBM. (2023). *What are large language models (LLMs)?* Retrieved from IBM: <https://www.ibm.com/topics/large-language-models>

Ippolito, P. (2024, 12 14). *Introduction to Autoencoders: From The Basics to Advanced Applications in PyTorch*. Retrieved from datacamp: <https://www.datacamp.com/tutorial/introduction-to-autoencoders>

Kalirane, M. (2023, 04 05). *Gradient Descent vs. Backpropagation: What's the Difference?* Retrieved from Analytics Vidhya: <https://www.analyticsvidhya.com/blog/2023/01/gradient-descent-vs-backpropagation-whats-the-difference/>

Kerner, M. (2024, 05). *What are large language models (LLMs)?* Retrieved from techtarget: <https://www.techtarget.com/whatis/definition/large-language-model-LLM>

- Marr, B. (2023, 08 23). *The Difference Between Generative AI And Traditional AI: An Easy Explanation For Anyone*. Retrieved from Forbes:
<https://www.forbes.com/sites/bernardmarr/2023/07/24/the-difference-between-generative-ai-and-traditional-ai-an-easy-explanation-for-anyone/>
- Merritt, R. (2022, 03 25). *What Is a Transformer Model?* Retrieved from Nvidia:
<https://blogs.nvidia.com/blog/what-is-a-transformer-model/>
- Meta . (2024). The Llama 3 Herd of Models.
- Meta. (2023). LLaMA: Open and Efficient Foundation Language Models.
- Mistral AI. (n.d.). *Cheaper, Better, Faster, Stronger*. Retrieved from
<https://mistral.ai/news/mixtral-8x22b/>
- Mistral AI. (n.d.). *Mistral AI*. Retrieved from <https://mistral.ai/>
- OpenAI. (2023). *GPT-4 Technical Report*. Retrieved from arxiv.com
- OpenAI. (2023). Improving Image Generation with Better Captions.
- OpenAI. (n.d.). *DALL·E: Creating images from text*. Retrieved from
<https://openai.com/index/dall-e/>
- OpenAI. (n.d.). *OpenAI*. Retrieved from OpenAI: <https://openai.com/>
- OpenAI. (n.d.). *Video generation models as world simulators*. Retrieved from
<https://openai.com/index/video-generation-models-as-world-simulators/>
- Toloka. (2023, 08 22). *History of generative AI*. Retrieved from Toloka:
<https://toloka.ai/blog/history-of-generative-ai/>
- Weng, L. (2021, 07 11). *What are Diffusion Models?* Retrieved from Github:
<https://lilianweng.github.io/posts/2021-07-11-diffusion-models/>
- Wohlfarth, A. (2024, 03 25). *Generative AI vs. Traditional AI: Understanding the Differences*. Retrieved from OurCrowd:
<https://www.ourcrowd.com/learn/generative-ai-vs-traditional-ai>

